PHYSICS

SECTION-I (i)

1. Ans (A)

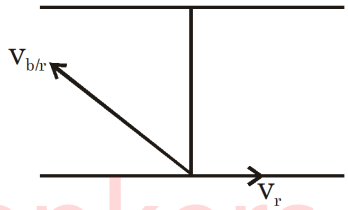
$$\frac{1}{a}=\frac{1}{4}v$$

$$\frac{dt}{dv}=\frac{1}{4}v$$

$$\int_{t_1}^{t_2} dt=\int_{2}^{4}\frac{v}{4}dv$$

$$t_2-t_1=\frac{1}{4}\times\frac{1}{2}[16-4]=\frac{3}{2}$$

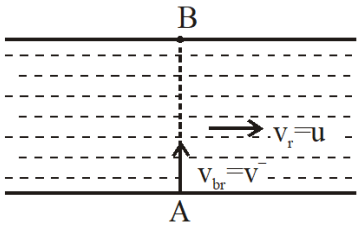
2. Ans (C)



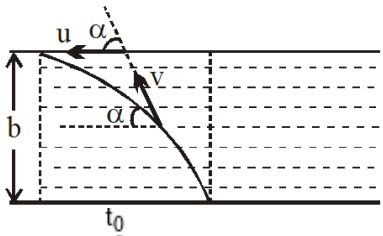
$v_r > v_{b/r}$, so zero drift in not possible.

(C) boatman should be mantain velocity v of boat at certain angle greater than 90° with direction of river flow to minimize drifting & than walk rest of distance on other bank.

3. Ans (A)



In the river frame dock will move towards left with speed u



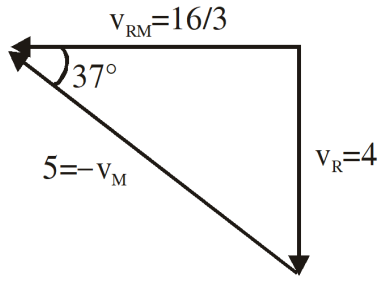
$$ut_0=v\int_0^{t_0}\cos\alpha\,dt \qquad \text{.....(1)}$$

$$v-u\cos\alpha=-\frac{ds_{rel}}{dt}$$

$$\int_0^{t_0}(v-u\cos\alpha)\,dt=\int_b^0 -ds_{rel} \qquad \text{.....(2)}$$

$$vt_0-\frac{u^2t_0}{v}=b,\ t=\frac{bv}{v^2-u^2}$$

4. Ans (C)



5. Ans (B)

Tension = T − mg = ma

T = m(g + a)

T = 10(10 + 2)

T = 10 × 12 = 120

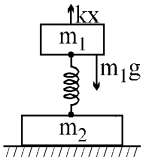
Upper support 120 + 80 = 200

(80) weight of latter monkey

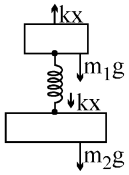
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6. **Ans (A)**

Initially $m_1g = kx$



When support is removed, spring force does not change.

New FBD 

For m_1 : $m_1g - kx = m_1a_1 \quad \Rightarrow a_1 = 0$

For m_2 : $m_2g + kx = m_2a_2 \quad \Rightarrow a_2 = \dfrac{(m_1 + m_2)g}{m_2}$

PHYSICS

SECTION-I (ii)

7. **Ans (A,C)**

Using $C = \dfrac{1}{\sqrt{\mu_0 \epsilon_0}}$ & $R = \sqrt{\dfrac{\mu_0}{\epsilon_0}}$ we can check the correctness.

(A) $\mu_0 I^2 = \epsilon_0 V^2$

$\dfrac{\mu_0}{\epsilon_0} = \dfrac{V^2}{I^2} = R^2$

$\therefore R^2 = R^2$ correct

(B) $\epsilon_0 I = \mu_0 V$

$\dfrac{\epsilon_0}{\mu_0} = \dfrac{V}{I}$

$\dfrac{1}{R^2} = R$ not correct

(C) $I = \epsilon_0 cV$

$\dfrac{I}{V} = \epsilon_0 c = \dfrac{\epsilon_0}{\sqrt{\mu_0 \epsilon_0}}$

$\dfrac{1}{R} = \sqrt{\dfrac{\epsilon_0}{\mu_0}} = \dfrac{1}{R}$ correct

(D) $\mu_0 CI = \epsilon_0 V$

$\dfrac{\mu_0 C}{\epsilon_0} = \dfrac{V}{I}$

$\dfrac{\mu_0}{\epsilon_0} \dfrac{1}{\sqrt{\mu_0 \epsilon_0}} = R$

$\therefore \dfrac{1}{\epsilon_0}\sqrt{\dfrac{\mu_0}{\epsilon_0}} = R \Rightarrow \dfrac{R}{\epsilon_0} = R$ incorrect

8. **Ans (A,B,C,D)**

$\dfrac{dv(t)}{dt} = 6.0 - 3v(t) = 0$ (for terminal speed)

(A) $6 = 3v$

$v = 2$ m/s

(B) initial acceleration at $t = 0$ is $= 6 - 3(0)$

$= 6$ m/s^2

(C)

$\dfrac{dv}{dt} = 6 - 3v \Rightarrow \int_0^v \dfrac{dv}{6-3v} = \int_0^t dt \Rightarrow \left|\dfrac{-1}{3}\ell n(6-3v)\right|_0^v = t$

$\ell n \dfrac{6-3v}{6} = -3t \Rightarrow 6 - 3v = 6e^{-3t}$

$6[1 - e^{-3t}] = 3v \Rightarrow v = 2(1 - e^{-3t})$

$v = 2(1 - e^{-3t})$ m/s

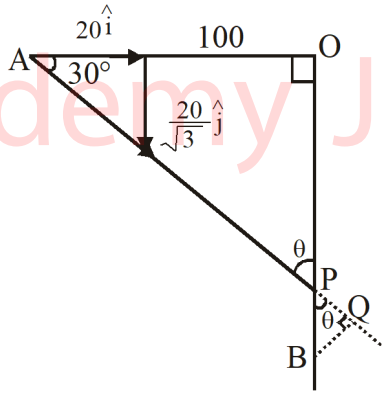
(D) $a = 3$ m/s^2

$3v = 6 - 3$

$3v = 3$

$v = 1$ m/s

9. **Ans (B,C)**



In frame of B

$\vec{v}_{A/B} = 20\hat{i} - \dfrac{20}{\sqrt{3}}\hat{j}$

$|\vec{v}_{A/B}| = \dfrac{40}{\sqrt{3}}$ m/s

$\dfrac{OP}{100} = \tan 30 = \dfrac{1}{\sqrt{3}}$

$OP = \dfrac{100}{\sqrt{3}}$

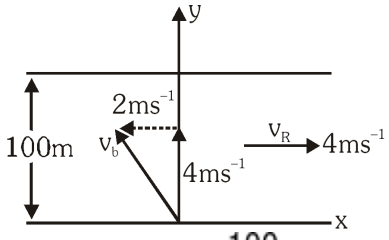
Minimum distance $\perp$ = BQ = BP sin q

$BP = \dfrac{200}{\sqrt{3}}$; $\theta = 60°$

In frame of B velocity of A is not along B, so they does not collide.

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10. Ans (A,B,C)



Time of crossing = $\frac{100}{4}$ = 25s

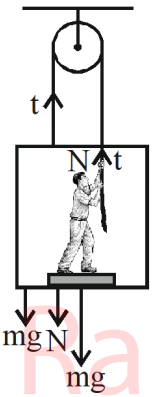
Absolute velocity of boat (i.e. w.r.t. ground)

= $|\vec{v}_{br} + \vec{v}_r|$

$|\vec{v}_{br} + \vec{v}_r| = \left|-2\hat{i} + 4\hat{j} + 4\hat{i}\right| = (2\hat{i} + 4\hat{j}) = \sqrt{20} = 2\sqrt{5}ms^{-1}$

Drift of the boat = (4.2) × 25 = 50m

11. Ans (A,B)



Man N + T = mg

N + T = 500 .....(i)

Weghing machine T = mg + N

T − N = mg

T − N = 200 ....(ii)

N + T = 500

T − N = 200

2T = 700

T = 350

N + 350 = 500

N = 150

Reading 15 kg

T = 350

12. Ans (A,B,C)

F − f = ma

30 − 10 = m × 10 μmg = 10

20 = 10m μ20 = 10

m = 2kg $\mu = \frac{1}{2}$

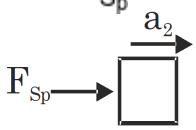
block start at force = 10. Limiting friction is 10.

13. Ans (B,C,D)

Spring force does not change instantaneously.

Thus for m_1 ; $a_1 = a_0$

for m_2 $F_{Sp} = m_2 a_2$ .....(i)



instantaneously aftr F_2 is withdrawn

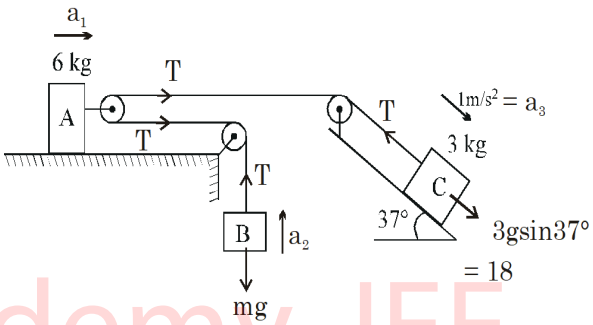
Initially $F_{Sp} - F_2 = m_2 a_0$

$F_{Sp} = F_2 + m_2 a_0$ .....(ii)

from (i) and (ii) $a_2 = \frac{F_2}{m_2} + a_0$

14. Ans (A,B,C)

M_B = 15 kg, T = 15 , a_A = 5 m/s^2, a_B = 9 m/s^2



By string constrain

$2Ta_1 + Ta_2 - Ta_3 = 0$

$2a_1 + a_2 = 1$ .... (i)

by F = ma

18 − T = 3 × 1 ⇒ T = 15

$2T = 6a_1 \Rightarrow a_1 = \frac{2 \times 15}{6} = 5m/s^2$

by equation (i) 2(5) + a_2 = 1 ⇒ a_2 = −9

T − mg = ma_2 ⇒ 15 − 10 m = m (−9)

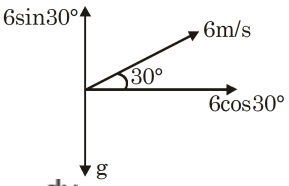
⇒ m = 15 kg

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PHYSICS

SECTION-I (iii)

16. Ans (D)



$a = \frac{dv}{dt} = -2t$

$v = -t^2 + 3$

$s = 3t - \frac{t^3}{3}$

when particle returns to ground

$s = 0, -\frac{t^3}{3} + 3t = 0, \ t = 3s$

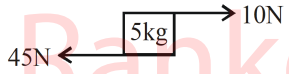
So time of flight = 3s

Range = $6 \cos 30° \times 3 = 9\sqrt{3}$

19. Ans (A)

At F = 10N, motion between blocks starts

20. Ans (C)



So the block will never move

21. Ans (A)

$30 - 10 = 10\ a$

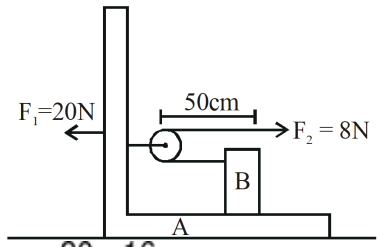
$20 = 10\ a$

$a = 2$

PHYSICS

SECTION-II (i)

1. Ans (0.5)



$a_A = \frac{20-16}{1} = 4 m/s^2$

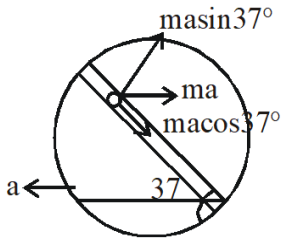
$a_B = \frac{8}{1} = 8 m/s^2$

$a_{B/A} = 4 m/s^2$

$0.5 = 1/2 \times 4t^2$

$t = \frac{1}{2} sec = 0.5 sec$

2. Ans (10)



$ma \cos 37 - \mu\, ma \sin 37 - \mu mg = ma_0$

$25 \times \frac{4}{5} - \frac{2}{5} \times 25 \times \frac{3}{5} - \frac{2}{5} \times 10 = a_0$

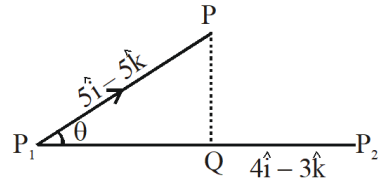
$20 - 6 - 4 = 90$

$a_0 = 10 \ m/s^2$

PHYSICS

SECTION-II (ii)

3. Ans (100)



$PP_1 = 5\sqrt{2}$

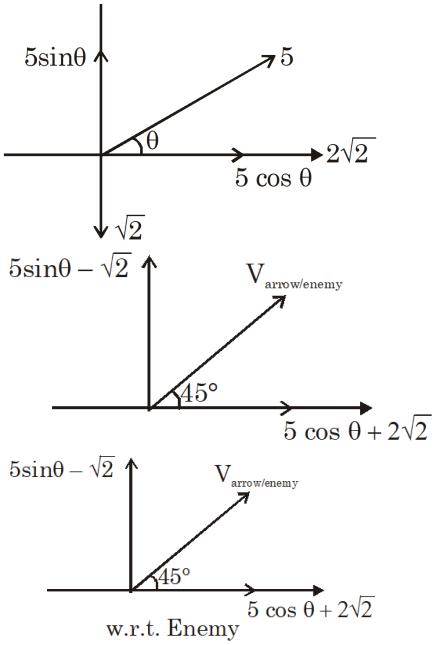
$\cos\theta = \frac{25+15}{\left(5\sqrt{2}\right) \times 5} = \frac{7}{5\sqrt{2}}, \ \sin\theta = \frac{1}{5\sqrt{2}}$

$PQ = (PP_1)\sin\theta = 5\sqrt{2} \times \frac{1}{5\sqrt{2}} = 1$

$t = \frac{PQ}{v} = \frac{1000}{10} = 100 \ sec$

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4. **Ans (82)**



w.r.t. Enemy

$5\cos\theta + 2\sqrt{2} = 5\sin\theta - \sqrt{2}$

$5\sin\theta - 5\cos\theta = 3\sqrt{2}$

$\frac{1}{\sqrt{2}}\sin\theta - \frac{1}{\sqrt{2}}\cos\theta = \frac{3}{5}$

$\sin(\theta - 45) = \sin 37°$

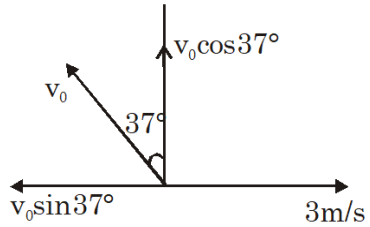
$\theta = 45° + 37°$

$\theta = 82°$

PHYSICS

SECTION-III

1. **Ans (5)**



$v_{ball/g} = v_{ball/boat} + v_{boat/g}$

In horizontal direction velocity of ball w.r.t. ground = 0

$v_0 \sin 37° = 3$

$v_0 = 3 \times \frac{5}{3} = 5$ m/s

2. **Ans (5)**

For collision

$u = \sqrt{u_B^2 + u_C^2} = \sqrt{\left(\frac{72}{5}\times\frac{5}{18}\right)^2 + \left(\frac{54}{5}\times\frac{5}{18}\right)^2} = \sqrt{4^2+3^2} = 5$ m/s

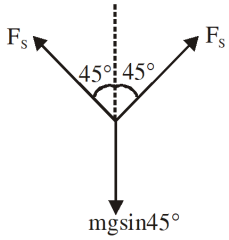
3. **Ans (8)**

$a = -3$

$u = \sqrt{\left(25\times\frac{4}{5} - 8\right)^2 + \left(25\times\frac{3}{5}\right)^2} = \sqrt{(12)^2 + (15)^2} = 3\sqrt{41}$

at $t = \frac{3\sqrt{41}}{3}$ so it will stop relative motion before 10 sec. So it will have same velocity as plank.

4. **Ans (5)**



$\frac{2F_S}{\sqrt{2}} = \frac{mg}{\sqrt{2}}$; $F_S = \frac{mg}{2}$ ∴

acceleration $= \frac{F_S}{m} = \frac{g}{2} = 5$ m/s^2

5. **Ans (0)**

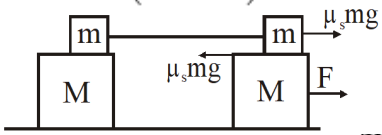


$f_k = 6$ N which will act till velocity is not zero.

Let velocity becomes zero at t, then $0 - 4.5 = \int_0^t (3-3t)dt \Rightarrow t = 3$ so at t = 3, v = 0

6. **Ans (2)**

$F = 2\mu_s mg\left(\frac{m+M}{2m+M}\right)$



$msmg = (M+2m)a \Rightarrow a = \frac{\mu_s mg}{M+2m}$

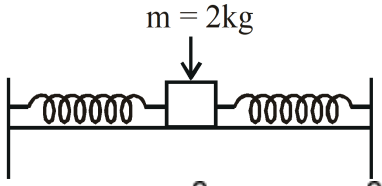
Also $F - \mu_s mg = Ma \Rightarrow 2\mu_s mg\frac{(m+M)}{(M+2m)}$

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PHYSICS

SECTION-IV

1. **Ans (A->S,B->S,C->Q,D->T)**



force = $100 \times \frac{2}{100} + 100 \times \frac{2}{100}$

$4 = 2a$; $a = 2$

(C) $100 \times \frac{4 \times 1}{100} + 100 \times \frac{4}{100}$

$8 = 2a$; $a = 4$

(D) $100 \times \frac{1}{100} + 100 \times \frac{1}{100}$

$2 = 2a$; $a = 1$

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PHYSICS

SECTION-I (i)

1. **Ans (D)**

About centre of circle :

$\omega_C = 2\omega = 0.8$ rad/sec

$v = \omega_C \cdot R = 0.4$ m/s

$a_C = \omega_C^2 R = 0.32$ m/sec

2. **Ans (A)**

Time of travel from A to B, $t = \sqrt{\frac{2h}{g}}$ In this time, the particle must make an integral number of rotations say n. $\Rightarrow 2\pi rn = ut \Rightarrow 2\pi rn = u\sqrt{\frac{2h}{g}}$

$\Rightarrow n = \frac{u}{2\pi r}\sqrt{\frac{2h}{g}}$

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3. **Ans (D)**

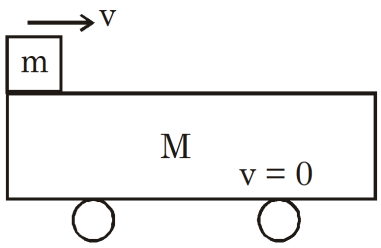
Suppose velocity of block at the end of slop = v

Then by work energy theorem

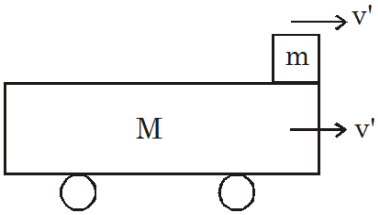
$mgh = \frac{1}{2}mv^2$

Now

Initially



Finally



Since there is no external force on (cart + block) system

$P_i = P_f$

$mv = (M + m)v'$

$v' = \frac{mv}{M+m}$

Now work done by friction = change in kinetic energy of system

$\therefore -\mu mg\ell = \frac{1}{2}(M+m)v'^2 - \frac{1}{2}mv^2$

$\mu mg\ell = \frac{1}{2}mv^2 - \frac{1}{2}(m+M)\left(\frac{m^2v^2}{(m+M)^2}\right)$

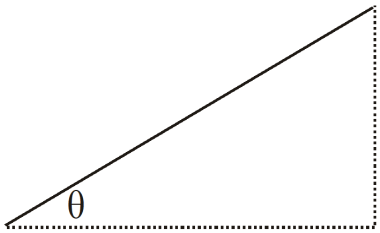
$\mu mg\ell = \frac{1}{2}\left(\frac{mM}{m+M}\right)v^2$;

$\mu mg\ell = \left(\frac{M}{m+M}\right)mgh; h = \mu\left(\frac{m+M}{M}\right)\ell$

4. Ans (C)

The speed at angle q is given by conservation of energy

$\Rightarrow \frac{1}{2}mv^2 = mgh \Rightarrow mv^2 = mg\,R\cos\theta \Rightarrow v = \sqrt{2gR\cos\theta}$ R will cancel out in the final answer. So we have the following setup of projectile motion.



$v = \sqrt{2gR\cos\theta}$

$v_x = v\cos\theta$

$v_y = v\sin\theta$

The time of flight is twice the time to get to the top

$\Rightarrow t = 2\left(\frac{v_y}{g}\right)$

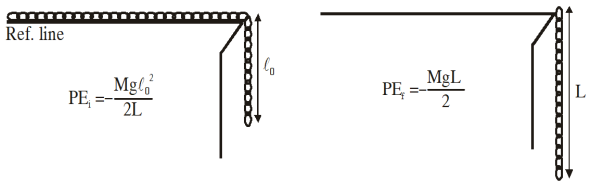
$\Rightarrow x = v_x t = v_x\left(\frac{2v_y}{g}\right) = \frac{2v_xv_y}{g} = \frac{2v_xv_y}{g} = \frac{2(v\cos\theta)(v\sin\theta)}{g} = \frac{2v^2\sin\theta\cos\theta}{g} = \frac{2(2gR\cos\theta)\sin\theta\cos\theta}{g} = 4R\cos^2\theta\sin\theta$

Maximise this $\Rightarrow$ take the derivative and equate it to zero

$\frac{dx}{d\theta} = -8R\cos\theta\sin^2\theta + 4R\cos^3\theta = 0 \Rightarrow 2\sin^2\theta = \cos^2\theta \Rightarrow \tan^2\theta = \frac{1}{2} \Rightarrow \tan\theta = \frac{1}{\sqrt{2}}$

5. Ans (C)

Potential energy of centre of mass is equal to potential energy of a system.



Loss in potential energy = gain in kinetic energy

$-\frac{Mg}{2}\left[\frac{\ell_0^2}{L} - L\right] = \frac{1}{2}MV^2 \Rightarrow V = \sqrt{\frac{g}{L}\left(L^2 - \ell_0^2\right)}$

6. Ans (B)

Loss in potential energy = gain in kinetic energy

$mg\left(\frac{h}{\cos\theta} - h\right) = \frac{1}{2}mv^2$

7. Ans (D)

$\frac{dy}{dx} = \tan\theta = x$

tangential acceleration = $g\sin\theta$

PHYSICS

SECTION-I (ii)

8. Ans (A,B,C)

$\vec{r} = ut[\cos\omega t\hat{i} + \sin\omega t\hat{j}]$

$\Rightarrow \vec{v} = \frac{d\vec{r}}{dt} = u[\cos\omega t\hat{i} + \sin\omega t\hat{j}] + ut[-\omega\sin\omega t\hat{i} + \omega\cos\omega t\hat{j}]$

$|a_1| = \frac{d|\vec{v}|}{dt} = \frac{\omega^2 tu}{\sqrt{1+\omega^2t^2}}$

9. Ans (A,B,C)

$\vec{F} = \frac{3\vec{r}}{r^3} = \frac{3\hat{r}}{r^2}$

$\vec{F}$ is a constant force and constant force is always conservative.

11. Ans (A,B,C)

(a) $ma = 3\mu mg - \mu mg \Rightarrow a = 2\mu g$

(b) From work-energy principle

$\frac{1}{2}k(x_0^2 - x_1^2) - mmg(x1 + x0) = 0$

$\Rightarrow x_0 - x_1 = 2\mu mg/k \Rightarrow x_1 = \mu mg/k$

(c) Speed will be maximum where net force is zero

$\mu mg = kx \Rightarrow x = \mu mg/k$ (extension)

Now from work-energy principle

$\frac{1}{2}k\left(\frac{\mu mg}{k}\right)^2(3^2 - 1^2) - \frac{2\mu mg}{k}(\mu mg) = \frac{1}{2}mv^2$

Solving we get

$v = 2\mu g\sqrt{\frac{m}{k}}$

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PHYSICS

SECTION-I (iii)

12. Ans (B)

by definition

$$U_r - U_\infty = -\int_\infty^r \left(\frac{-2\alpha}{r^3} + \frac{\beta}{r^2}\right) \cdot dr$$

$$= -\int_\infty^r -\frac{2\alpha}{r^3} \cdot dr - \int_\alpha^r \frac{\beta}{r^2} \cdot dr$$

$$= +\frac{2\alpha}{(-2)r^2}\Big|_\infty^r + \beta\frac{1}{r}\Big|_\infty^r$$

$$\Rightarrow -\alpha\left(\frac{1}{r^2} - 0\right) + \beta\left(\frac{1}{r} - 0\right) \Rightarrow -\frac{\alpha}{r^2} + \frac{\beta}{r}]$$

13. Ans (C)

At equilibrium $\vec{F} = 0 \Rightarrow -\frac{2\alpha}{r_0^3} + \frac{\beta}{r_0^2} = 0 \Rightarrow r_0 = \frac{2\alpha}{\beta}$

Ionization energy $E_0 = -\Delta U = -(U_\infty - U_{r_0})$

$$\Rightarrow \left[0 - \left(-\frac{\alpha}{r_0^2} + \frac{\beta}{r_0}\right)\right]$$

$$E_0 = \frac{\alpha}{(2\alpha/\beta)^2} - \frac{\beta}{(2\alpha/\beta)} \Rightarrow \frac{\beta^2}{4\alpha} - \frac{\beta^2}{2\alpha} = -\frac{\beta^2}{4\alpha}$$

$$= |E_0| = \frac{\beta^2}{4\alpha}$$

14. Ans (B)

Work required $\Rightarrow U_2 - U_1$ = work required.

$$W = U_2 - U_1 = -\frac{2\alpha}{r_0^2} + \frac{\beta}{r_0} =$$

$$\left[-\frac{2\alpha}{\left(r_0/\sqrt{2}\right)^2} + \frac{\beta}{\left(r_0/\sqrt{2}\right)^2}\right] - \left[-\frac{2\alpha}{\left(\sqrt{2}r_0\right)^2} + \frac{\beta}{\left(\sqrt{2}r_0\right)^2}\right]$$

]

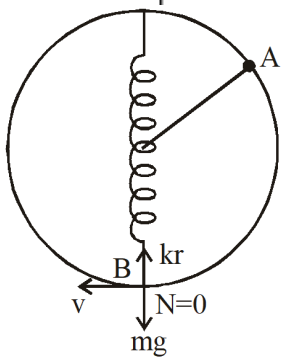
PHYSICS

SECTION-II

1. Ans (500)

Initial length of spring = r (relaxed)

$$kr - mg = \frac{mv^2}{r} \quad \text{... (i)}$$



COE at A & B

$$mgr\,(1 + \cos 60°) - \frac{1}{2}kr^2 = \frac{1}{2}mv^2 \Rightarrow mv^2 = 3mgr$$

From equaiton (i)

$kr - mg = 3mg$

$kr = 4mg$

$$k = \frac{2mg}{r} = 500$$

2. Ans (729)

$$\frac{dv}{dt} = \frac{v^2}{R} \Rightarrow \int_{v_0}^{v} \frac{dv}{v^2} = \frac{1}{R}\int_0^t dt$$

$$\Rightarrow \left(-\frac{1}{v}\right)_{v_0}^{v} = \frac{t}{R} \Rightarrow v = \frac{v_0}{1 - \frac{v_0}{R}t}$$

$$\Rightarrow \frac{ds}{dt} = \frac{v_0}{1 - \frac{v_0}{R}t} \Rightarrow \int_0^{2\pi R} ds = \int_0^t \frac{v_0}{\left(1 - \frac{v_0}{R}t\right)} dt$$

$$\Rightarrow 2\pi R = -R\left[\ell n\left(1 - \frac{v_0}{R}t\right)\right]_0^t$$

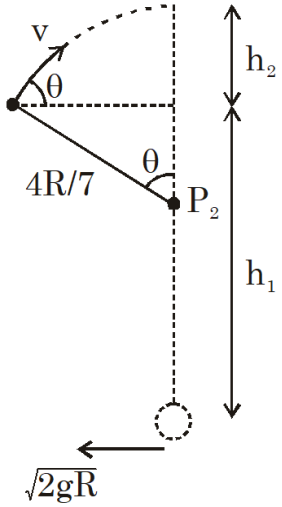
$$\Rightarrow 2\pi = -\ell n\left(1 - \frac{v_0}{R}t\right) \Rightarrow 1 - \frac{v_0}{R}t = e^{-2\pi}$$

$$\Rightarrow t = \frac{R}{v_0}\left(1 - e^{-2\pi}\right)$$

$$\Rightarrow a = 1, \beta = 2 \Rightarrow (\alpha + \beta)^6 = (1 + 2)^6 = 729$$

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3. **Ans (27)**



At lowest point velocity of bob = $\sqrt{2gR}$

Angle where the string slack & bob start performing projecitle motion = q

$$mg\cos\theta = \frac{mv^2}{\frac{4R}{7}} \quad \text{... (i)}$$

Applying work energy theorem

$$-mg\left(\frac{4R}{7}\right)(1+\cos\theta) = \frac{1}{2}mv^2 - \frac{1}{2}m\left(\sqrt{2gR}\right)^2$$

.... (ii)

Solving (i) & (ii)

$$\cos\theta = \frac{1}{2} \quad \text{and} \quad V = \sqrt{\frac{2gR}{7}}$$

$$h_1 = \frac{4R}{7}(1+\cos\theta)$$

$$h_2 = \frac{v^2\sin^2\theta}{2g}$$

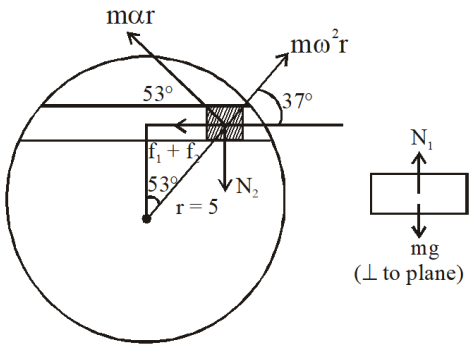
Maximum height from lowest point = $h_1 + h_2$

PHYSICS

SECTION-III

1. **Ans (2)**

w.r.t table :-



$f_1 = \mu N_1$; $f_2 = \mu N_2$

also at time t

$\omega = \omega_0 + at = t$

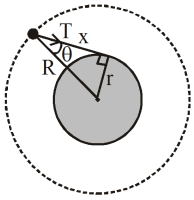
block start slide if resultant of $-(m\vec{a}_c + m\vec{a}_T) + \vec{N}_2$ become greater than maximum total friction

$\Rightarrow m\omega^2 r\cos 37° - m\alpha\, r\cos 53° > \mu\,(m\omega^2 r\sin 37° + m\alpha r\sin 53°) + \mu mg$

$$\Rightarrow \frac{4t^2}{5} - \frac{3}{5} > \frac{1}{2}\left[\frac{3}{5}t^2 + \frac{4}{5} + \frac{g}{r}\right]$$

$\Rightarrow t^2 > 4 \Rightarrow t > 2$ sec.

2. **Ans (1)**



$F_{CP} = ma_C = 1\,(a_C)$

$F_C = T\cos\theta = a_C$

$a_C = 2\,[x/R]$

$$\because x = \sqrt{R^2 - r^2} \Rightarrow \sqrt{\frac{4r^2}{3} - r^2} \qquad x = \frac{r}{\sqrt{3}}$$

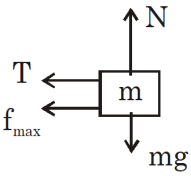
$$\therefore a_C = \frac{2\sqrt{R^2 - r^2}}{R}$$

$a_C = 1\ m/s^2$

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3. **Ans (3)**

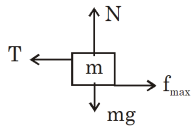
$T = mg$



For r_{max},

$T + f_{max} = m\omega^2 r_{max}$

$mg + \mu_S mg = m\omega^2 r_{max}$ .... (i)



For r_{min},

$T - f_{max} = m\omega^2 r_{min}$

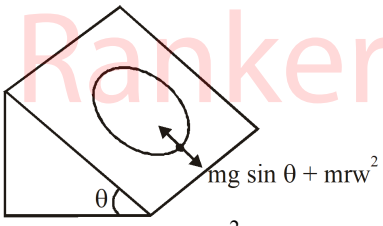
$mg - \mu_S mg = m\omega^2 r_{min}$ .... (ii)

4. **Ans (3)**

(a) $m_1 g \sin\theta = (m_1+m_1)a_1$

(b) $(m_2 - m_1)\, g \sin \theta/_2 = (m_1 + m_2)\, a_2$

5. **Ans (6)**



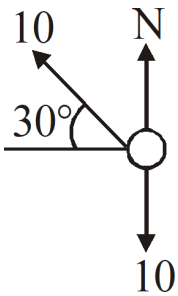
$f \geq mg \sin\theta + mrw^2$

$\mu\, mg \cos\theta \geq mg \sin\theta + mrw^2$

or $\mu \geq \tan\theta + \dfrac{w^2 r}{g\cos\theta}$

6. **Ans (5)**

$T\cos\alpha = m\omega^2 (0.1)\cos\alpha$



$T = 1 \times 100 \times 0.1 = 10$

$10\sin 30^\circ + N - 10 = 0 \Rightarrow N = 5$

7. **Ans (1)**

When the block has fallen by 60°, $\frac{1}{2}mv^2 = mgR(1-\cos\theta) \Rightarrow v = \sqrt{2gR \times \frac{1}{2}} = \sqrt{gR}$

Applying NLM along the radial direction $mg\cos\theta + kx = \dfrac{mv^2}{R} \Rightarrow x = \dfrac{1}{2}m$

Extension in spring = 0.5 m ⇒ Natural length = (R − x) = 1m

PHYSICS

SECTION-IV

1. **Ans (A->Q,B->PQT,C->PT,D->P)**

(A) Friction along tangent only.

(B) Friction provide centripetal and tangential acceleration.

(C) Friction provide only centripetal acceleration.

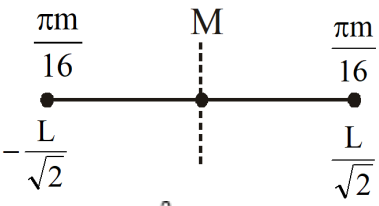
(D) If speed is not equal to $\sqrt{rg\tan\theta}$ then friction is along the incline of road to avoid slipping & if speed is equal to $\sqrt{rg\tan\theta}$, no friction.

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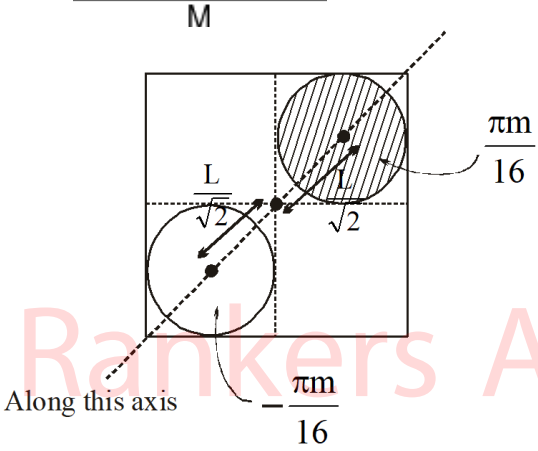
PHYSICS

SECTION-I (i)

1. Ans (B)



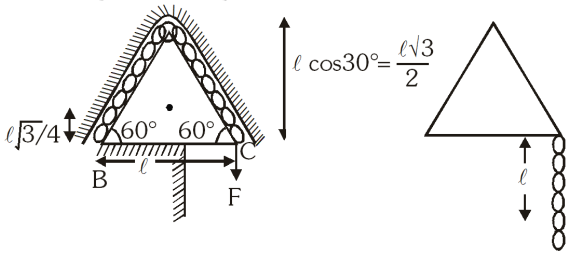
$$X_{Cr} = \frac{M\pi\left(\frac{L}{2}\right)^2}{4L^2} \times \frac{\left(L\hat{i} + L\hat{j}\right)}{M}$$



2. Ans (D)

Change in position of CM in vertical direction

$$(\Delta h) = \left(\ell\frac{\cos 30^\circ}{2}\right) + \ell$$

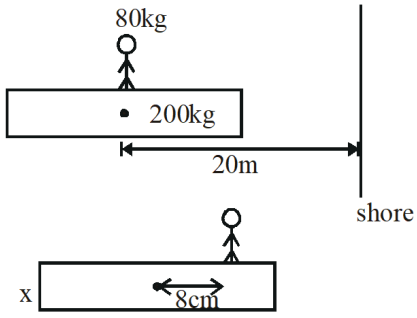


∴ Work done by gravity

$$= + mg\Delta h = + mg\left(\frac{\ell\sqrt{3}}{4} + \ell\right) = + mg\ \ell\left(\frac{\sqrt{3}+4}{4}\right)$$

3. Ans (C)

Let the displacement of boat be x towards the shore



$$80\,(x + 8) + 200\,x = 0$$

$$\Rightarrow x = \frac{-640}{280} = -\frac{16}{7}$$

$$\therefore\ (\Delta x)_{mon} = 8 - \frac{16}{7} = \frac{40}{7}$$

∴ New distance from shone

$$= 20 - \frac{40}{7} = \frac{100}{7} = 14.28\text{m} \approx 14.3\text{ m}$$

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PHYSICS

SECTION-I (ii)

5. Ans (B,D)

$W_{man} = \Delta KE$

From momentum conservation $mv_{man} = Mv$

For (A) : Work done by man on boat = ΔKE_{boat}

$$= \frac{1}{2}Mv^2$$

For (B) :

$$W_{man\ total} = \Delta KE_{system} = \frac{1}{2}m\frac{M^2v^2}{m^2} + \frac{1}{2}Mv^2$$

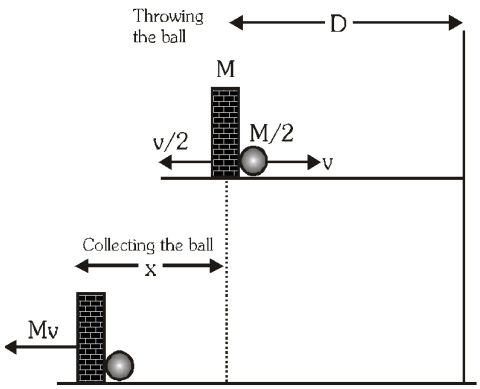
For (C) : $V_{CM} = 0$ {momentum conservation}

For (D) :

$$W_{man\ on\ himself} = \Delta KE_{man} = \frac{1}{2}\frac{M^2}{m}v^2$$

6. Ans (A,D)

From conservation of linear momentum velocity of man =



From conservation of linear momentum, momentum of man + ball system after he collects the ball

$$p_2 = \frac{Mv}{2} + \frac{M}{2}v = Mv \quad ...(i)$$

For (A) : Velocity of man + ball system

$$= \frac{\text{momentum}}{\text{mass}} = \frac{Mv}{M + \frac{M}{2}} = \frac{2v}{3}$$

For (B) : Linear impulse by the ball on man

= change in momentum of the man

$$= M\left(\frac{2v}{3}\right) - \frac{Mv}{2} = \frac{Mv}{6}$$

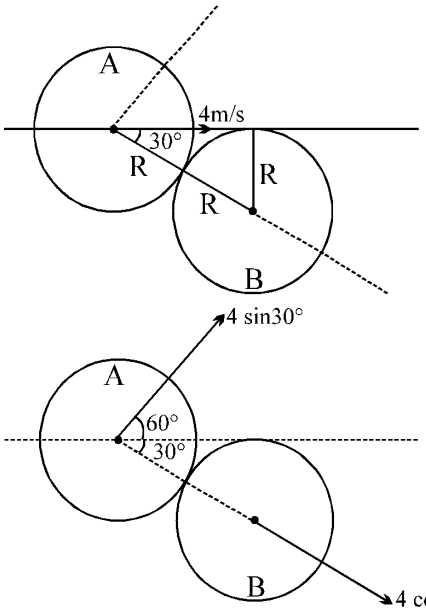
For (D) : $t = \frac{x}{v/2} = \frac{2D + x}{v} \Rightarrow x = 2D$.

So time $t = \frac{x}{v/2} = \frac{4D}{v}$

7. Ans (B,C,D)

Linear momentum conservation along horizontal can be made only if there is no friction or non–impulsive friction.

9. Ans (A,C)



$v_A = 4 \sin 30°[\cos 60\,\hat{i} + \sin 60\,\hat{j}]$; $v_A = \hat{i} + \sqrt{3}\hat{j}$

$\overrightarrow{J_{A\text{ on }B}} = m\overrightarrow{V_{B_f}} - \overrightarrow{V_{B_i}} = m[4\cos 30°(\cos 30°\hat{i} - \sin 30°\hat{j}) - 0] = (3mi - \sqrt{3}mj)$kg-m/s

10. Ans (B,C)

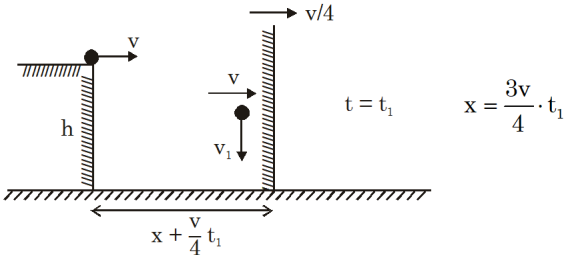
$$h' = \frac{e^2v^2\sin^2\alpha}{2g} = e^2\frac{2gh}{2g} \times \frac{3}{4}$$

$$h' = \frac{3h}{4}e^2$$

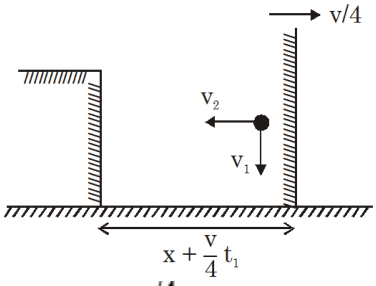
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11. Ans (A,C)

Just before collision with wall



Just after collision with wall

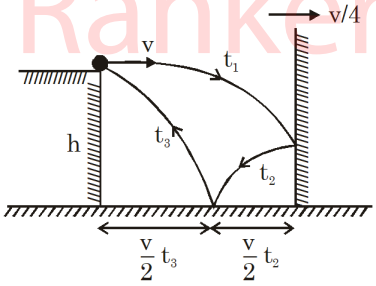


$$e = 1 = \frac{v_2 + v/4}{v - v/4}$$

$$\Rightarrow v_2 = \frac{v}{2}$$

Due to collision of particle with wall, there is no effect on vertical motion of particle.

∴ Total time $T = 2\sqrt{\frac{2h}{g}}$



$$t_3 = \sqrt{\frac{2h}{g}}$$

$$t_1 + t_2 = \sqrt{\frac{2h}{g}} = t_3$$

$$\frac{v}{2} \cdot t_3 + \frac{v}{2} \cdot t_2 = x + \frac{v}{4} \cdot t_1$$

$$\frac{v}{2} \cdot t_3 + \frac{v}{2}(t_3 - t_1) = x + \frac{v}{4} \cdot t_1$$

$$v \cdot t_3 - \frac{v}{2} \cdot t_1 = x + \frac{v}{4} \cdot t_1$$

$$vt_3 = x + \frac{3}{4} \cdot v \cdot t_1$$

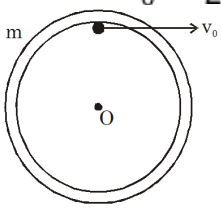
$$vt_3 = 2x$$

$$x = \frac{v}{2} \cdot t_3 = \frac{v}{2}\sqrt{\frac{2h}{g}}$$

$$x = v\sqrt{\frac{h}{2g}}$$

12. Ans (A,C)

Straight line, $\frac{2\pi r}{v_0}, \frac{\sqrt{3}v_0}{2}$

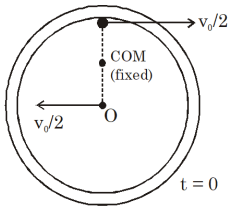


For given system (ring + particle) ⇒ Fext = 0

$$\vec{v}_{cm} = \text{constant} = \frac{mv_0}{2m} = \frac{v_0}{2}$$

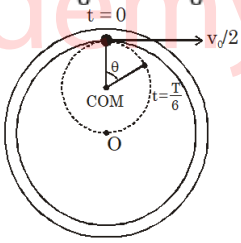
∴ Path of COM is straight line

In COM frame ⇒



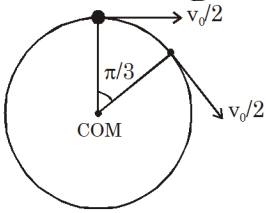
In COM frame particle moves in a circle with constant speed $\frac{v_0}{2}$. Velocity of tyre again becomes zero after one complete revolution.

$$T = \frac{2\pi r/2}{v_0/2} = \frac{2\pi r}{v_0}$$



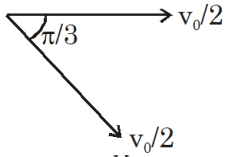
At $t = \frac{T}{6}$

$$\theta = \omega \cdot \frac{T}{6} = \frac{v_0}{2 \cdot \frac{r}{2}} \cdot \frac{T}{6} = \frac{v_0}{6r} \cdot \frac{2\pi r}{v_0} = \frac{\pi}{3}$$



$$\vec{v}_{Particle} = \vec{v}_{Particle,\,COM} + \vec{v}_{COM}$$

=

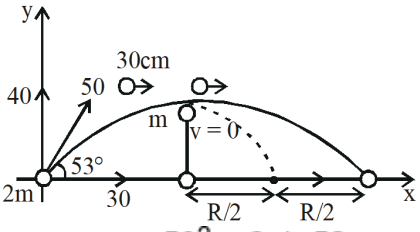


$$\vec{v}_{Particle} = \frac{v_0}{2}\sqrt{3}$$

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13. Ans (A,B)

(i) , (ii)



(i) Range $R = \frac{50^2 \times 2\sin 53^\circ . \cos 53^\circ}{10} = 250 \times 2 \times .6 \times .8$

$R = 240$ m

$\therefore d = R + \frac{R}{2} = \frac{3R}{2} = 360$ m

(ii) From $P_f = P_i$ in horizontal

$2m \times 30 = mv$

$\Rightarrow v = 60$ m/s

Just before explosion

$K_i = \frac{1}{2}.2m.v_x^2 \quad 2m = 24 \Rightarrow m = 12$

$= \frac{1}{2} \times 24 \times (30)^2 = \frac{1}{2} \times 21600J = 10800J$

Just after explosion

$k_f = \frac{1}{2}mv^2 = \frac{1}{2} \times 12 \times 60^2 = 21600J$

$\therefore$ Energy released = $k_f - k_i = 21600 - 10800 = 10800$ J

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PHYSICS

SECTION-I (iii)

15. Ans (B)

$\frac{h_1}{\sqrt{3}} = \tan 60^\circ \Rightarrow h_1 = 3$m

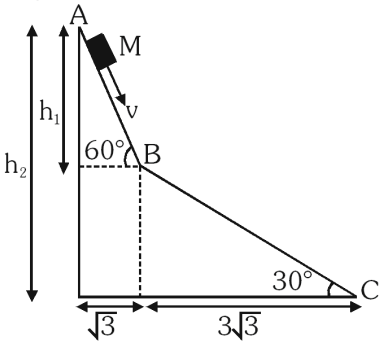
$\frac{h_2 - h_1}{3\sqrt{3}} = \tan 30^\circ$

$\Rightarrow h_2 - h_1 = 3$

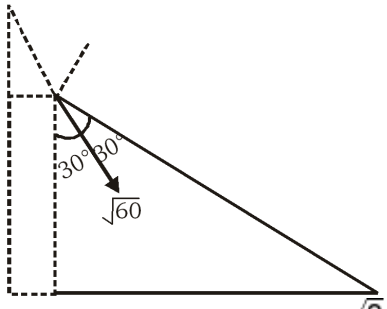
$\Rightarrow h_2 = 6$m

Velocity of block just before collision at B

$= \sqrt{2gh} = \sqrt{2 \times 10 \times 3} = \sqrt{60}$ ms^{-1}



For totally inelastic collision velocity of block along normal to BC becomes zero and since there is no impulse along BC so momentum (velocity) along BC remains unchanged. speed of block just after collision

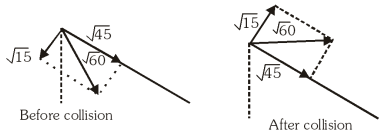


$v_B = \sqrt{60}\cos 30^\circ = \sqrt{60} \times \frac{\sqrt{3}}{2} = \sqrt{45}$ms^{-1}

16. Ans (B)

$v_c^2 = v_B^2 + 2g(h_2 - h_1) \Rightarrow v_c = \sqrt{45 + 2 \times 10 \times 3} = \sqrt{105}$ ms^{-1}

17. Ans (C)



vertical component of velocity = $\sqrt{15}\sin 60^\circ - \sqrt{45}\sin 30^\circ = 0$

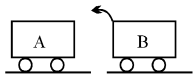
PHYSICS

SECTION-II

1. Ans (1.2)

$v_B = 55/36$m/s, $v_A = 11/6$ m/s, solved

Given $v_{D/C} = 6$ m/sec

Case Ist 

C.O.L.M.

$O = M_D \vec{v}_D + m_B \vec{v}_B$

$\vec{v}_D = -6\hat{i} + \vec{v}_B$

$\therefore$ i.e. $v_D = -5\hat{i}$ toword A

Case II[nd] In this condition when dog & trolly A will move with common speed again

C.O.L.M. $-5i \times 4 = (20 + 4)\, v_{common}$

$v_{common} = -5/6\hat{i}$

so will jumping from A $\longrightarrow$ B

$\vec{v}_D = 6\hat{i} + \vec{v}_A$

$-(24) \times \frac{5}{6}\hat{i} = \left(\left(6\hat{i} + \vec{v}_A\right) \times 4 + \vec{v}_A \cdot 20\right.$

$v_A = -\frac{11}{6}\hat{i}$

Case III[rd]

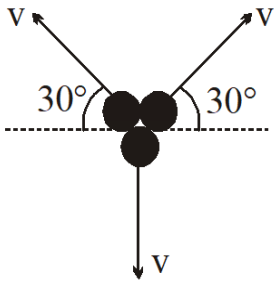
$v_D = 6 - \frac{11}{6}\hat{i} = \frac{25}{6}\hat{i}$

$M_D\vec{v}_D + M_B\vec{v}_B = (M_B + M_D)\, v_{common}$

$\therefore \quad v_{common} = \frac{11}{6}$m/sec Ans.

2. Ans (50)

$v_1 = v_2 = v_3$



$h = vt + \frac{1}{2}gt^2 = v \cdot 2 + \frac{1}{2} \cdot g(2)^2$

$h = -\frac{v}{2} \cdot t_1 + \frac{1}{2}gt_1^2 = - \cdot 4 + \frac{1}{2} \cdot g \cdot 4^2$

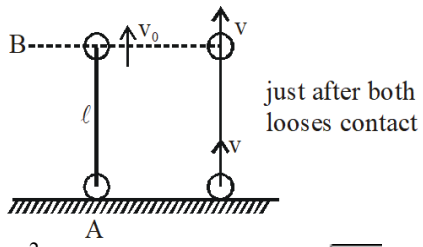
3. Ans (10)

$50\,(L - x) - 150\,x = 0 \quad \Rightarrow 50\,L - 200\,x = 0 \quad \Rightarrow x = \frac{L}{4}$

Similarly in y direction : $y = \frac{L}{4}$

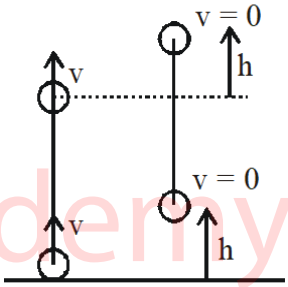
$S = \sqrt{x^2 + y^2} \Rightarrow S = \frac{L}{2\sqrt{2}} = \frac{20\sqrt{2}}{2\sqrt{2}} = 10$

4. Ans (5)



$v_0^2 = 6\,g\ell - 2g\ell \Rightarrow v_0 = \sqrt{4g\ell}$

$2mv = mv_0 \Rightarrow v = \frac{v_0}{2} = \sqrt{g\ell}$



just after both looses contact Þ $v_{cm} = \frac{mv + mv}{2m} = v = \sqrt{g\ell}$ & $a_{cm} = g$

Now just after both looses contact

$\Rightarrow 0^2 = v_{cm}^2 - 2gh$

$\Rightarrow h = \frac{\ell}{2}$ = displacement of each block after both looses contact.

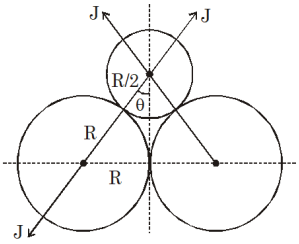
Displacement of com till only one block moves upward is

$\Rightarrow (\Delta x_{cm})_1 = \frac{m\ell}{2m} = \frac{\ell}{2}$

$\therefore$ Total upward displacement of com is $= \frac{\ell}{2} + \frac{\ell}{2} = \ell$

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5. **Ans (1)**



J = impulse during collision between a larger ball and smaller ball

For smaller ball

$2J\cdot\cos q = mv_0$ ... (i)

For larger ball, J sin q = mv ... (ii)

From (i) & (ii)

$$v = \frac{v_0}{2}\tan\theta$$

$$= \frac{\sqrt{5}}{2}\left[\frac{R}{\sqrt{\left(\frac{3R}{2}\right)^2 - R^2}}\right] = \left(\frac{\sqrt{5}}{2}\right)\left(\frac{2}{\sqrt{5}}\right)$$

= 1 m/s

PHYSICS

SECTION-IV

1. **Ans (A->PQRS,B->PQRS,C->PQRS,D->QS)**

For (A) :

$\Rightarrow a_{CM} = 0$

$F_{net_{external}} = 0 \Rightarrow$ momentum is conserved

$\Rightarrow$ work done by external = 0 $\Rightarrow$ Mechanical energy is conserved

Potential energy gets converted to K.E. $\Rightarrow$ KE $\neq$ constant

For (B) :

$\Rightarrow a_{CM} = 0$

$F_{net_{external}} = 0 \Rightarrow$ momentum is conserved

$\Rightarrow$ work done by external = 0 $\Rightarrow$ Mechanical energy is conserved

KE gets stored in the masses during collision as PE $\Rightarrow$ KE $\neq$ constant

For (C) :

$\Rightarrow a_{CM} = 0$

$F_{net_{external}} = 0 \Rightarrow$ momentum is conserved

$\Rightarrow$ work done by external = 0 $\Rightarrow$ Mechanical energy is conserved

KE gets stored in the masses during collision as PE $\Rightarrow$ KE $\neq$ constant

For (D) :

$F_{net_{external}} \neq 0 \Rightarrow a_{CM} \neq 0$

$\Rightarrow$ momentum of system is not conserved

K.E. gets conserved in PE of spring $\Rightarrow$ KE $\neq$ constant.

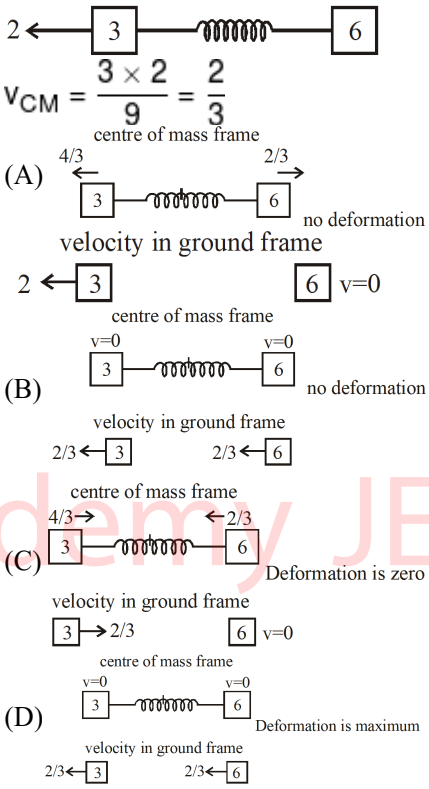
2. **Ans (A->PRT,B->QS,C->PST,D->PST)**

No external force along horizontal in A, C, D so center of mass will remain at rest and linear momentum will always remain constant.

In B work is done by F so mechanical energy increases.

In C, D internal forces of man does work so mechanical energy increases

3. **Ans (A->PQRS,B->PRS,C->PQ,D->PQ)**



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PHYSICS

SECTION-I (i)

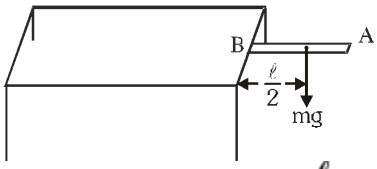
1. **Ans (D)**

For the semicricular lamina of mass m, the moment of inertia about an axis through C is I_C = .

Let I_{CM} = moment of inertia about an axis through its centre of mass.

$I_C = I_{CM} + md^2 \Rightarrow I_{CM} = \frac{1}{2}mr^2 - m\left(\frac{4r}{3\pi}\right)^2 = \left(\frac{1}{2} - \frac{16}{9\pi^2}\right)mr^2$

2. **Ans (A)**



Torque acting about the point B = $(mg)\frac{\ell}{2}$

If I is the moment of inertia of the stick about B and a is the angular acceleration then

$I\alpha = (mg)\frac{\ell}{2} \Rightarrow \alpha = \frac{mg\ell}{2I}$

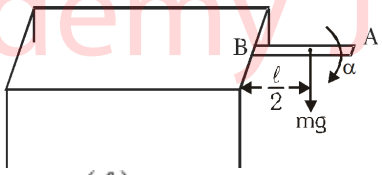
Acceleration of centre of mass $a = \alpha\left(\frac{\ell}{2}\right) = \frac{mg\ell^2}{4I}$

Moment of inertia of this stick about its edge $I = \frac{m\ell^2}{3}$

$\Rightarrow \quad a = \frac{mg\ell^2}{4\left(\frac{m\ell^2}{3}\right)} = \frac{3}{4}g$

OR

For angular motion of the stick



$\tau = mg\left(\frac{\ell}{2}\right) = I\alpha$

moment of intertia of stick about B is $I = \left(\frac{m\ell^2}{3}\right)$

$\Rightarrow mg\left(\frac{\ell}{2}\right) = \left(\frac{mg\ell^2}{3}\right)\alpha \Rightarrow \alpha = \left(\frac{3g}{2\ell}\right)$

Acceleration of centre of mass = $\alpha\left(\frac{\ell}{2}\right) = \left(\frac{3g}{2\ell}\right) = \left(\frac{\ell}{2}\right) = \frac{3}{4}g$

3. **Ans (C)**

For toppling $mg\sin\theta\frac{3R}{8} \geqslant mg\cos\theta R$

$\tan\theta \geqslant \frac{8}{3}$

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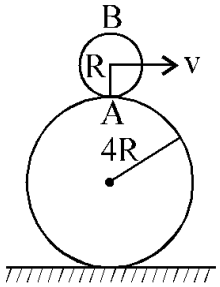
4. **Ans (A)**

$\vec{a}_A = \vec{a}_{A/cm} + \vec{a}_{cm}$

$\vec{a}_{cm} = \frac{v^2}{5R}\downarrow$

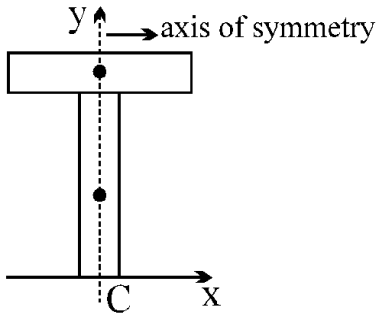
$\vec{a}_{A/cm} = R\omega^2 = \frac{v^2}{R}\uparrow$

$\vec{a}_A = \left(\frac{v^2}{R} - \frac{v^2}{5R}\right)\uparrow = \frac{4v^2}{5R}\uparrow$



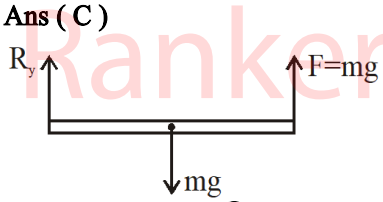
5. **Ans (C)**

Only translation is possible if force is applied at center of mass



$cm = \frac{2m.l + m.2l}{3m} = \frac{4}{3}l$

6. **Ans (C)**



$mg\ell - mg\frac{\ell}{2} = \frac{m\ell^2}{3}\alpha$ ... (i)

$\alpha = \frac{3g}{2\ell}$

$R_y + F - mg = m\frac{\ell}{2}\alpha$ ... (ii)

$R_y = m\frac{\ell}{2} \times \frac{3g}{2\ell} = 15\ N$

7. **Ans (C)**

$\vec{L} = \vec{r} \times \vec{p}$

$= \left[\sqrt{\ell - R^2}\,(-\hat{k}) + R\hat{i}\right] \times \left[Mv\hat{j}\right]$

$= Mvl\left[\frac{\sqrt{l^2 - R^2}}{l}\hat{i} + \frac{R}{l}\hat{k}\right]$

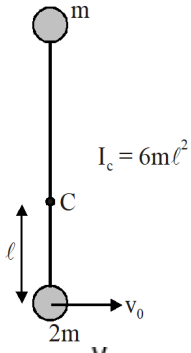
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PHYSICS

SECTION-I (ii)

8. **Ans (B,C)**

For (B) : $2mv_0\ell = 6m\ell^2\omega$



$\therefore\ \omega = \frac{v_0}{3\ell}$

$T = 2m\omega^2\ell = \frac{2mv_0^2}{9\ell}$

For (C) : $I_A = 2m(3\ell)^2$; $I_B = m(3\ell)^2$

9. **Ans (B,C)**

Balancing torque about support,

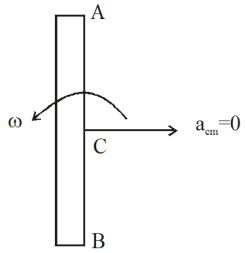
$85g \times 1.5 + 15\ g \times 0.5 = T \times 0.5$

$\Rightarrow T = 2700\ N$

Balancing force in vertical direction,

$N = 15g + 85g + 2700 = 3700$

10. **Ans (A,B,C,D)**



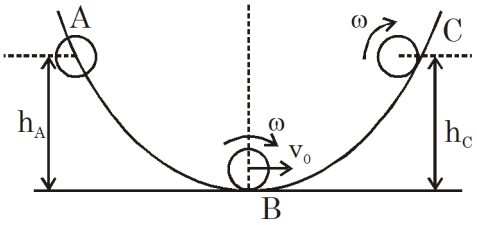
$\omega = \frac{11 + 3}{7/2} = 4$

So accelerations of A & B is

$|a_A| = |a_B| = \omega^2\frac{L}{2} = 4^2 \times \frac{7}{2} = 56$



11. Ans (A,B)



$mgh_A = K_B = (K_{trans})_B + (K_{rot})_B$

$mgh_A = \frac{1}{2}mv_0^2 + \frac{1}{2}I_0\omega^2 = K_B$ ... (i)

$\frac{1}{2}mv_0^2 + \frac{1}{2}I_0\omega^2 = \frac{1}{2}I_0\omega^2 + mgh_C$, where

$\frac{1}{2}I_0\omega^2 = K_C$

$mgh_C = \frac{1}{2}mv_0^2$ ... (ii)

From (i) & (ii)

$h_A > h_C$; $K_B > K_C$

and $h_A > h_C$; $K_C > K_A$

12. Ans (A,B,C)

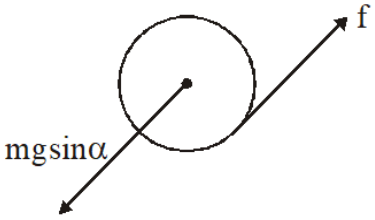
For (A) $a = \frac{1}{M}\left(Mg\sin\alpha - \frac{Mg\sin\alpha}{3}\right) = \frac{2}{3}g\sin\alpha$

For (B) $V^2 = 0 + 2 \times \frac{2}{3}g\sin\alpha l \Rightarrow V = \sqrt{\frac{4}{3}gl\sin\alpha} \Rightarrow \omega = \frac{v}{R} = \sqrt{\frac{4gl\sin\alpha}{3R^2}}$

For (C & D)

For pure roling $a = \alpha R$

$\frac{Mg\sin\alpha - f}{M} = \frac{f \times R^2}{\frac{MR^2}{2}} \Rightarrow Mg\sin a$

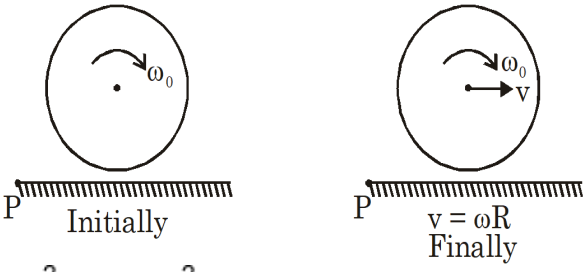


$-f = 2f$

$\Rightarrow f = \frac{Mg\sin\alpha}{3}$

$\mu\, mg\cos\alpha = \frac{mg\sin\alpha}{3} \Rightarrow \mu = \frac{\tan\alpha}{3}$

13. Ans (A,B)



$\frac{mR^2}{2}\omega_0 = \frac{mR^2}{2}\left(\frac{v}{R}\right) + mvR$

$\frac{mR^2}{2}\omega_0 = \frac{3}{2}mvR$

$v = \frac{R\omega_0}{3}$; $\omega = \frac{v}{R} = \frac{\omega_0}{3}$

$\frac{R\omega_0}{3} = \mu gt \Rightarrow t = \frac{R\omega_0}{3\mu g}$

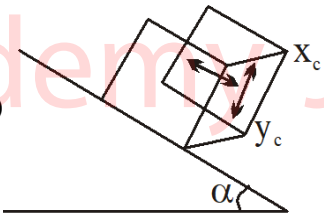
$s = ut + \frac{1}{2}at^2 = \frac{1}{2} \times \mu g\frac{R^2\omega_0^2}{9\mu^2g^2} = \frac{R^2\omega_0^2}{18\mu g}$

PHYSICS

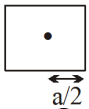
SECTION-I (iii)

15. Ans (A)

(i)



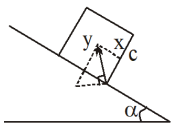
At toppling cm is just a box base line.



$x_c = \frac{\frac{a}{2} \times m + \frac{a}{2} \times m + 0 \times m}{3m} = \frac{a}{3}$

$y_{cm} = \frac{a}{2}$

$\tan(90° - \alpha) = \frac{y_c}{x_c} = \frac{3}{2} \Rightarrow \tan\alpha = \frac{2}{3}$



(ii) for it to not slide $3mg\sin\alpha \le 3m\, mg\cos\alpha$

$\tan\alpha \leqslant \mu$

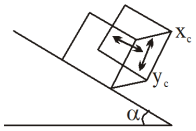
$\frac{2}{3} \leqslant \mu \Rightarrow \mu \ge 0.66$

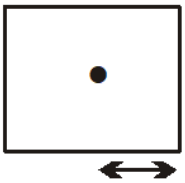
(iii) $a_n = g\sin\alpha$

$\Rightarrow a = \frac{1}{2}g\sin\alpha t^2 \Rightarrow t = \sqrt{\frac{2a}{g\sin\alpha}}$

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16. Ans (B)

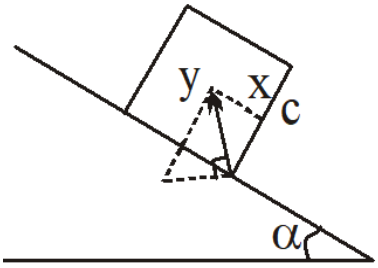
(i)  at toppling cm is just a box base line.



$x_c =$

$$\frac{\frac{a}{2}\times m+\frac{a}{2}\times m+0\times m}{3m}=\frac{a}{3}$$

$$y_{cm}=\frac{a}{2}$$



$\tan(90^\circ -$

$\alpha) = \frac{y_c}{x_c} = \frac{3}{2} \Rightarrow \tan\alpha = \frac{2}{3}$

(ii) for it to not slide $3mg\sin\alpha \le 3m\,mg\cos\alpha$

$$\tan\alpha \leqslant \mu$$

$$\frac{2}{3} \leqslant \mu$$

$\Rightarrow \quad \mu \ge 0.66$

(iii) $a_n = g\sin\alpha$

$$\Rightarrow \quad a=\frac{1}{2}g\sin\alpha t^2 \quad \Rightarrow t=\sqrt{\frac{2a}{g\sin\alpha}}$$

17. Ans (C)

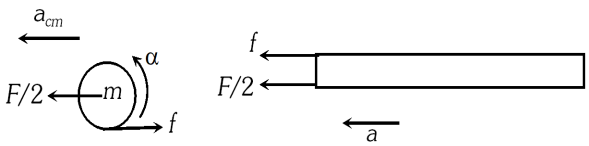
$\frac{F}{2} - f = ma_{cm}$ … (i) $\quad fR = \frac{1}{2}mR^2\alpha$ … (ii)

$\frac{F}{2} + f = 2ma$ … (iii)

For pure rolling $a_{cm} - R\alpha = a$ ...(iv)

On solving $f = \frac{F}{14}, a_{cm} = \frac{3F}{7m}, a = \frac{2F}{7m}$

$\therefore$



22. Ans (A)

Let ball leaves with velocity u & disc rotates with ω clockwise direction

Angular moment conservation about centre of disc

$\vec{L}_i = \vec{L}_f$

$$O = muR - \left(\frac{MR^2}{2}\right)\omega$$

$$\omega = \frac{2mu}{MR} \quad \text{... (i)}$$

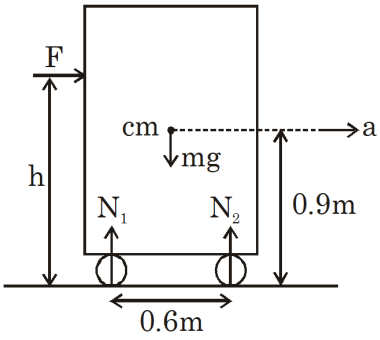
Energy conservation

$$\frac{1}{2}mv^2 = \frac{1}{2}mu^2 + \frac{1}{2}I\omega^2 \quad \text{... (ii)}$$

$$u = v\sqrt{\frac{M}{M+2m}},\ \omega = 2\frac{mv}{MR}\sqrt{\frac{M}{M+2m}}$$

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26. Ans (B)



F = ma

$a = 5m/s^2$

$N_1 + N_2 = mg = 200$ .... (i)

$(\vec{\tau}_{net})_{cm} = 0$

Case-1 : $F(h - 0.9) + N_1(0.3) - N_2(0.3) = 0$

$$\frac{100(h-0.9)}{0.3} = -(N_1 - N_2)$$

$$N_2 - N_1 = \frac{1000}{3}(h-0.9) \quad \text{.... (ii)}$$

From (i) & (ii)

$$2N_1 = 200 - \frac{1000}{3}(h-0.9)$$

$$N_1 = 100 - \frac{500}{3}(h-0.9)$$

For not tipping, $N_1 > 0$

$h < 0.9 + 0.6$

$h < 1.5$ m

$(\vec{\tau}_{net})_{cm} = 0$

Case-2 : $F(0.9 - h) + N_1(0.3) - N_2(0.3) = 0$ .... (iii)

From (i) & (iii)

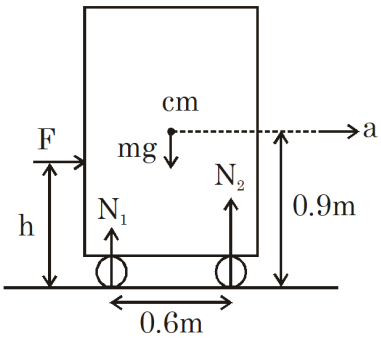
$$N_1 = 100 - \frac{500}{3}(0.9-h)$$

For not tipping, $N_1 > 0$

$$\frac{300}{500} > 0.9 - h$$

$h > 0.9 - 0.6$

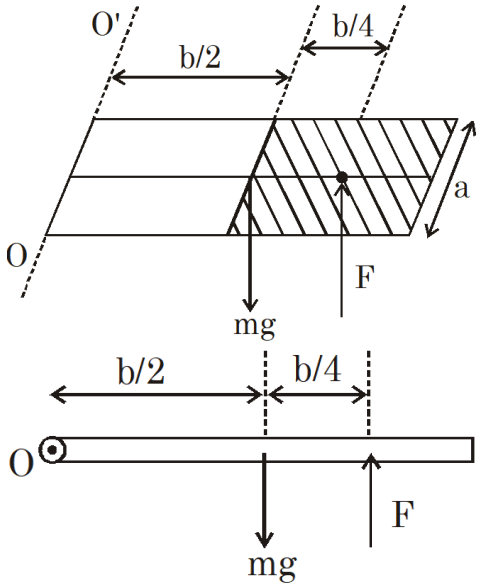
$h > 0.3$ m



PHYSICS

SECTION-II

1. Ans (10)



Taking torque about axis OO' & putting it equal to zero

$(\vec{\tau}_{net})_0 = 0$

$$mg\frac{b}{2} - F\frac{3b}{4} = 0$$

$$F = \frac{2mg}{3} \quad \text{.... (i)}$$

$$F = n(2mv)\left(\frac{ab}{2}\right) \quad \text{.... (ii)}$$

From (i) & (ii), putting the values we get

v = 10 m/s

2. Ans (36.00)

$$\frac{a\ell^4}{36}$$

$$I_0 = \int dmx^2 = \int_0^{\ell} \lambda dx x^2 = \frac{a\ell^4}{4}$$

$$\text{Now CM} = d = \frac{\int_r^{\ell} \lambda dx x}{\int_0^{\ell} \lambda dx} = \frac{2\ell}{3}$$

$$\therefore I_{CM} = I_0 - md^2 = \frac{a\ell^4}{4} - md^2$$

$$m = \int \lambda dx = \frac{a\ell^2}{2}$$

$$\therefore \frac{a\ell^4}{4} - \left(\frac{a\ell^2}{2}\right)\frac{4}{9}\ell^2 = \frac{a\ell^4}{36}$$

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PHYSICS

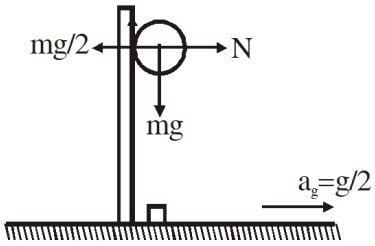
SECTION-III

1. **Ans (8)**

$\tau = I\alpha$; $\tau = mg\left(\frac{4R}{3\pi}\right)$; $I = \frac{MR^2}{2}$,

$\alpha = \frac{8g}{3\pi R} \Rightarrow x = 8$

2. **Ans (8)**



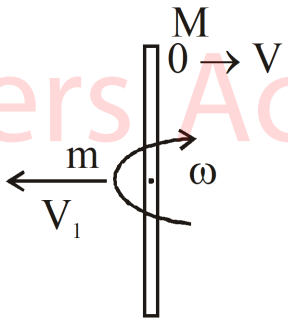
For the given sphere :

$mg - f_s = m\,(a_{cm})_{rel}$ ...(1)

$N = \frac{mg}{2}$; $f_s \times R = \frac{2}{5}mR^2\alpha$; $a = \alpha R$

3. **Ans (9)**

(i) $L_i = L_f$; $\frac{m\,l^2}{12}\omega = MV \times \frac{l}{2} \Rightarrow V = \frac{m\omega l}{6M}$



(ii) $P_i = P_f$; $0 = MV - mV_1 \Rightarrow V = \frac{mV_1}{M} \Rightarrow \frac{\omega l}{6} = V_1$

(iii) $e = 1 = \frac{V-(-V_1)}{\omega\frac{l}{2}-0} \Rightarrow V + V_1 = \frac{\omega l}{2}$; $\frac{m\omega l}{6M} + \frac{\omega l}{6}$

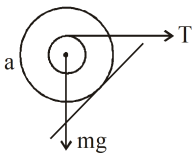
$= \frac{\omega l}{2}$; $\left(\frac{m}{M}+1\right) = 3$

$\Rightarrow \frac{m}{M} = 2 \Rightarrow M = \frac{m}{2} = 9$ kg.

4. **Ans (4)**

$mv_0 \cdot 2R = \left[\frac{3}{2}mR^2 + m(2R)^2\right]\omega$; $\omega = \frac{4v_0}{11R} = 4$ rad/s

5. **Ans (2)**



$mg\sin\theta - T\cos\theta = ma \therefore a = 2m/s^2$

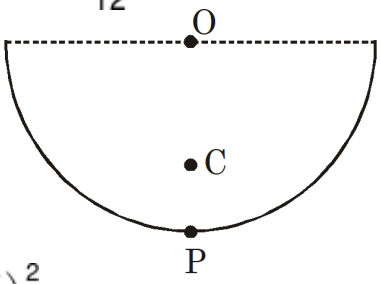
PHYSICS

SECTION-IV

1. **Ans (A->Q,B->P,C->S,D->R)**

(A) $I = \frac{ML^2}{3}\sin^2 30^\circ = \frac{ML^2}{12}$

$I_O = MR^2$



$I_O = I_C + M\left(\frac{2R}{\pi}\right)^2$

$I_P = I_C + M\left(R - \frac{2R}{\pi}\right)^2$

$I_P - I_0 = -\frac{3}{11}MR^2$

$I_P = \frac{8}{11}MR^2$

(C) $I_{base} = \frac{Mh^2}{6} = \frac{M}{6}\left(\frac{R\sqrt{3}}{2}\right)^2 = \frac{MR^2}{8}$

(D) $I = I_{(+)} + I_{(-)}$

$= \frac{M(2R)^2}{2} - \frac{M}{4}\left(\frac{R^2}{2} + R^2\right)$

$= 2MR^2 - \frac{M}{8}3R^2$

$= \frac{13}{8}MR^2$

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PHYSICS

SECTION-I (i)

1. Ans (B)

$\tan\theta = \frac{AP}{CP}$; $\ell n(\tan\theta) = \ell n(AP) - \ell n(CP)$

$\frac{\sec^2\theta}{\tan\theta}\theta d\theta = \frac{d(AP)}{AP} - \frac{d(CP)}{CP} = (\alpha_y - \alpha_x)\Delta T$

. So dq is negative.

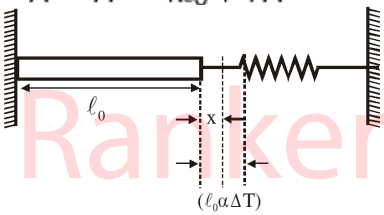
2. Ans (A)

Change in length of rod = (l0aDT−x)

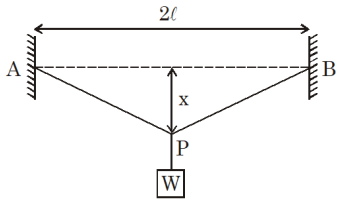
$kx = \left(\frac{\ell_0\alpha\Delta T - x}{\ell_0}\right) yA$

$\Rightarrow x[k\ell_o + YA] = \ell_o\alpha\Delta TYA$

$\therefore \frac{F}{A} = \frac{kx}{A} = \frac{k\ell_o\alpha\Delta TY}{k\ell_o + YA}$



3. Ans (C)



$AP = BP = (x^2 + \ell^2)^{1/2}$

$\Delta\ell = 2(x^2 + l^2)^{1/2} - 2\ell$

$= 2\ell\left[\left(1 + \frac{x^2}{\ell^2}\right)^{1/2} - 1\right] = 2\ell\left[1 + \frac{1}{2}\frac{x^2}{\ell^2} - 1\right]$

($\because (1 + x)n = 1 + nx$ if $x << 1$)

$= \frac{x^2}{\ell}$

$\because$ Strain $= \frac{\text{Change in length}}{\text{Original length}}$

$= \frac{(x^2/\ell)}{2\ell} = \frac{x^2}{2\ell^2}$

4. Ans (B)

Suppose m kg steam is required per hour

Heat is released by steam in following three steps

(i) When 150°C steam $\xrightarrow{Q_1}$ 100°C steam $Q_1 = mc_{steam}\Delta\theta = m \times 1\ (150 - 100) = 50$ m cal

(ii) When 150°C steam $\xrightarrow{Q_2}$ 100°C water $Q_2 = mL_v = m\times 540 = 540$ m cal

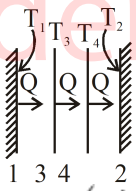
(iii) When 100 °C water $\xrightarrow{Q_2}$ 90°C water $Q_3 = mc_W\Delta\theta = m \times 1 \times (100 - 90) = 10$ m cal

Hence total heat given by the steam $Q = Q_1 + Q_2 + Q_3 = 600$ mcal ....(i)

Heat taken by 10 kg water $Q' = mc_W\Delta\theta = 10 \times 10^3 \times 1 \times (80 - 20) = 600 \times 10^3$ cal

Hence $Q = Q' \Rightarrow 600\ m = 600 \times 10^3 \Rightarrow m = 10^3$ gm = 1 kg

5. Ans (B)



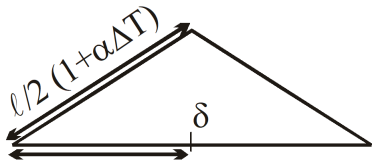
$Q = \sigma\left(T_1^4 - T_3^4\right) W/m^2 = \sigma\left(T_3^4 - T_4^4\right) W/m^2 = \sigma\left(T_4^4 - T_2^4\right) W/m^2$

$3Q = \sigma\left(T_1^4 - T_2^4\right) = Q_0 W/m^2$

$\eta = \frac{Q}{Q_0} = \frac{1}{3}$

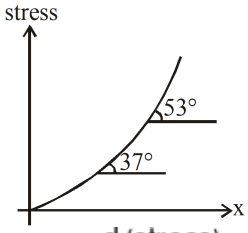
6. Ans (A)

$\delta^2 = (\ell/2(1 + \alpha\Delta T))^2 - (\ell/2)^2$



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7. Ans (A)



$\tan\theta = \frac{d(\text{stress})}{dx}$ (slope)

$= \frac{dF/A}{dx} = \frac{(dm)\,a/A}{dx} = \frac{(\rho \cdot d(\text{vol.}))\,a}{A\,dx}$

$\tan\theta = \rho a$

$\frac{\tan 37^\circ}{\tan 53^\circ} = \frac{\rho_1}{\rho_2} = \frac{9}{\rho_2} \Rightarrow \rho_2 = 16\ \text{gm/cm}^3$

8. Ans (B)

$\frac{dT}{dt} = \frac{\sigma e A}{ms}\left(T^4 - T_S^4\right)$

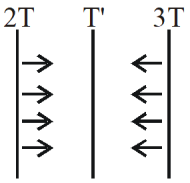
$\frac{dT}{dt} \propto \frac{A}{m}$ $(m = \rho V)$

for disc (pizzas) $m \propto A$ so $\frac{dT}{dt}$ is same for both

for gulab jamuns (sphere) → small (A, m); big (4A, 8m)

$\left(\frac{dT}{dt}\right)_{small} > \left(\frac{dT}{dt}\right)_{big}$ (for Gulab jamun)

9. Ans (C)



Let temperature of middle plate = T'

$k(2T)^4 + k(3T)^4 = 2k(T')^4$

$T' = \left(\frac{97}{2}\right)^{1/4} T$

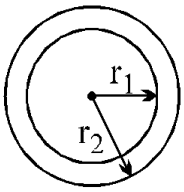
10. Ans (A)

Since $t = \frac{\rho L}{2K\theta}(x_2^2 - x_1^2)$ $\therefore$ $t = \frac{\rho L}{2K\theta}(x^2 - y^2) = \frac{\rho L(x+y)(x-y)}{2K\theta}$

11. Ans (D)

$\int dR = \int \frac{dr}{4\pi r^2 K} = \frac{1}{4\pi K}\left[\frac{r_2 - r_1}{r_2 r_1}\right]$ [R = thermal resistance]

$\dot{Q} = \frac{4\pi K\Delta\theta}{\left(\frac{r_2 - r_1}{r_1 r_2}\right)} = \frac{4\pi K\Delta\theta}{\left(\frac{t}{r^2}\right)}$

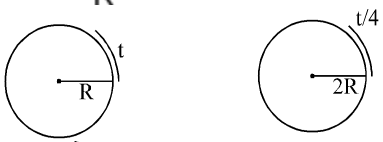


$\left[m = \rho \times \frac{4}{3}\pi r^3\right]$

$\frac{mL}{\text{time}} = \frac{4\pi K\Delta\theta}{\left(\frac{t}{r^2}\right)}$

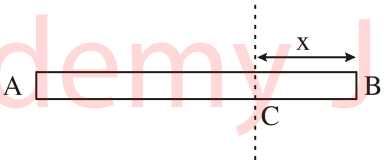
$\frac{\rho L}{\text{time}} = \left(\frac{K}{tr}\right) \times \text{constant}$

$\text{time} \propto \frac{tr}{K}$



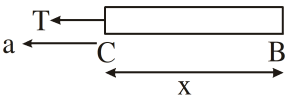
$\frac{25}{16} = \frac{\frac{t}{4} 2r K_S}{tr K_L} = \frac{1}{2}\frac{K_S}{K_L}$; $\frac{K_L}{K_S} = \frac{8}{25}$

12. Ans (B)



at any distance x from edge let there be a cross section for which FBD is given:

for complete rod : $ma = F \Rightarrow a = \frac{F}{m}$ ...(1)



for portion BC : $m_{BC}\, a = T \Rightarrow T = \left(\frac{mx}{L}\right)a$; $T = \left(\frac{mx}{L}\right)\frac{F}{m} = \frac{F}{L}x \Rightarrow$ stress $= \Rightarrow \frac{Fx}{AL}$

(C) $\frac{d\ell}{dx} = \frac{\text{stress}}{Y} = \frac{\frac{Fx}{LA}}{Y} = \frac{Fx}{LAY}$

$\Rightarrow \Delta\ell = \int_0^{\ell} \frac{Fx}{LAY}dx \Rightarrow \frac{\Delta\ell}{L} = \frac{F}{2AY}$

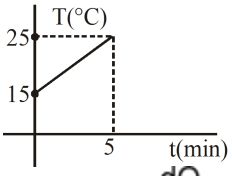
(D) Energy stored $= \int_0^{L} \frac{1}{2}\frac{F^2}{L^2A^2Y}x^2\,dx\,A = \frac{\frac{1}{2}F^2A}{L^2A^2Y}\left(\frac{L^3}{3}\right) = \frac{F^2L}{6AY}$

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PHYSICS

SECTION-I (ii)

13. Ans (A,C)



In final state, $\frac{dQ_e}{dt} = k(45-15) = 12$

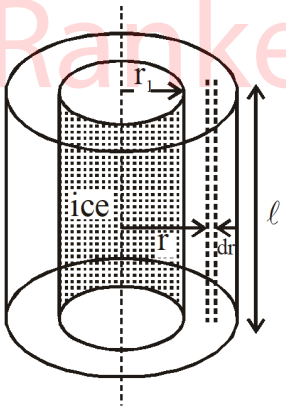
$K = \frac{2}{5}$; at 20°C, $\frac{dQ_e}{dt} = k(20-15) = \frac{2}{5} \times 5 =$ 2watt.

m.s. $\frac{dT}{dt} = 12 - \frac{2}{5}(T-15) \Rightarrow$ m.s. $\frac{dT}{dt} = 18 - \frac{2}{5}T$

m.s. $\int_{15}^{25} \frac{dT}{18-\frac{2}{5}T} = \int_0^{300} dt$

$\Rightarrow s = \frac{240}{\ell n\left(\frac{3}{2}\right)} J/kg-^\circ C$

14. Ans (A,B)



$R_{eq} = \int_{r_1}^{r_2} \frac{dr}{k.2\pi r\ell} = \frac{\ell n2}{k.2\pi\ell}$

$\therefore$ Heat current $= \frac{T_S - T_{ice}}{R_{eq}} = \frac{40}{\ell n2} k2\pi\ell = 80\pi$ J/sec

Also, $\frac{dm_{ice}}{dt} L_f = 80\pi$

$\Rightarrow \frac{dm_{ice}}{dt} = \frac{80\pi}{80 \times 4.2 \times 1000} = \frac{\pi}{4200}$ kg/sec

17. Ans (A,D)

Heat required to melt ice = 50 × 1/2 × (0 −(10)) + 50 × 80 = 4250 cal.

heat av. from condension steam = 10 × 540 = 5400 cal

⇒ whole ice can melt. To rais temperature of system from 0° to 100°.

Heat required = 50 × 1 × (100−0) + 1.5 × 1 × (100−0)

= 5150 cal. × not possible.

So, finally the substance is in from of water

5400 + 10 × 1 × (100 − T) = 4250 + 50 × 1 (T − 0) + 1.5 × 1(T − 0)

2150 = 61.5 T

$T = \frac{2150}{61.5} = 35^\circ C$

18. Ans (B,D)

$\frac{P}{4\pi R_S^2} = \sigma T_S^4$

$\frac{P}{4\pi d^2} \times \pi R_E^2 = 4\pi R_E^2 \sigma T_E^4$

19. Ans (B,D)



(a) As thermal expansion of isotropic solids is similar to true photographic enlargement, expansion of a cavity is same as if it had been a solid body of the same material :

i.e. $\Delta V = V\gamma\ \Delta\theta$

As here V, γ and $\Delta\theta$ are same for both solid and hollow spheres treated (cavity) ;

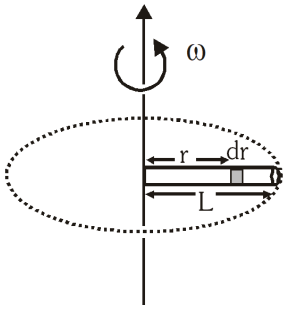
so the expansion of both will be equal.

(b) If same heat is given to the two spheres due to lesser mass, rise in temperature of hollow sphere will be more [as $\Delta\theta = \frac{Q}{mc}$] and hence its expansion will be more [as $\Delta V = V\gamma\Delta\theta$].

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20. Ans (A,B,C)

(a) Consider an element of length dr at a distance r from the axis of rotation as shown in figure. The centripetal force acting on this element will be $dT = dmr\omega^2 = (\rho A dr)\, r\omega^2$. As this force is provided by tension in the rod (due to elasticity), so the tension in the rod at a distance r from the axis of rotation will be due to the centripetal force due to all elements between x = r to x = L



$$\text{i.e., } T = \int_r^L \rho A\omega^2 r\,dr = \frac{1}{2}\rho A\omega^2\left[L^2 - r^2\right] \quad ...(i)$$

$$\text{So here } T = \frac{1}{2} \times 10^4 \times \pi \times 10^{-2} \times (400)^2 \left[\left(\frac{1}{2}\right)^2 - r^2\right]$$

$$= 8\pi \times 10^6 \left[\frac{1}{4} - r^2\right] N$$

(b) Now if dy is the elongation in the element of length dr at position r then strain

$$\frac{dy}{dr} = \frac{\text{stress}}{Y} = \frac{T}{AY} = \frac{1}{2}\frac{\rho\omega^2}{Y}\left[L^2 - r^2\right] dr$$

So the elongation of the whole rod

$$\Delta L = \frac{\rho\omega^2}{2Y}\int_0^L \left(L^2 - r^2\right) dr = \frac{1}{3}\frac{\rho\omega^2 L^3}{Y}$$

$$= \frac{1}{3} \times \frac{10^4 \times (400)^2 (0.5)^3}{2 \times 10^{11}} = \frac{1}{3} \times 10^{-3} m$$

PHYSICS

SECTION-I (iii)

21. Ans (D)

Radiation

22. Ans (B)

$$U = 4\sigma T^3 \frac{1}{N\left(\frac{2}{\epsilon} - 1\right) + 1}$$ Here N = 50, T = 500 K

& $\epsilon = 0.2$

23. Ans (A)

$$UA\,\Delta T = kA\frac{\Delta T}{\Delta \ell}$$

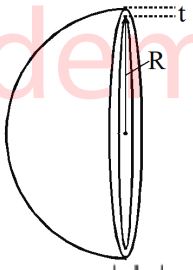
$$\Rightarrow k = U \times \Delta \ell$$

$$\text{Here } U = 4\sigma T^3 \frac{1}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1}$$

PHYSICS

SECTION-II

1. Ans (300)



$$|dP| = \beta\left|\frac{dv}{V}\right|$$

Balancing force,

$$dP\pi R^2 = \sigma 2\pi Rt \Rightarrow \sigma = \frac{dP}{2}\frac{R}{t} = \frac{\beta\gamma\Delta\theta R}{2t}$$

where β = bulk modulus

γ = coefficient of thermal expansion

$\Delta\theta$ = change in temperature

R = radius of sphere

t = thickness of sphere

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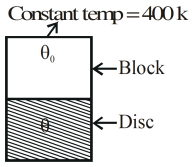
2. **Ans (4500)**

Temperature as a function of time $T = \frac{1}{10}t^\circ C$

$$\frac{dQ}{dt} = kA\frac{dT}{dx} \Rightarrow \int_0^Q dQ = \int_0^{(10\times60)} \left(\frac{kA}{0.4}\right) 0.1t\,dt$$

$$= \frac{100 \times 1 \times 10^{-4} \times 0.1t^2}{0.8}\Bigg|_0^{600} = 450$$

3. **Ans (1663)**



θ = temp of disc

θ_0 = constant temp

Heat input to disc = $\frac{kA(\theta_0 - \theta)}{L}$

Heat utilised by disc = $ms\frac{d\theta}{dt}$

S = Specific heat of disc

$$ms\frac{d\theta}{dt} = \frac{kA(\theta_0 - \theta)}{L}$$

$$\Rightarrow \int_{300}^{350} \frac{d\theta}{(\theta_0 - \theta)} = \frac{kA}{msL}\int_0^t dt$$

$$\Rightarrow t = \frac{msL}{kA}\ell n\left(\frac{\theta_0 - 300}{\theta_0 - 350}\right)$$

$\Rightarrow$ t = 166.32 sec

4. **Ans (3500)**

Let the equivalent resitance of the wall be R_{eq}

$\therefore$ At steady state, $P = \frac{\Delta T_1}{R_{eq}}$ ...(i),

$\Delta T_1 = [15 - (-10)]^\circ C$

when second heater is switched on $2P = \frac{\Delta T_2}{R_{eq}}$ ...(ii),

$\Delta T_2 = [T_{in} - (-15)]^\circ C$ Solving (i), (ii), T_{in} = 35°C

PHYSICS

SECTION-III

1. **Ans (4)**

$$mg = \sigma\left(\pi\left(\frac{d_1}{2}\right)^2\right)$$

$$mg + \frac{mv^2}{\ell} = \sigma\left(\pi\left(\frac{d_2}{2}\right)^2\right)$$

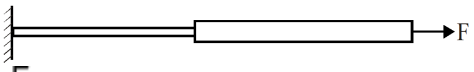
So, $1 + \frac{v^2}{\ell g} = \left(\frac{d_2}{d_1}\right)^2 \Rightarrow 1 + \frac{36}{12} = \left(\frac{d_2}{2mm}\right)^2$

$\Rightarrow d_2 = 4mm$

2. **Ans (1)**

$V_0(1 + \gamma \times 1) - V_0 = h(A_0) \Rightarrow h = \frac{V_0\gamma}{A_0} = \frac{1.8 \times 10^{-4}}{1.8 \times 10^{-4}} = 1$ cm

3. **Ans (6)**



$$\frac{F}{A_1} = y\frac{\Delta\ell_1}{2\ell}$$

$$\frac{F}{A_2} = y\frac{\Delta\ell_2}{\ell}$$

$$\Delta\ell_1 + \Delta\ell_2 = \frac{27}{4}mm$$

$$\frac{2F\ell}{A_1 y} + \frac{F\ell}{4A_1 y} = \frac{27}{4}mm$$

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PHYSICS

SECTION-I (i)

1. **Ans (B)**

$i^2R = nC_P\frac{dT}{dt} + mgv$

$C_P = \frac{7}{2}R$ & $PV = nRT \Rightarrow PAv = nR\frac{dT}{dt}$

2. **Ans (D)**

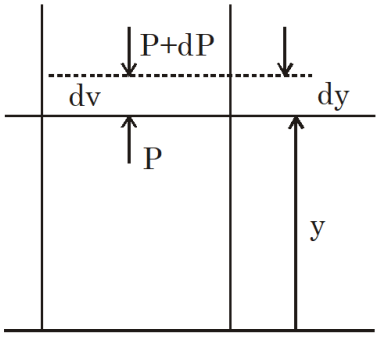
$\gamma = \frac{dV}{VdT}$

For ideal gas PV = nRT .....(i)

At constant pressure, PdV = nRdT ...(ii)

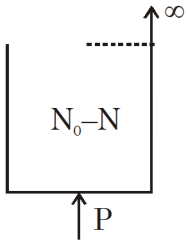
Dividing (ii) by (i) $\frac{dV}{V} = \frac{dT}{T} \Rightarrow \frac{dV}{VdT} = \frac{1}{T}$

3. **Ans (C)**



Let total number of molecules are N_0 and molecules below height y are N.

for volume of gas, from height h to ∞



$P \times A = (N_0 - N)mg$ ... (i)

Where m is mass of 1 molecule

for volume dV,

PdV = dN kT (T is constant)

$P \times Ady = dN\ kT$

from (i)

$(N_0 - N)\ mg\ dy = dN\ kT$

$\frac{dN}{N_0 - N} = \frac{mg}{kT}dy$

for y = 0, N = 0

$\int_0^N \frac{dN}{N_0 - N} = \frac{mg}{kT}\int_0^y dy$

$-[\ell n\,(N_0 - N)]_0^N = \frac{mg}{kT}y$

$\left(\frac{N_0 - N}{N_0}\right) = e^{-mgy/kT}$

$1 - \frac{N}{N_0} = e^{-mgy/kT}$

$\frac{N}{N_0} = 1 - e^{-mgy/kT}$

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4. Ans (C)

$$\text{COP} = \frac{Q_{\text{extracted}}}{\text{work done}} = \frac{T_1}{T_2 - T_1} = \frac{270}{30} = 9$$

$$\Rightarrow \text{work} = \frac{1000}{9}\text{J}$$

5. Ans (B)

$$Q_1 - Q_2 = \frac{Q_2}{2} - Q_3,$$

$$\frac{Q_1}{Q_2} - 1 = \frac{1}{2} - \frac{Q_3}{Q_2} = \eta - 1 = \frac{1}{2} - \frac{\eta'}{2}$$

6. Ans (D)

$$\eta = 1 - \frac{270}{300} = \frac{1}{10}$$

Efficiency of refrigerator = $0.5\eta = \frac{1}{20}$

If Q is the heat/s transferred at higher temperature then $\frac{W}{Q} = \frac{1}{20}$

or Q = 20 W = 20 kJ,

and heat removed from lower temperature = 19 kJ.

7. Ans (D)

$$W = \frac{\phi C}{\text{cop}} = \frac{60 \times 10^3}{60 \times 1.2}\text{J/s} = \frac{10^3}{1.2}\text{J/s}$$

No of units consumed

$$= \frac{\left(\frac{10^3}{1.2}\right)(4)(30)}{1000} = 100 \text{ units}$$

Cost of energy = (6) (100) = 600 Rs

$\therefore$ (D)

8. Ans (C)

$$\eta_A = \frac{T_1 - T_2}{T_1} = \frac{W_A}{Q_1} \Rightarrow \eta_B = \frac{T_2 - T_3}{T_2} = \frac{W_B}{Q_2}$$

$$\therefore \frac{Q_1}{Q_2} = \frac{T_1}{T_2} \times \frac{T_2 - T_3}{T_1 - T_2} = \frac{T_1}{T_2} \quad \therefore \quad W_A = W_B$$

$$\therefore T_2 = \frac{T_1 + T_3}{2} = \frac{800 + 300}{2} = 550\text{K}.$$

10. Ans (A)

A / (A + B)

PHYSICS

SECTION-I (ii)

11. Ans (A,D)

As, $W_{by} = Q = 0$, So U = const.

$$2 \times \frac{5}{2}R \times T_0 + 3 \times 3R \times 2T_0 = (2 \times \frac{5}{2}R + 3 \times 3R)\, T$$

$$23RT_0 = 14\,RT \Rightarrow T = \frac{23T_0}{14}$$

$$\text{By } \frac{n_1 + n_2}{1 - \gamma_{mix}} = \frac{n_1}{1 - \gamma_1} + \frac{n_2}{1 - \gamma_2}$$

$$\Rightarrow \gamma_{mix} = \frac{19}{14}$$

12. Ans (A,B)

Total displacement of mercury against atmospheric process = $\left[\ell + \frac{\ell}{2} + \ell\right]$

$$PE_{\text{initial}} = \left[\left(\rho S\frac{\ell}{2}\right)\frac{\ell}{4}\right] \times 2 + 0$$ {Assuming zero at ground level};

$$PE_{\max} \left[\left(\rho S\frac{\ell}{2}\right) \times 2 + \rho S\ell\right] \ell$$

14. Ans (A,D)

We know

$\gamma_{He} = \frac{5}{3}$, $\gamma_{O_2} = \frac{7}{5}$

$PV^{\gamma} = \text{const.}$

$$P_{He}\,(LA)^{5/3} = P'_{He}\left(\frac{L}{2}A\right)^{5/3}$$

$$P'_{He} = P_{He}(2)^{5/3} \quad \ldots (1)$$

$$P_{O_2}\,(LA)^{7/5} = P'_{O_2}(L'A)7/5$$

$$P'_{O_2} = P_{O_2}\left(\frac{L}{L'}\right)^{7/5} \quad \ldots (2)$$

Now $P_{O_2} = P_{He} \Rightarrow P'_{O_2} = P'_{He}$

$$\Rightarrow \frac{P_{O_2}}{P'_{O_2}} = \frac{P_{He}}{P'_{He}} \Rightarrow \left(\frac{L'}{L}\right)^{7/5} = \frac{1}{2^{5/3}}$$

$$(L')^{7/5} = \frac{L^{7/5}}{2^{5/3}}$$

$$L' = \frac{L}{2^{5/3}x^{5/7}} = \frac{L}{2^{25/21}}$$

so volume of $O_2 = L'A = \frac{LA}{2^{25/21}}$

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15. Ans (A,D)

For iso-thermal process PV = constant and for adiabatic process PV^{γ} = constant where $\gamma > 1$

First we assuming that BC is isothermal & CA is adiabatic

$\Rightarrow P_0V_0 = 3P_0V_C \Rightarrow V_C = \frac{V_0}{3}$ [for process BC]

$3P_0\left(\frac{V_0}{3}\right)^{\gamma} = \frac{V_0P_0}{2} \Rightarrow \gamma = \frac{\ln 6}{\ln 3}$ [for process CA]

Now we are assuming that process

BC is adiabatic & AC is isothermal

$3P_0V_C = \frac{P_0V_0}{2} \Rightarrow V_C = \frac{V_0}{6}$ [for process CA]

$3P_0\left(\frac{V_0}{6}\right)^{\gamma} = P_0V_0^{\gamma} \Rightarrow 6^{\gamma} = 3$ [for process BC] here $\Rightarrow V = \frac{\ln 3}{\ln 6}$ $\gamma < 1$ [not possible]

- Process AC is adiabatic & $V = \frac{\ln 6}{\ln 3}$

16. Ans (A,B,C)

The cylinder is perfectly conducting and heat has been provided slowly therefore in the absence of any temperature gradient entire heat given to gas is absorbed by ice. Phase changes take place at constant temperature and pressure.

(a) $W_{gas} = 0$ (b) $Q_{gas} = 0$

(c) Heat absorbed by ice water system to reach 100°C

$= mL + 2ms_W(100) = 10 \times 80 + 2 \times 10 \times 1 \times 100 =$ 2800 Cal

$= nC_p\Delta T = 3 \times \frac{7}{2}R \times (100) = 3 \times \frac{7}{2} \times \frac{25}{3} \times \frac{1}{42} \times$ 100 =6250/3 Cal

Total heat = 2800 + 6250/3 = 14650/3 Cal

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PHYSICS

SECTION-I (iii)

17. Ans (B)

Initially $V_A = V_B = V_0$

$T_A = T_0$, $T_B = T_0$ (always)

$m_A = m$, $P_i = P$

Finally, $V_A = \frac{V_0}{2}$, $V_B = \frac{3}{2}V_0$

$P_A = P_B$ (at equilibrium)

For A

process is adiabatic

$TV^{\gamma-1}$ = constant

$\gamma = 1.5$ (given)

$TV^{1/2}$ = constant

$T_0V_0^{1/2} = T\left(\frac{V_0}{2}\right)^{1/2}$

$T_A = T_0\sqrt{2}$

Now, $P_A = P_B$

$\frac{nRT_A}{V_A} = \frac{nRT_B}{V_B}$

$m\frac{T_0\sqrt{2}}{\frac{V_0}{2}} = m_B\frac{T_0}{\frac{3V_0}{2}}$

$[m_B = 3\sqrt{2}\,m]$

18. Ans (B)

$\Delta V = nC_V\Delta T$

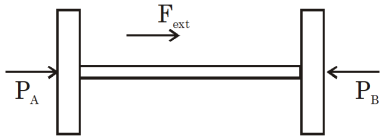
$= n \times \frac{R}{\gamma-1} \times (T_2 - T_1)$

$= \frac{nR\left(\sqrt{2}-1\right)T_0}{0.5}$

$= PV_0 \times 2\left(\sqrt{2}-1\right)$

19. Ans (C)

Since piston is slowly moving towards left



According to work energy theorem.

$W_B + W_{ext} + W_A = 0$

$W_B = -(W_A - W_{ext})$

Since $W_A < 0$, $W_{ext} < 0 \Rightarrow |W_B| = |W_A| + |W_{ext}|$

$|W_B| > |W_A|$

20. Ans (C)

At equilibrium,

Compressive force, $F = p_f \times S$

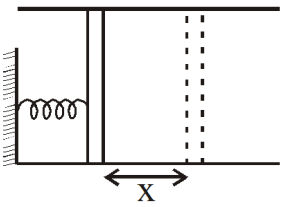
For A $\Rightarrow$

$$\frac{PV_0}{T_0} = \frac{P_f \times \frac{V_0}{2}}{T_0\sqrt{2}} \Rightarrow P_f = P \times 2\sqrt{2} = 2^{3/2}P$$

PHYSICS

SECTION-II

1. Ans (100)



$P_2V_2 = P_1V_1$

$V_1 = 20000$cc.

$P_1 = 10^5$ Pa

$V_2 = Ax$

$P_2 = P_0 + \frac{kx}{A}$

$\Rightarrow \quad 105 \times 20000 \times 10^{-6} = \left(10^5 + \frac{1000x}{100 \times 10^{-4}}\right) 10^{-4} \times 100x$

$\Rightarrow \quad 2000 = 1000\,[x\,(1 + x)]$

$x^2 + x - 2 = 0$

$x = \quad = 1$ m

$x = 100$ cm.

2. Ans (20)

Let initial pressure be P_0

If level A is maintained $\Rightarrow$ volume remains same

Temperature double $\Rightarrow$ pressure doubles

$\therefore 2P_0 - P_0 = rg \times 20$ cm

$\therefore P_0$ corresponds to 20 cm of Hg

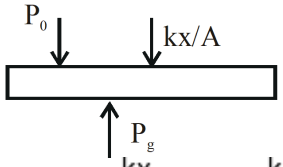
3. Ans (375)

Let us denote the final volume and pressure by V' = 1.25 V_0 and P', respectively. Taking helium to be an ideal gas, we can write that $P_0V_0 = P'V'$ under isothermal conditions, so that P' = 0.8 P_0. Since the external pressure is 0.5 P_0, the net outward pressure on the surface of the balloon is 0.3 P_0. Thus the net pressure of the balloon is 0.3 P0. Thus the net pressure force on half of the balloon is 0.3 $P\pi R^2$, where R is the final radius of the balloon, given by 1.25 V = $4\pi R^3/3$. But this must just balance the elastic force $\sigma\Delta A$, where $\Delta A = 2\pi Rt$ is the cross-sectional area of the balloon material at its mid-plane. Here t is the thickness of the balloon material, obtained from ρ = m/ $(4\pi R^2 t)$. Combining the equations gives $\sigma = \frac{9P_0V_0\rho}{16m}$

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4. Ans (364)

1.6 m, 364 K



$P_g = P_0 + \frac{kx}{A} = P_0 + \frac{kh}{16A}$

For adiabatic process $P_1V_1^{\gamma} = P_2V_2^{\gamma}$

$$P_0(hA)^{\gamma} = P_g\left[\left(\frac{h}{2} + \frac{h}{16}\right)A\right]^{\gamma}$$

$$10^5(hA)^{\gamma} = \left[10^5 + \frac{3700 \times h}{16 \times 27 \times 10^{-4}}\right]\left[\frac{9h}{16}A\right]^{\gamma}$$

$$\frac{64}{27} = 1 + \frac{370 \times h}{16 \times 27}$$

$$\frac{37}{27} = \frac{370 \times h}{16 \times 27} \Rightarrow h = \frac{16}{10}$$

PV = nRT

$P_0hA = nR \times 273$ (initial)

$$\left\{P_0 + \frac{kh}{16A}\right\}\left\{\frac{Ph}{16}A\right\} = nRT \quad \text{(final)}$$

divide

$$\left\{1 + \frac{kh}{16A \times P_0}\right\}\frac{9}{16} = \frac{T}{273}$$

$$T = 273 \times \frac{9}{16}\left[1 + \frac{370}{270}\right]$$

$$T = 273 \times \frac{4}{3} = 364k$$

5. Ans (2)

$$\frac{Q_h}{\omega} = \frac{T_h}{T_h - T_C}$$

$$= \frac{300}{30} = 10$$

6. Ans (6)

$$\frac{3}{130}W$$

$$\frac{0.15}{P} = \frac{260}{40} \Rightarrow P = \frac{6}{260}W$$

7. Ans (2400)

2400 joule per cycle

W = 1200 J, T_1 = 800 K, T_2 = 400 K

$$\therefore \eta = 1 - \frac{T_2}{T_1} = \frac{W}{Q_1} \Rightarrow 1 - \frac{400}{800} = \frac{1200}{Q_1} \Rightarrow 0.5 = \frac{1200}{Q_1}$$

Heat energy supplied by source $Q_1 = \frac{1200}{0.5} =$ 2400 joule per cycle

8. Ans (137)

1.88°C

$$\frac{dQ_{ext/dt}}{P} = \frac{260}{14}$$

$$\frac{dQ}{dt}ext + P = C(1-0)$$

$$\frac{274P}{14} = C$$

Now $\frac{\frac{dQ}{dt}ext}{P} = \frac{260}{T+13}$

$$2\left(\frac{dQ}{dt}ext + P\right) = C(T-0)$$

$$2P\left(\frac{60}{T+13} + 1\right) = C(T)$$

$$2\left(\frac{260 + T + 13}{T+13}\right) = \frac{274}{14}T$$

$273 \times 14 + 14T$

$= 137\ T2 + 13 \times 137T$

$137\ T2 + 1767\ T - 3822 = 0$

$$T = \frac{-1767 + \sqrt{1767^2 + 4 \times 137 \times 3822}}{2 \times 137}$$

$= +1.88°C$

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PHYSICS

SECTION-III

1. **Ans (5)**

Process A ⟶ B

$W_{AB} = \int P\,dv = \int \frac{3}{2}T^{1/2}\,dv = \int \frac{3}{2}T^{1/2} \times \frac{1}{3}RT^{-1/2}\,dT$

On solving, $W^{AB} = \frac{115.2}{2} \times R = 480$ J

Process B ⟶ C

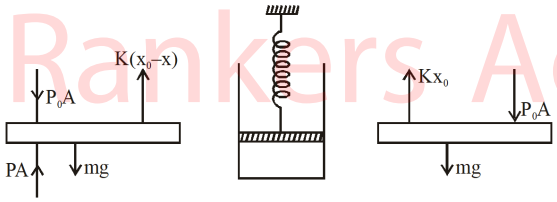
$U = \frac{1}{2}V^{1/2} \quad \frac{3}{2}RT = \frac{1}{2}V^{1/2} \Rightarrow 3PV^{1/2} = 1$

$\therefore P = \frac{1}{3\sqrt{V}}$

New $W_{BC} = \int P\,dv = \int_{100}^{1600} \frac{1}{3\sqrt{V}}\,dv = \frac{2}{3}\sqrt{V} = \frac{2}{3}[40 - 10] = \frac{2}{3} \times 30 = 20$ J

Total work W = 480 + 20 = 500 J

2. **Ans (8)**



$P_0A + mg = kx_0$

for very slow process

$PA = mg + p_0A - K(x_0 - x)$

$\Rightarrow P = \frac{Kx}{A} = \frac{KV}{A^2}$

Thus process followed by gas is $PV^{-1} = \frac{K}{A^2} =$ constant.

$\therefore$ Molar heat capcity for process is

$= C = C_v + \frac{R}{2} = 2R \quad \Rightarrow \quad Q = n.C\Delta T$

Since $\frac{P}{V} = \frac{AK}{A^2} \Rightarrow T = \frac{KV^2}{nRA^2}$

$\Rightarrow \Delta T = \frac{K}{nR}\left(1 - \frac{1}{9}\right)\ell^2 \quad \Rightarrow Q = \frac{16K}{9}\ell^2$

PHYSICS

SECTION-IV

1. **Ans (A->R,B->PQ,C->PQ,D->ST)**

For process-1,

Gas undergoes compression work done by gas is − ve

for DU & DQ nothing can be comented.

For process-2,

Gas undergoes expansion Þ work done by gas is +ve

Also temperature of gas increases.

$\therefore$ ΔQ is +ve & ΔU is +ve

For process-3

Process is isobaric

For process-4

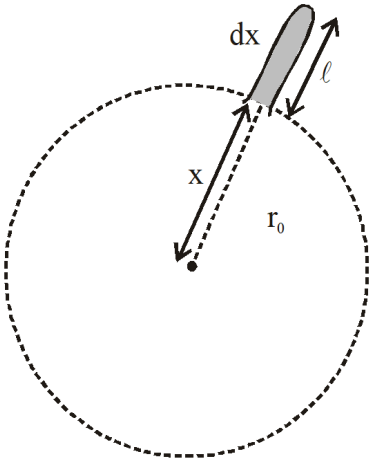
Process is iso-choric

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PHYSICS

SECTION-I (i)

1. Ans (A)



For the section of length dx,

$dT = \rho A\, dx\, \omega^2 x$

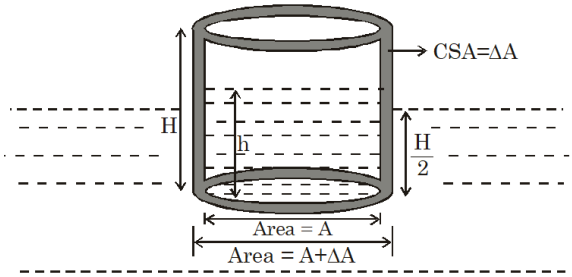
$\therefore$ T at centre $= \int_{r_0}^{\left(\frac{\ell}{2}+r_0\right)} \rho A\omega^2 x dx = \frac{1}{2}\rho A\omega^2\left[\left(\frac{\ell}{2}+r_0\right)^2 - r_0^2\right]$

$= \frac{1}{2}\rho A\omega^2\left[\frac{\ell^2}{4} + r_0^2 + \ell r_0 - r_0^2\right] =$

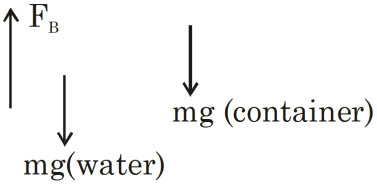
$\frac{1}{2}\rho A\omega^2\left[\frac{\ell^2 + 4\ell r_0}{4}\right] = \frac{1}{8}\rho A\omega^2 \ell\,[\ell + 4r_0]$

$\therefore$ Pressure $= \frac{T}{A} = \frac{1}{8}\rho\omega^2\ell\,[\ell + 4r_0]$

2. Ans (A)



For equilibrium $\Rightarrow$ FBD $\Rightarrow$



$F_B = mg_{(Water)} + mg_{(container)}$

$\rho_W (A + \Delta A)\frac{H}{2}g = \rho_W hAg + \rho_C \rho_W H\Delta Ag$

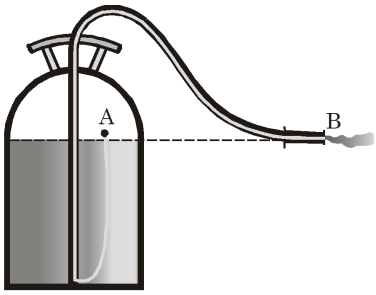
$hA = \left(\frac{AH}{2}\right) + \left(\frac{\Delta AH}{2}\right) - (\rho_C \Delta AH)$

$h = \frac{H}{2} + \frac{H\Delta A}{A}(0.5 - \rho_C)$

This shows that $h > \frac{H}{2}$ if $\rho_C < 0.5$

So Ans. (A)

3. Ans (B)



Writing Bernoulli's equation along flow line at point A & B

$P_{tank} + P_0 + 0 + \frac{1}{2}\rho V_A^2 = P_0 + 0 + \frac{1}{2}\rho V_B^2$

$\Rightarrow V_B = \sqrt{\frac{2P_{tank}}{\rho}}$

[V_A << V_B as area of nozzle is very less as compared to area of tank]

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4. Ans (C)

As the height decreases, the rate of flow with which the water is coming out decreases.

5. Ans (B)

$F = \frac{\Delta p}{\Delta t} = \frac{m(v_1 + v_2)}{\Delta t} = V\rho(v_1 + v_2)$ where $\frac{m}{\Delta t} = V$ = volume/second.

6. Ans (A)

Let y be the height of liquid at some instant.

Then $-\frac{dy}{dt}$ = constant (given)

From equation of continuity,

$(\pi x^2)\left(-\frac{dy}{dt}\right) = a\sqrt{2gy}$ (a = area of hole)

Here, $\pi\left(\frac{-dy}{dt}\right)$, a and g are constants

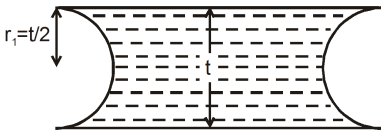
Hence, squaring the equation, we get $y = kx^4$.

10. Ans (A)

Pressure inside the film is less than outside by an amount, $P = T\left[\frac{1}{r_1} + \frac{1}{r_2}\right]$, where r_1 and r_2 are the radii of curvature of the meniscus. Here $r_1 = t/2$ and $r_2 = \infty$, then the force required to separate the two glass-plates, between which a liquid film is enclosed (figure) is,

$F = P \times A = \frac{2AT}{t}$,

where t is the thickness of the film, A = area of film.



$F = \frac{2A^2T}{At} = \frac{2A^2T}{V} = \frac{2 \times (40 \times 10^{-4})^2 \times (70 \times 10^{-3})}{0.05 \times 10^{-6}} = 45$ N

11. Ans (A)

By force balance on the wire,

$2T\cos\theta\,\ell = \lambda\,\ell\,g$

$\frac{2Ty}{\sqrt{a^2+y^2}}\ell = \lambda\ell g$

$\left[T = \frac{\lambda ag}{2y}\right]$

PHYSICS

SECTION-I (ii)

12. Ans (B,C)

$F_H = (\rho gR)\pi R^2 = \rho g\pi R^3 = \frac{3Mg}{2}$

$Fv = (M + m)g$

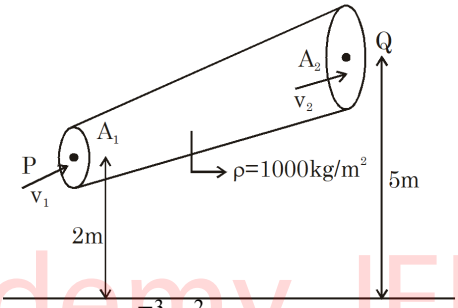
$F_{net} = \sqrt{[(M+m)g]^2 + \left(\frac{3Mg}{2}\right)^2}$

$F_{net} = Mg\sqrt{\frac{9}{4}\left(1 + \frac{m}{M}\right)^2}$

13. Ans (B,C,D)

$P_{new} = \rho\left(g + \frac{g}{2}\right)h$ & $B_{new} = P_{new} \times V$

14. Ans (A,B)



$A_1 = 4 \times 10^{-3}$ m^2

$A_2 = 8 \times 10^{-3}$ m^2

$v_1 = 1$ m/sec

$\rho_W = 10^3$ kg/m^3

Continuity equation at P & Q

$A_1v_1 = A_2v_2$

$(4 \times 10^{-3})(1) = (8 \times 10^{-3})(v_2)$

$v_2 = \frac{1}{2}$ m/sec

Work done per unit volume by pressure = ΔP = pressure difference

Applying Bernoulli's theorem between P & Q

$P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2$

$P_1 - P_2 = \rho g(h_2 - h_1) + \frac{\rho}{2}\left(v_2^2 - v_1^2\right) = 10^4(5-2) + 500\left(\frac{1}{4} - 1\right)$

$= 30000 - \frac{1500}{4} = 29625$ J/m^3

Work done per unit volume by gravity force = $\frac{mgh}{v} = (\rho gh)$

So, work done/volume = $\rho g(h_1 - h_2)$

$= 1000 \times 10[2 - 5]$

$= -30000$ J/m^3

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15. **Ans (B,C)**

$$\frac{F}{2\pi x\ell} = \eta\frac{dv}{dx}$$

$$\frac{F}{2\pi\ell}\int_{R}^{2R}\frac{dx}{x} = \eta\int_{-v_0}^{2v_0}dv$$

$$\frac{F}{2\pi\ell}\times \ell n2 = \eta\times 3v_0$$

$$\frac{F}{\ell} = \frac{6\pi v_0\eta}{\ell n2}$$

$$\frac{F}{2\pi\ell}\times \ell n\left(\frac{x}{R}\right) = \eta v_0$$

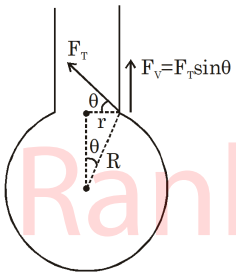
$$\Rightarrow \frac{3}{\ell n2}\ell n\left(\frac{x}{R}\right) = 1$$

$$\Rightarrow \ell n\left(\frac{x}{R}\right) = \ell n2^{1/3} \Rightarrow x = 2^{1/3}R$$

PHYSICS

SECTION-I (iii)

16. **Ans (C)**



$F_T = T\cdot 2\pi r$

$F_V = F_T\sin\theta$

$= T\cdot 2\pi r\cdot\frac{r}{R}$

$= \frac{2\pi r^2T}{R}$

17. **Ans (A)**

Drop detaches from dropper

When :

$F_V = mg$

$$\frac{2\pi r^2T}{R} = \rho\cdot\frac{4}{3}\pi R^3g$$

$$R = \left(\frac{3r^2T}{2g\rho}\right)^{1/4}$$

$$= \left[\frac{3\times\left(5\times10^{-4}\right)^2\times(0.11)}{2\times10\times1000}\right]^{1/4}$$

$\approx 1.4\times10{-}3$ m

18. **Ans (B)**

Surface energy of drop after detachment = T × surface area

$= T\times 4\pi R^2 = 0.11\times 4\times\pi\times(1.4\times10^{-3})^2$

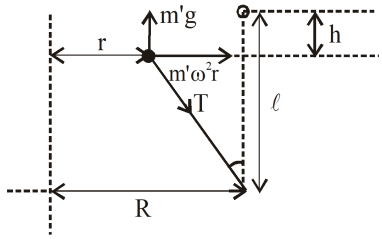
$\approx 2.7\times10^{-6}$ Joule

19. **Ans (D)**

Force exerted by liquid = (mass of liquid displaced) $\times\sqrt{g^2+(\omega^2r)^2}$

$= \rho\omega v\times\sqrt{g^2+(\omega^2r)^2}$

20. **Ans (B)**



$R - r = \ell\sin\theta$ .....(i)

$\ell - h = \ell\cos\theta$ .....(ii)

$m' = (\rho\omega - \rho_{body})v$ & $m'g = T\cos\theta$ .....(iii)

$m'\omega^2r = T\sin\theta$ .....(iv)

from (i), (ii), (iii) & (iv)

$$\frac{\omega^2r}{g} = \tan\theta = \frac{R-r}{\ell-h}$$

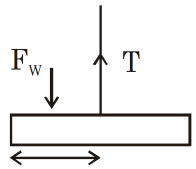
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21. Ans (A)

$h_1 = \frac{2(3+\pi)}{15\pi} = 0.26$; $h_2 = \frac{3+\pi}{10\pi} = 0.195$

Plug opens when level of water reaches 'h'.

- FBD of plug

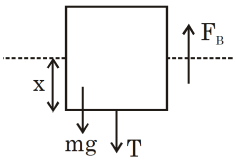


$F_W = T \Rightarrow P.\pi R^2 = T$

$T = \rho_W ghA = 1000 \times 10 \times h \times \pi\left(\frac{5}{100}\right)^2$

$T = 25\ \pi h$

- FBD of float



$F_B = mg + T$

$\rho_W gxA = 3 \times 10 + T$

$10000x\pi \times \left(\frac{10}{100}\right)^2 = 30 + 25\ \pi h$

$100\ \pi x = 30 + 25\ \pi h$

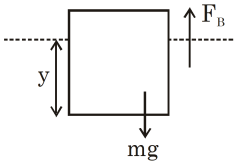
$100\pi\ (h - 0.1) = 30 + 25\ \pi h$

$75\ \pi h = 30 + 10\ \pi$

$h = \frac{(3+\pi)\,10}{75\pi} = \left(\frac{2(3+\pi)}{15\pi}\right) m$

- To again close the plug

$mg = F_B$ (for the float) $\longrightarrow$ [When water has risen to height y for the float]



$F_B = mg$

$\rho_W gyA = 3g$

$1000 \times 10 \times y \times \pi \times \left(\frac{10}{100}\right)^2 = 3g$

$100py = 3g$

$y = \frac{30}{100\pi} = \frac{3}{10\pi} m$

So $h = 0.1\ m + \frac{3}{10\pi} m = \left(\frac{1}{10} + \frac{3}{10\pi}\right) m$

$h = \left(\frac{\pi+3}{10\pi}\right) m$

PHYSICS

SECTION-II (i)

1. Ans (0.1)

$\rho_1 g h_1 = \rho_2 g h_2$

$\frac{h_2}{h_1} = \frac{\rho_1}{\rho_2} = \frac{\rho_1}{\rho_1(1-\gamma\Delta T)}$

$Y = 10^{-5}/°C$, $\Delta T = 100°C$, $h_1 = 300$ cm

$h_2 - h_1 = 0.1$ cm ans.

2. Ans (4.5)

$P_\ell V_\ell = P_f V_f$

$P_0 (V) = P_f \left(\frac{2V}{3}\right)$

$P_f = \frac{3}{2}P_0 = P_0 + \rho_w gh$

$\frac{P_0}{2} = \rho_w gh$

$\frac{\rho_w g \times 9}{2} = \rho_w gh$

[h = 4.5 m]

PHYSICS

SECTION-II (ii)

3. Ans (100)

$T_{max} = 7 \times 10^6 \times 10^{-6} N = 7N$

$\therefore F_{B_{min}} = (15 - 7) = 8N$

$\therefore$ Water level its be lowered = 2 cm

$\therefore$ Area occupied by water level at the top = 100 cm^2

$\therefore 2t = 100 \times 2 \Rightarrow t = 100$ sec

4. Ans (110)

$I\alpha = (V\rho g)\frac{\ell}{4}$

where $V = pr^2 \ell$

so $\alpha = \frac{\pi r^2 \rho g \ell^2}{4I} = 35\pi = 110$ rad/sec^2

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5. **Ans (2500)**

$A_1V_1 = A_2V_2$

$10 \times 1 = 5\ V_2$

$V_2 = 2$ m/s

$$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2$$

$$4000 + \frac{1}{2} \times 10^3(1)^2 = P_2 + \frac{1}{2} \times 1000(2)^2$$

$P_2 = 2500$

6. **Ans (40)**

$$mg = \rho_w \frac{v}{3}g = \rho_A \frac{v}{2}g \quad ...(i)$$

$$\rho_{mixture} = \frac{\rho_w V_0 + \rho_A V_0}{2V_0} = \frac{\rho_w + \rho_A}{2}$$

$$mg = \left[\frac{\rho_w + \rho_A}{2}\right] \times \left[\frac{v}{n}\right] g \quad ...(ii)$$

From equation (i) & (ii)

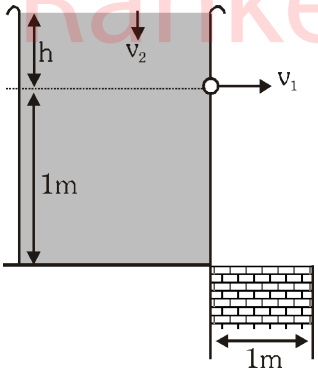
$$\rho_w \frac{v}{3}g = \frac{vg}{2n}\left[\rho_w + \frac{2\rho_w}{3}\right] ;$$

$$\frac{1}{3} = \frac{1}{2n}\left[\frac{5}{3}\right] \Rightarrow n = \frac{5}{2}$$

PHYSICS

SECTION-III

1. **Ans (5)**



For minimum velocity at orifice

$$\Rightarrow (v_{min}) \times \sqrt{\frac{2h}{g}} = 1 \Rightarrow v_{min} = \sqrt{5}\ m/s$$

This will give us minimum height for this velocity

$$\Rightarrow \sqrt{2gh_{min}} = \sqrt{5} \Rightarrow h_{min} = \frac{1}{4}m = 0.25m$$

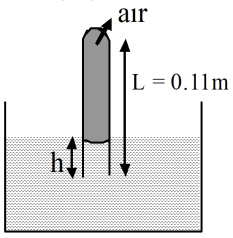
$v_1 = \sqrt{2gh}$ & $v_2 = -dh/dt$

By equation of continuity $av_1 = Av_2$

$$\Rightarrow a\sqrt{2gh} = A\left(-\frac{dh}{dt}\right) \Rightarrow \int_{0.81}^{0.25} \frac{dh}{\sqrt{h}} = \int_0^t \frac{\left(a\sqrt{2g}\right)}{A} dt$$

$\Rightarrow t = 125$ sec $= (5)^3 = (a)^3 \Rightarrow \alpha = 5$

2. **Ans (1)**



If final pressure of gas in tube is P_2 then

$P_2 - \frac{2T}{r} = P0$ (as levels inside and outside are same)

i.e. $P_2 = P_0 + \frac{2T}{r}$

but as temperature remains constant

$\therefore$ $P_1V_1 = P_2V_2$

$$P_0AL = \left(P_0 + \frac{2T}{r}\right)A(L - h)$$

On solving, h = 0.01 m

3. **Ans (4)**

$$p = \frac{2s}{r} = \frac{BdV}{V}$$

$$V = \frac{4}{3}\pi r^3$$

$$dV = 4\pi r^2 dr$$

$$\frac{2s}{r} = B \times \frac{4\pi r^2 dr}{4/3\pi r^3}$$

$$dr = \frac{2s}{3B} = \frac{2 \times 0.075}{3 \times 1.25 \times 10^8} = \frac{0.15}{3 \times 1.25} \times 10^{-8} =$$

4Å

4. **Ans (6)**

$$\sigma = \mu \frac{\partial u}{\partial y} = 1.5 \times 4 - 2y = 1.5 \times 4 = 6$$

PHYSICS

SECTION-IV

1. **Ans (A->PQ,B->PQ,C->PRST,D->PRST)**

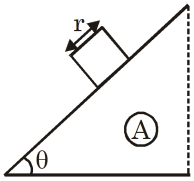
If $dgh_1 > P$ $\Rightarrow$ height of liquid will not cross h_1

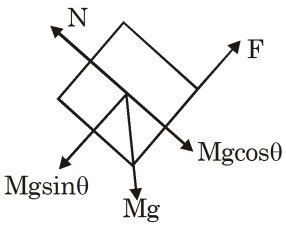
$dgh_1 < P \Rightarrow$ height of liquid will cross h_1

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2. Ans (A->PQRST,B->PQT,C->RS,D->Q)

(A) $\theta < \theta_R$





Block will not slide

Block in equilibrium

Force exerted by A = Mg

$N = mg\cos\theta$

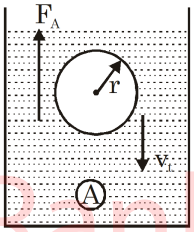
$f = mg\sin\theta$

$F_A =$

work done by $F_A = 0$ (displacement is zero)

Hence, mechanical energy as well as KE remains const.

(B)



Body falling with terminal velocity

KE is const.

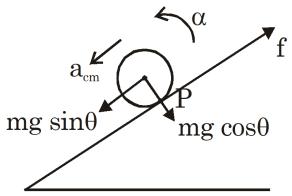
Hence, $F_{net} = 0$

$F_A = mg$

If density of ball increased then mass will increase and it will start accelrating since it

acquire new terminal velocity.

(C)



$mg\sin\theta - f = ma_{cm}$ .....(i)

$f(R) = \left(\frac{2}{5}MR^2\right)\alpha$ .....(ii)

$\alpha = \frac{a_{cm}}{R}$ .....(iii)

On solving

$f = \frac{2}{7}mg\sin\theta$

$F_A = \sqrt{f^2 + N^2} =$

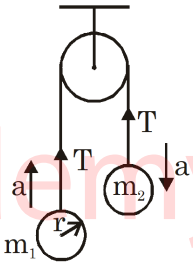
$\sqrt{mg\cos\theta^2 + \left(\frac{2}{7}mg\sin\theta\right)^2} < mg$

velocity of pt. P = 0

Hence, $\vec{s}_P = 0$

work done by $F_A = 0$.

(D)



$m_1 < m_2$

$m_2g - T = m_2a$

$T - m_1g = m_1a$

$a = \frac{(m_2 - m_1)g}{m_1 + m_2}$

$T = \left(\frac{2m_1m_2}{m_1 + m_2}\right)g \neq mg$

If density of body is changed, then mass will change and so will acceleration.

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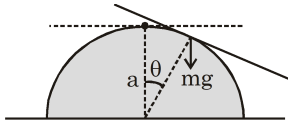
PHYSICS

SECTION-I (i)

1. **Ans (D)**

$\omega = \frac{2\pi}{T}, v_0 = \omega A$ and $a = \omega^2 A$ [m, T and v_0 are given]

2. **Ans (A)**



$I\alpha = \tau = mga\sin\theta$

$\frac{m(2\ell)^2}{12}\alpha = mga\theta$

$\alpha = \left(\frac{3ga}{\ell^2}\right)\theta = \omega^2\theta$

$T = \frac{2\pi}{\omega} = \frac{2\pi\ell}{\sqrt{3ga}}$

3. **Ans (B)**

$\frac{17\pi}{12}\sqrt{\frac{m}{K}}$

4. **Ans (C)**

$(2\pi a/b)(m/k)^{1/2}$

PHYSICS

SECTION-I (ii)

5. **Ans (A,B)**

$x = 2 + 5\sin\left(\pi t + \frac{\pi}{6}\right)$

$x - 2 = 5\sin\left(\pi t + \frac{\pi}{6}\right)$

equilibrium position x = 2

$v_{max} = A\omega = 5\pi$ m/s

6. **Ans (A,B,D)**

$8 = \frac{1}{2}kA^2$

A = 6 cm

$k = \frac{16}{(6\times10^{-2})^2} = \frac{16}{36}\times10^4 = \frac{40}{9}\times10^3 = 4.44$ kN/m

mass con't be found k_{max} at mean.

8. **Ans (A,B,D)**

$v_{rms} = \sqrt{\frac{v_0^2\int_0^T\cos^2(\omega t+\phi)\,dt}{\int_0^T dT}} = \frac{v_0}{\sqrt{2}}$

9. **Ans (B,C,D)**

$U = 5x(x-4)$

$\Rightarrow F = -\frac{dU}{dx} = -10x + 20$

when a = 0

$\Rightarrow$ x = 2m velocity is maximum

$F = -10x + 20 \longrightarrow$ SHM eq^m

$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{0.1}{10}} = \pi/5$ sec

10. **Ans (A,B,C)**

$\frac{\pi}{10} - T = \frac{2}{\pi}\sqrt{\frac{3\times6}{9\times k}} \Rightarrow \frac{1}{10} = 2\sqrt{\frac{2}{k}} \Rightarrow \frac{1}{100} = 2\left(\frac{2}{k}\right)$

k = 800

$x_1 + x_2 = 6 \Rightarrow 3(x_1) = 6x_2 \Rightarrow x_2 = 2m, x_1 = 4$ cm

$\frac{1}{2}kx_2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \Rightarrow m_1v_1 = m_2v_2$

$3v_1 = 6v_2 \Rightarrow \frac{1}{2}\times800\times\frac{6\times6}{100\times100} = 2.4$ kg m/s

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11. Ans (C,D)

$T = 2\pi\sqrt{\frac{\ell}{g_{eff.}}}$

$g_{eff} = g + a \quad \Rightarrow \quad T \downarrow \Rightarrow \omega\uparrow$

$\Omega = \theta_0\omega$

$\omega\uparrow \Rightarrow \theta_0 \downarrow$

12. Ans (A,D)

(A) $kx = 3\,mg$;

$k(x + x_0) - T - mg = ma$ .....(1)

$T - 2\,mg = 2\,ma$ .....(2)

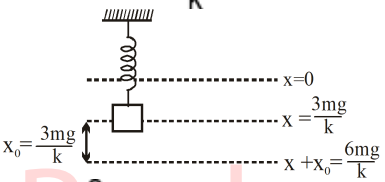
On solving,

$T = \frac{2m}{3}k(x + x_0)$

$\therefore$ for $x_0 = \frac{3mg}{k}$

$T = 4\,mg$

(D) If $x_0 = \frac{3mg}{k}$



If $x_0 > \frac{3mg}{k}$ then string will become lack when 'm' comes to rest at top most extreme possible.

13. Ans (A,B)

For (A) :

$T = \frac{T_1}{2} + \frac{T_2}{2} = \pi\sqrt{\frac{m}{k}} + \pi\sqrt{\frac{m}{4k}} = \frac{3\pi}{2}\sqrt{\frac{m}{k}}$

For (B) : $\frac{1}{2}k(A)^2 = \frac{1}{2}4k(x)^2$

For (C) : Not possible $\left[x = \frac{-A}{2}\right]$

For (D) : $TE = \frac{1}{2}kA^2$

14. Ans (A,D)

Maximum velocity for C.M. $= \frac{v}{2}$

mechanical energy is equal to maximum K.E.

$= \frac{1}{2}I\omega^2$ (about O)

$= \frac{1}{2}\left(2mr^2\right)\left(\frac{v}{2r}\right)^2 = \frac{mv^2}{4}$

& time period is given by $2\pi\sqrt{\frac{I}{mgd}}$

$= 2\pi\sqrt{\frac{2mr^2}{mgr}} = 2\pi\sqrt{\frac{2r}{g}}$

15. Ans (A,B)

$T_1 = 2\pi\sqrt{\frac{I}{mg\ell}}$

$= 2\pi\sqrt{\frac{2\ell}{3g}}$

$T_2 = 2\pi\sqrt{\frac{\frac{1}{3}m\ell^2 + m\ell^2}{(2m)g\cdot\left(\frac{3\ell}{4}\right)}}$

$= 2\pi\sqrt{\frac{\frac{4m\ell^2}{3}}{\frac{3mg\ell}{2}}} = 2\pi\sqrt{\frac{8}{9}\frac{\ell}{g}}$

$\frac{T_2}{T_1} = \frac{\sqrt{\frac{8}{9}}}{\sqrt{\frac{2}{3}}} = \frac{2}{13}$

Conserving angular momentum about hinge, before and after the collision.

$\frac{1}{3}m\ell^2\omega = \frac{4}{3}m\ell^2\omega' \Rightarrow \omega' = \frac{\omega}{4}$

$\omega = (\theta_0)\sqrt{\frac{3g}{2\ell}}$

$\omega' = (\theta_0)_{new}\sqrt{\frac{9g}{8\ell}}$

$\frac{\omega'}{\omega} = \frac{(\theta_0)_{new}}{(\theta_0)}\sqrt{\frac{3}{4}} \Rightarrow \frac{1}{4}\sqrt{\frac{4}{3}} = \frac{(\theta_0)_{rew}}{(\theta_0)}$

$(\theta_0)_{new} = \frac{\theta_0}{\sqrt{12}}$

17. Ans (A,C)

$mgy = \frac{1}{2}ky^2 \Rightarrow y = \frac{2mg}{k}$

$W_{N.C} + W_{mg} + W_{spring} = 0$

$W_{NC} = -mg(y_{eq}) + \frac{1}{2}ky_{eq}^2$

$y_{eq} = \frac{mg}{k}$

$W_{NC} = -\frac{(mg)^2}{2k}$

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18. **Ans (A,B,C,D)**

If particle is at extreme position then in one second it will cover phase angle of 60° or 300° for displacement of $\frac{A}{2}$

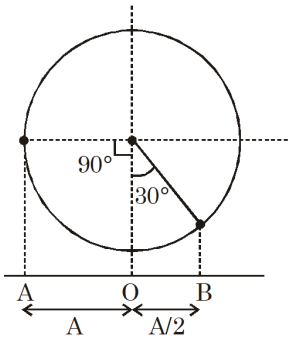
Hence time period can be 6 second or 1.2 seconds

If particle is at mean position then in one seconds it will cover phase angle of 30° or 150° for displacement $\frac{A}{2}$

Hence time period can be 12 sec or 2.4 sec

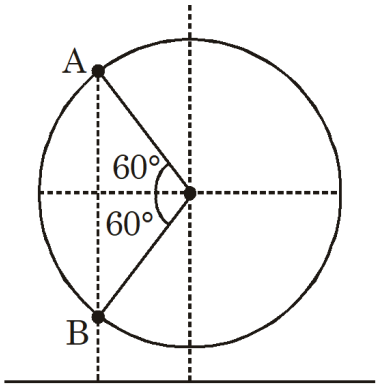
19. **Ans (B,D)**

Draw phasor diagram



$\phi = 90° + 30° = 120° = \frac{2\pi}{3}$ rad

When both meet, the situation on phasor diagram can be drawn as

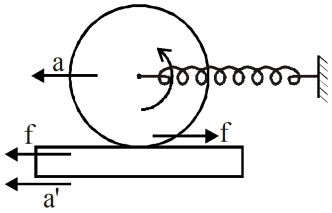


Angle rotated by A = 60° = $\frac{\pi}{3}$

$\phi = \omega t \Rightarrow \frac{\pi}{3} = \omega t$

$t = \frac{\pi}{3\omega}$

20. **Ans (B,C)**



$kx - f = Ma$

$f = Ma'$

$a' = a - Ra$

$fR = \frac{1}{2}MR^2 a$

$\frac{f}{M} = a - \frac{2f}{M}$

$\frac{3f}{M} = a$

$a' = \frac{3f}{M} - \frac{2f}{M} = \frac{f}{M}$

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21. Ans (A,B,C,D)

(a) The block moves with the board in what we take as the positive x direction, stretching the spring until the spring force $-kx$ is equal in magnitude to the maximum force of static friction: $kx = \mu_s n = \mu_s mg$

This occurs at $x = \frac{\mu_s mg}{k}$

(b) Since v is small, the block is nearly at the rest at this break point. It starts almost immediately to move back to the left, the forces on it being $-kx$ and $+\mu kmg$. While it is sliding the net force exerted on it can be written as

$$F_{net} = -kx + \mu_k mg = -kx + \frac{k\mu_k mg}{k} = -k\left(x - \frac{\mu_k mg}{k}\right)$$

$$= -kxrel$$

where xrel is the excursion of the block away from the point $\frac{\mu_k mg}{k}$.

Conclusion : the block goes into simple harmonic motion centered about the equilibrium position where the spring is stretched by $\frac{\mu_k mg}{k}$

(c) The amplitude of its motion is its original displacement, $A = \frac{\mu_s mg}{k} - \frac{\mu_k mg}{k}$, because the block has been pulled out to $x = \frac{\mu_s mg}{k}$, then it goes into simple harmonic motion centered about $x = \frac{\mu_k mg}{k}$

It first comes to rest at spring extension $\frac{\mu_k mg}{k} - A = \frac{(2\mu_k - \mu_s)\, mg}{k}$.

Almost immediately at this point it latches onto the slowly moving board to move with the board. The board exerts a force of static friction on the block, and the cycle continues.

(d) The time during each cycle when the block is moving with the board is $\frac{2A}{v} = \frac{2(\mu_s - \mu_k) mg}{kv}$.

The time for which the block is springing back is one half a cycle of simple harmonic motion, $\frac{1}{2}\left(2\pi\sqrt{\frac{m}{k}} = \pi\sqrt{\frac{m}{k}}\right)$ (because the block slides from +A to −A during its SHM). We ignore the times at the end points of the motion when the speed of the block changes from v to 0 and from o to v. Since v is small compared to $\frac{2A}{\pi\sqrt{m/k}}$, these times are negligible. Then the period is

$$T = \frac{2(\mu_s - \mu_k) mg}{kv} + \pi\sqrt{\frac{m}{k}}$$

22. Ans (A,D)

(a) p/5, (b) 2cm

24. Ans (A,D)

(a) $\frac{1}{2\pi}\sqrt{\frac{2C}{MR^2}}$, (b) $\theta_{new} = \frac{\theta_0}{\sqrt{3}}$; $f' = \frac{1}{2\pi}\sqrt{\frac{2C}{3MR^2}}$

$\tau = C\theta = I\alpha$

$C\theta = MR^2\alpha$

$\alpha = -\frac{\frac{2C}{MR^2}}{\omega^2}$; $f = \frac{1}{2\pi}\sqrt{\frac{2C}{MR^2}}$

at $t = \frac{2\pi}{\omega}$ at mean position

$I_{new} = \frac{3}{2}MR^2$

$C\theta = -MR^2\alpha$

$-\frac{\frac{2C}{3MR^2}}{\omega^2} = \alpha$; $f = \frac{1}{2\pi}\sqrt{\frac{2C}{3MR^2}}$

$\frac{1}{2}C\theta^2 = \frac{1}{2}I\omega^2$

$\frac{1}{2}C\theta_{new}^2 = \frac{1}{2}I_{new}\omega_{new}^2$

$\frac{\theta_{new}^2}{\theta^2} = \frac{I_{new}\omega_{new}^2}{I\omega^2} = \frac{I_{new}}{F}\frac{\frac{3}{2}MR^2}{\frac{MR^2}{2}} \times \frac{I^2}{I_{new}^2}$

$\frac{\theta_{new}^2}{\theta^2} = \frac{I_{new}}{I} \times \frac{I^2}{I_{new}^2} = \frac{I}{I_{new}} = \frac{\frac{1}{2}MR^2}{\frac{3}{2}MR^2}$

$\frac{\theta_{new}^2}{\theta^2} = \frac{1}{\sqrt{3}}$

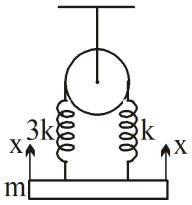
$\theta_{new} = \frac{\theta}{\sqrt{3}}$

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PHYSICS

SECTION-II

1. **Ans (26.00)**



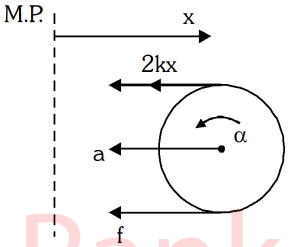
$3kx_1 + kx_2 = ma$

$\frac{x_1 + x_2}{2} = x - 6kx1 = ma$

$\frac{-3kx}{m} = a$

$a = \sqrt{\frac{3 \times 676}{3}} = 26$

2. **Ans (3)**



$2kx + f = Ma$ ....(1)

$(2kx - f)R = \frac{MR^2}{2}\alpha$ ....(2)

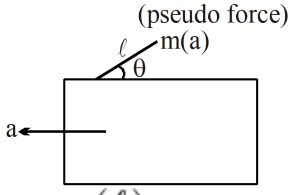
$a = Ra$ ....(3)

$f = \frac{2kx}{3}$

$F_{net} = -(2kx + f) = -\frac{8kx}{3}$

$T = 2x\sqrt{\frac{3M}{8k}} = \pi\sqrt{\frac{3M}{2k}}$

3. **Ans (1)**



$\tau = ma\left(\frac{\ell}{2}\right)\sin\theta$

$\alpha = \frac{\left(ma\frac{\ell}{2}\right)\sin\theta}{\left(\frac{m\ell^2}{3}\right)}$

$\alpha = \frac{3a}{2\ell}\sin\theta$

for slightly open θ is very small

$\sin\theta \approx \theta$

$\alpha = \left(\frac{3a}{2\ell}\right)\theta$

$\frac{d^2\theta}{dt^2} = (\omega^2)\theta$

$\theta = \theta_0 \cos(\omega t)$

time taken to shut $= \frac{T}{4} = \frac{2\pi/\omega}{4} = \frac{\pi}{2\sqrt{\frac{3a}{2\ell}}} = \pi\sqrt{\frac{\ell}{6a}}$

distance travelled by car $= 0 + \frac{1}{2}at^2 = \frac{1}{2}a\,p^2\,\frac{\ell}{6a} = \frac{\pi^2\ell}{12} = \frac{(10)(1.2)}{12} = 1$ m

4. **Ans (2)**

$T = 2\pi\sqrt{\frac{1\ell}{1g}}$

$\Rightarrow x + y = 1 + 1 = 2$

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5. **Ans (3)**

$mg - T = ma_{cm}$

$T = 2ma_{cm}$

$mg - 2ma_{cm} = ma_{cm}$

$mg = 3ma_{cm}$

$a_{cm} = g/3$

In centre of mass frame

$$\omega = \sqrt{\frac{k}{\mu}} = \sqrt{\frac{k(2m+m)}{(2m \times m)}}$$

$$\omega = \sqrt{\frac{3k}{2m}}$$



maximum acceleration of mass m = $\frac{g}{3}$

$\frac{g}{3} = \omega^2 x_1$

x_1 is maximum displacement of mass m

$x_1 = \frac{g}{3\omega^2}$

maximum acceleration of mass m = $\frac{2g}{3}$

$\frac{2g}{3} = \omega^2 \times 2$

x_2 is maximum displacement of mass 2m

$x_2 = \frac{2g}{3\omega^2}$

total displacement

$$x = x_1 + x_2 = \frac{g}{3\omega^2} + \frac{2g}{3\omega^2}$$

$$= \frac{3g}{3\omega^2} = \frac{g}{\omega^2} = \frac{g}{3k} \times 2m$$

$= 10 \times 2 \times 3 = 2$ mela

net separation between

then is $= \left(\frac{1}{100} + \frac{2}{100}\right) m = \frac{3}{100} m = 3cm$

6. **Ans (2)**

From energy conservation

$$\frac{1}{2}kx^2 + 2\left(\frac{1}{2}I\omega^2 + \frac{1}{2}mv^2\right) + \frac{1}{2}mv^2 = 0$$

$\left(I = \frac{mR^2}{2}\right)$

$\Rightarrow \quad kx^2 + 4m\,v^2 = 0 \Rightarrow$

$$2kx \cdot \frac{dx}{dt} + (4m)2v\frac{dv}{dt} = 0$$

$$\Rightarrow a = -\left(\frac{k}{4m}\right)x$$

$$\Rightarrow w = \sqrt{\frac{k}{4m}}$$

PHYSICS

SECTION-IV

1. **Ans (A->PQRT,B->PQRST,C->P,D->PRST)**

(P) To have a turning point, total energy should be less than the energy barrier (or maximum potential energy shown in graph)

(Q) If a graph seems like a parabola, then about O, potential energy function is quadratic in variable plotted on x-axis.

(R) If $F \propto x$, or $\tau \propto \theta$, for small displacement about O, motion will be simple harmonic.

(S) If total energy < E_2, the particle will execute simple harmonic motion about O, if it is a position of stable equilibrium and if E_2 is less than maximum potential energy (PE of Barrier)

(T) Point O will be position of stable equilibrium if

$\frac{dU}{dr} = 0$ and $\frac{d^2U}{dr^2} < 0$.

Now checking the above points for (A)

(A) O is the position of stable equilibrium and graph is parabola near O,

$U \propto \theta^2 \Rightarrow \tau \propto \theta$

Motion will be SHM for small oscillation about O.

E_2 is greater than the barrier potential energy or maximum potential energy.

Checking the condition for (B)

$$U(x) = 1 - e^{-x^2} = 1 - \left(1 - x^2 + \frac{x^4}{2!} + \ldots.\right)$$

$$U(x) = x^2 - \frac{x^4}{2!} + \ldots.$$

Neglecting higher power $F(x) = -\frac{dU}{dx} = -2x$

About O motion is SHM for small oscillation

(C) At O, graph is flat, it is a position of neutral equilibrium,

(D) If total energy is E_3, particle can have turning point if it mover on the left side of graph.

For small displacement, $U \propto x^2$

$\Rightarrow \quad F \propto x \Rightarrow$ SHM

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PHYSICS

SECTION-I (i)

2. Ans (B)

$\mu_1 \sin i = \mu_2 \sin 90°$

$\tan i = \dfrac{\sqrt{\left(2^2 + \left(\sqrt{5}\right)^2\right)}}{4} \Rightarrow i = 37° \Rightarrow \mu_2 = \dfrac{6}{5}$

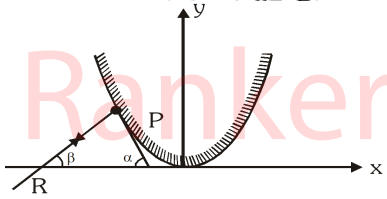
3. Ans (A)

Point of intersection of two curves, $y = \dfrac{x^2}{4}$ and $y = x + 3$ is, $(-2, 1)$

$\dfrac{dy}{dx} = \dfrac{x}{2}$ i.e., $\left(\dfrac{dy}{dx}\right)_{x=-2} = \dfrac{-2}{2} = -1$

$\tan \beta$ = slope of incident ray RP =1 or $\beta = 45°$

and $\tan \alpha = \left|\left(\dfrac{dy}{dx}\right)_{x=-2}\right| = 1 \Rightarrow \alpha = 45°$



From geometry we can show that $\angle i = \angle r = 0°$

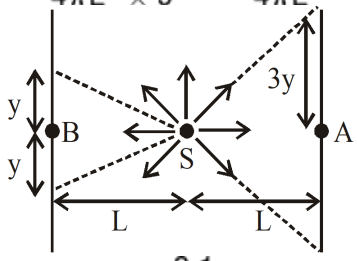
i.e., the ray is incident normally.

Hence, the desired unit vector : $A = \dfrac{1}{\sqrt{2}}\left(-\hat{i} - \hat{j}\right)$

4. Ans (A)

$I_{net} = I_A + I_B$

$I_B = \dfrac{I_0}{4\pi L^2 \times 9}, I_A = \dfrac{P_0}{4\pi L^2} = 8.1\mu$



$I_{net} = 8.1\ \mu + \dfrac{8.1\mu}{9}$

$I_{net} = 9\ \mu$

5. Ans (C)

For the image to coincide with object, image formed by concave mirror must be at 'a' in front of plane mirror.

$\therefore \dfrac{1}{v} + \dfrac{1}{u} = \dfrac{1}{f} \Rightarrow \dfrac{-1}{2a} - \dfrac{1}{4a} = \dfrac{1}{f} \Rightarrow f = -4$

$\therefore$ R = 8m

6. Ans (A)

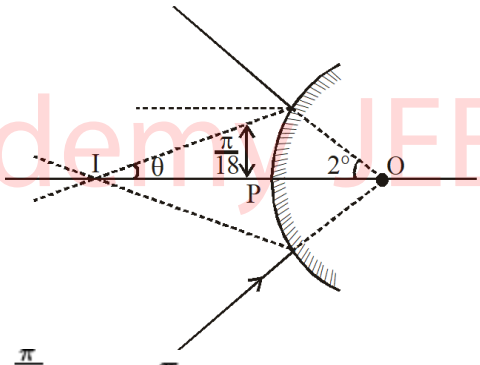
At t = 3 sec the radius becomes ∞ and the mirror behaves like plane mirror.

$\dfrac{V_I + V_0}{2} = V_M$

$\dfrac{V_I + 6}{2} = -3$

$V_I = -12$

7. Ans (D)



$\dfrac{\frac{\pi}{18}}{OP} = 2 \times \dfrac{\pi}{180}$

OP = 5

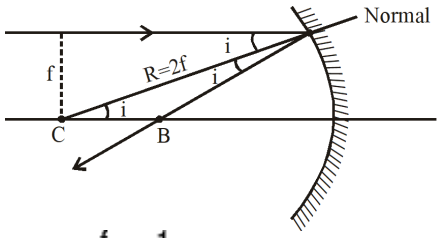
$\dfrac{1}{v} + \dfrac{1}{5} = \dfrac{1}{10}$

f = 10 v = −10

$\theta = \dfrac{\pi}{18 \times 10} = \dfrac{\pi}{180} = 1°$

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8. Ans (A)



$$\sin i = \frac{f}{R} = \frac{1}{2}\ ;\ i = 30^\circ$$

Not a paraxial ray

$$CB = \frac{R}{2\cos i} = \frac{f}{\left(\sqrt{3}/2\right)} = f\left(\frac{2}{\sqrt{3}}\right)$$

$$\frac{CB}{f} = \frac{2}{\sqrt{3}}$$

9. Ans (D)

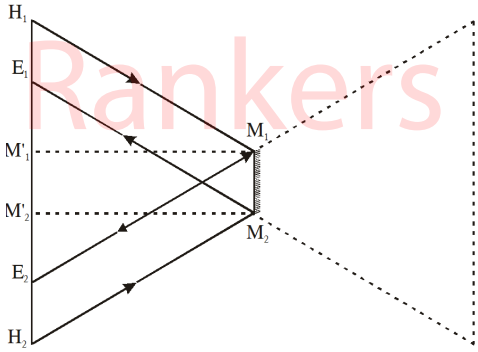
Let's say $H_1H_2 = D$, $E_1E_2 = d$

H_1H_2 = width of persons face.

E_1E_2 = separation between the eyes.

Let's say M_1M_2 = Width of mirror required.

Then following will be the ray diagram and calculations.



$$H_1M'_1 = \frac{H_1E_2}{2} \ \&\ H_2M'_2 = \frac{H_2E_1}{2}$$

$$H_1E_2 = D - \frac{1}{2}(D-d) = \frac{D+d}{2} = H_2E_1$$

$$M'_1M'_2 = D - H_1M'_1 - H_2M'_2$$

$$= D - \left(\frac{D+d}{2}\right) = \frac{D-d}{2}$$

10. Ans (B)

Reflection at concave mirror

$$v = \frac{uf}{u-f} = \frac{(-20)(-15)}{-20+15} = -60\ \text{cm}$$

$$m = \frac{h_I}{h_0} = -\frac{v}{u} \Rightarrow h_I = -h_0 \times \left(\frac{v}{u}\right) \Rightarrow h_I = -2 \times \left(\frac{-60}{-20}\right) = -6\ \text{mm}$$

Again reflection at convex mirorr

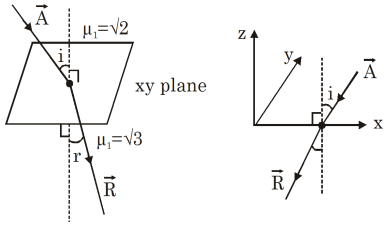
$$v = \frac{uf}{u-f} = \frac{10 \times 20}{10-20} = -20\ \text{cm}\ ;$$

$$h_I = -h_0 \times \frac{v}{u} = -(-8) \times \left(\frac{-20}{10}\right) = -16\ \text{mm}$$

11. Ans (D)

Ans.

$$\vec{r} = \frac{3}{5\sqrt{2}}\hat{i} + \frac{2\sqrt{2}}{5}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}\ \text{(angle of incidence } = 60^0;\ r = 45^0)$$



From the figure, it can be seen that normal to the boundary separating the two media will be in z-direction, as incident ray, refracted ray and normal are in the same plane

The angle made by the incident ray with the normal is given as

$$\cos i = \frac{A_z}{A} = \frac{10}{\sqrt{\left(6\sqrt{3}\right)^2 + \left(8\sqrt{3}\right)^2 + 100^2}} = \frac{10}{\sqrt{108+192+100}} = \frac{10}{20}$$

$$\Rightarrow \cos i = \frac{-1}{2} \Rightarrow i = \cos^1\left(\frac{1}{2}\right)$$

$$\Rightarrow i = 60^\circ$$

Applying Snell's law of refraction at the interface of two media

$n_1 \sin i = n_2 \sin r$

$$\sqrt{2}\sin 60^\circ = \sqrt{3}\sin r$$

$$\sin r = \frac{\sqrt{2}\times\sqrt{3}}{\sqrt{3}\times 2} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow r = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) \Rightarrow r = 45^\circ$$

As the X, Y components remain unchanged, so unit vector along direction of refracted ray is

$$\hat{R} = \frac{6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j}}{\sqrt{\left(6\sqrt{3}\right)^2 + \left(8\sqrt{3}\right)^2}}\sin 45 + \cos 45\left(-\hat{k}\right)$$

$$\hat{R} = \frac{3}{5\sqrt{2}}\hat{i} + \frac{4}{5\sqrt{2}}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}$$

$$\hat{R} = \frac{1}{5\sqrt{2}}\left(3\hat{i} + 4\hat{j} - 5\hat{k}\right)$$

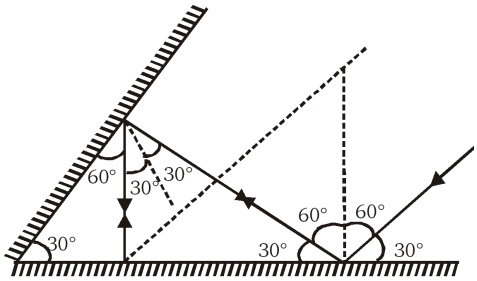
Hence, the answer is $\frac{1}{5\sqrt{2}}\left(3\hat{i} + 4\hat{j} - 5\hat{k}\right)$

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PHYSICS

SECTION-I (ii)

12. Ans (A,C)



13. Ans (B,C,D)

y component of velocity is changing

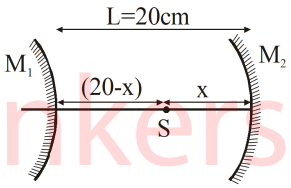
Mirror must be in x-z plane

Since ray was initially travelling towards +y axis

Normal must be along –ve y i.e. $-\hat{j}$

$\tan\theta = \frac{3}{4}$ angle of incidense = 37°

14. Ans (A,C)



For first reflection from convex surface

$u_1 = -(20 - x)$

$f_1 = +6$

$v_1 = \frac{u_1 f_1}{u_1 - f_1} = \frac{(20-x)6}{26-x}$

and for second reflection from concave surface,

$v_2 = (-x)$

$f_2 = -6$

$\frac{1}{-x} + \frac{1}{u_2} = \frac{1}{-6} \Rightarrow \frac{1}{u_2} = \frac{1}{x} - \frac{1}{6}$

$u_2 = \frac{6x}{6-x}$

But image of first reflection is object for second reflection,

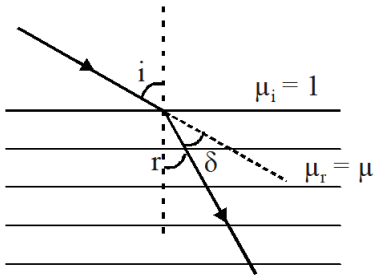
so, $u_2 = v_1 + 20$

or $\frac{6x}{6-x} = \left[\frac{(20-x)6}{26-x}\right] + 20 \Rightarrow x = 8$ cm

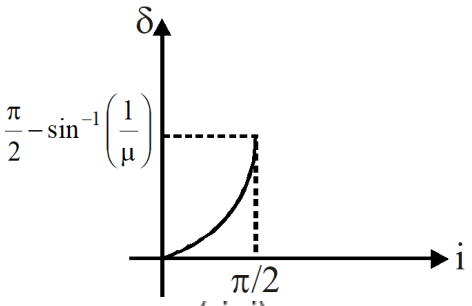
By principle of reversibility of light, it is also, true for first reflection on concave mirror and then on convex mirror.

15. Ans (A,B,D)

Light travels from rarer to denser medium



$\delta = i - r$



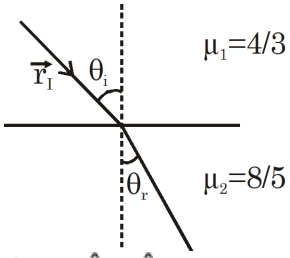
$\delta = i - \sin^{-1}\frac{(\sin i)}{\mu}$ non-linear increasing function of i.

slope of tangent : $\frac{d\delta}{d_i} = 1 - \frac{\cos i}{\sqrt{\mu^2 - \sin^2 i}}$

Slope of tangent increases as i increases

Hence 'c' is not the graph for light that travels from rarer to denser medium.

16. Ans (A,C)



$\vec{r}_I = +3\hat{i} - 4\hat{j}$

$\tan\theta_i = \frac{3}{4}$

$\theta_i = 37°$

$\mu_1 \sin\theta_i = \mu_2 \sin\theta_r$

$\frac{4}{3} \times \frac{3}{5} = \frac{8}{5}\sin\theta_r$

$\sin\theta_r = \frac{1}{2}$ $\theta_r = 30°$

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17. **Ans (B,D)**

Let the critical angle of interface between media 1 and 2 is C_1 and between 1 and 3 is C_2.

Then $\sin C_1 = \frac{n_2}{n_1}$ and $\sin C_2 = \frac{n_3}{n_1}$.

From TIR at second interface $90 - C_1 > C_2$. taking sin of both side we get

$\cos C_1 > \sin C_2$ or $\sqrt{1-\left(\frac{n_2}{n_1}\right)^2} > \frac{n_3}{n_1}$ or

$n_1^2 - n_3^2 > n_2^2$

18. **Ans (B,C)**

$m^2 = \left(\frac{v-f}{f}\right)^2$ = graph will be a parabola

$m = 0$ at focus $\Rightarrow$ $f = 30$

$b = 1 = m$ at $v = 0 = a$ as mag at $2f$

at $v = 15$ m $= \frac{1}{2}$; $m^2 = \frac{1}{4} \Rightarrow \frac{a}{4} = c$

19. **Ans (A,B,C)**

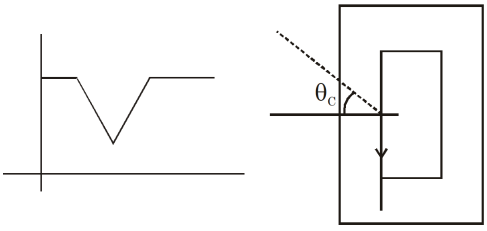
$x = C$ [$\because$ After critical angle reflection starts], $y = r - i = 90^\circ - C$ & $z = 180^\circ - 2C$

PHYSICS

SECTION-II

1. **Ans (53)**

$\theta \approx 53^\circ$



For refraction from plastic sheet

$i < \theta_C$

maximum angle

$i = \theta_C$

$\sin i = \sin\theta_C = \frac{\mu_r}{\mu_D} = \frac{1.2}{1.5}$

$\sin\theta_C = \frac{4}{5}$

$\theta_C = 53^\circ$

2. **Ans (6)**

$\angle ABC = \pi - 2\left(\hat{i} - \hat{r}\right)$

$2\angle ABC = \angle AOC$ [half angle theorem]

$2\pi - 4\hat{i} + 4r = (\pi + 2r)$

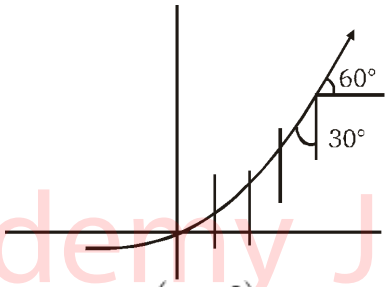
$2\hat{i} - r = \frac{\pi}{z} \Rightarrow r = 2\hat{i} - \frac{\pi}{2}$

$1\sin\hat{i} = \mu\sin\hat{r}$

$\sin\hat{i} = \mu\sin\left(2\hat{i} - \frac{\pi}{2}\right)$

$\sin\hat{i} = \frac{\sqrt{3}}{2} \Rightarrow \hat{i} = \frac{\pi}{3} \Rightarrow \frac{2\pi}{6} \Rightarrow k = 6$

3. **Ans (1)**



$\mu_1 \sin\theta_1 = c,$

$1\sin 90^\circ = \left(1 + y^2\right)\sin 30^\circ \Rightarrow y = 1$

4. **Ans (5)**

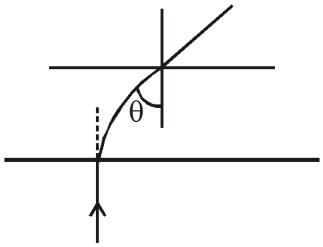
Let frequency of rotating mirror be f.

$\therefore \frac{2D}{v} \times 2\pi f = \frac{\theta}{2} \Rightarrow D = \frac{25}{4}$m

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5. **Ans (2)**

Apply snell's law, $\mu \sin\theta$ = constant



From Snell's law

$1 = \mu\sin\theta$

$\sin\theta = \frac{1}{\mu}$

$\frac{dy}{dx} = \tan\theta = \frac{\sin\theta}{\cos\theta}$

$\frac{dy}{dx} = \frac{1/\mu}{\frac{(\mu^2-1)^{1/2}}{\mu}}$

$\frac{dy}{dx} = e^{\frac{-x}{2d}}$

$\int dy = \int e^{\frac{-x}{2d}} dx$

$y = -2d e^{\frac{-x}{2d}} + C$

at x = 0, y = 0, C = 2d

at y = d

$y = -2d e^{\frac{-x}{2d}} + 2d$

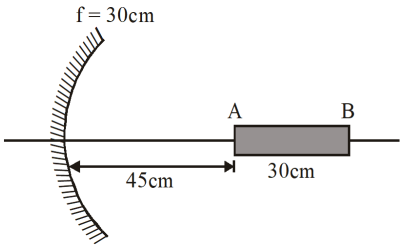
$-d = -2d e^{\frac{-x}{2d}}$

$\frac{1}{2} = e^{\frac{-x}{2d}}$

$\ell n2 = \frac{x}{2d}$

$2d\,\ell n2 = x$

6. **Ans (4)**



Let's find the image of end A.

$\Rightarrow \frac{1}{V_1} + \frac{1}{45} = \frac{1}{30}$

$\Rightarrow V_1 = 90$ cm ... (i)

$\Rightarrow$ Let's find the image of end B.

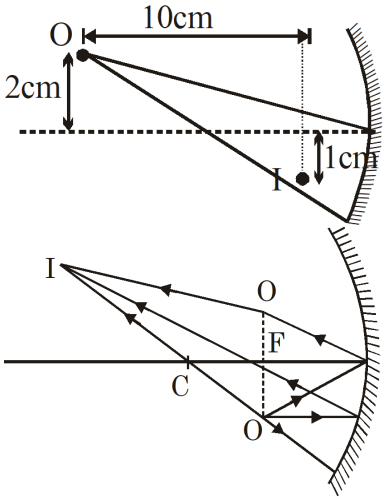
$\Rightarrow \frac{1}{V_2} + \frac{1}{(45+30)} = \frac{1}{30}$

$V_2 = 50$ cm ... (ii)

$\Rightarrow$ So length of image = $|V_1 - V_2| = 40$ cm

7. **Ans (3)**

Ans. 20 cm from object, 20/3 cm from object



PHYSICS

SECTION-IV

1. **Ans (A->R,B->S,C->P,D->Q)**

(C) Velocity of bird in water = $6 \times \frac{4}{3} = 8 \downarrow$

w.r.t fish = 8 + 4 = 12 $\downarrow$

(D) Velocity of bird in water after reflection from mirror = 8 $\uparrow$

w.r.t fish = 8 – 4 = 4 $\uparrow$

(A) Velocity of fish in air = $4 \times \frac{3}{4} = 3 \uparrow$

Velocity of fish w.r.t bird = 3 + 6 = 9 $\uparrow$

(B) Velocity of image of fish after reflection from mirror in air = $4 \times \frac{3}{4} = 3 \downarrow$

w.r.t bird = – 3 + 6 = 3 $\uparrow$

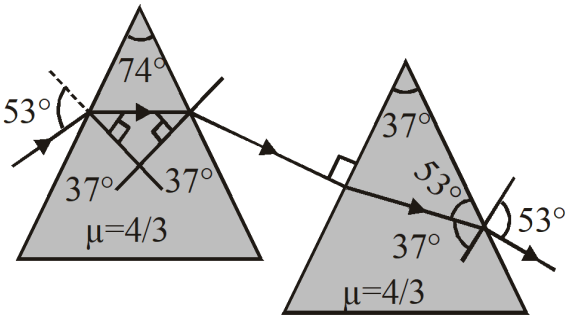
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PHYSICS

SECTION-I (i)

1. Ans (A)

Here we have drawn the ray diagram with all angles. The values of all angles can be found by using concepts of prism theory, refraction and basic geometry.



Deviation produced by 1st prism is, $\delta_1 = 53° + 53° - 74° = 32°$ clockwise

Deviation produced by 2nd prism is, $\delta_2 = 53° - 37° = 16°$ – clockwise

So, $\delta = 32° + 16° = 48°$ clockwise

2. Ans (A)

$\frac{1}{v} - \frac{1}{-15} = \frac{1}{5}$; $\frac{1}{v} - \frac{1}{5} - \frac{1}{15}$; v = 7.5; v' = – 12.5

$\frac{1}{v'} - \frac{1}{12.5} = \frac{1}{-15}$; $\frac{1}{v'} = \frac{2}{25} - \frac{1}{15}$; $\frac{1}{v'} = \frac{12-10}{150}$

v' = 75 cm; v'' = – 95 cm; $\frac{1}{v''} + \frac{1}{95} = \frac{1}{5}$; v'' = 5.3 cm

3. Ans (A)

$\frac{1}{f_{eq}} = \frac{1}{f_m} - \frac{2}{f_L}$

$\frac{1}{f_L} = (\mu - 1)\left[\frac{+1}{R}\right] = \frac{1}{2R}$

$\therefore \quad \frac{1}{f_q} = -\frac{1}{R} \Rightarrow f_{eq} = -R$ [concave mirror]

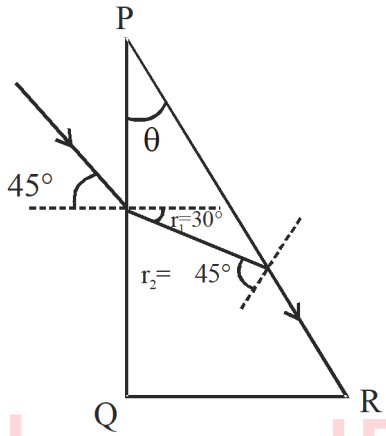
$\therefore$ Image of object kept at 2R will form at 2R

4. Ans (A)

This case of prism is unlike to the cases we are familiar with. In this case the light ray is coming from above, so our conventional formulas would not be applicable. In the ray diagram below we can say that:

$\Rightarrow (r_1 + 90°) + (90 - r2) + \theta = 180°$

$\Rightarrow r_2 - r_1 = \theta$. Rest of the solution is as below.



$1 \sin 45° = \sin r_1$

$r_2 - r_1 = \theta$

$\theta = 45° - 30°$

$\Rightarrow \theta = 15°$

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PHYSICS

SECTION-I (ii)

5. Ans (A,B,C,D)

$\delta_{min} = 2i - A \quad \Rightarrow \quad 60° = 2(60°) - A \quad \Rightarrow \quad A = 60°$

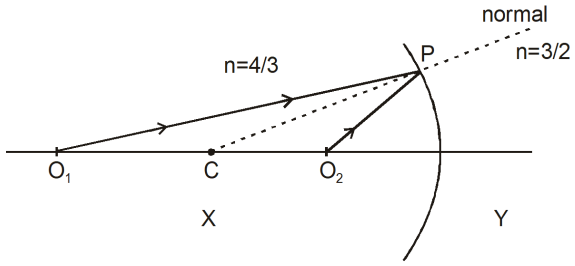
$\mu = \frac{\sin\left(\frac{A+\delta_{min}}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\sin\left(\frac{60°+60°}{2}\right)}{\sin\left(\frac{60°}{2}\right)} = \sqrt{3} \Rightarrow \delta$

$= i + e - A$

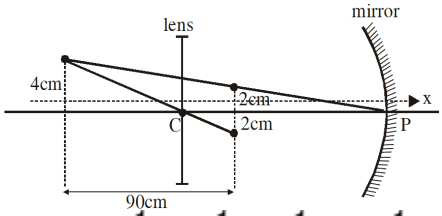
$65° = i_1 + 70° - 60° \quad \Rightarrow \quad i_1 = 55°$

6. Ans (A,B,D)

Let there be two point objects O_1 and O_2, Incident rays from O_1 and O_2 at point P shall both bend towards normal and hence the corresponding refracted rays shall intersect the principal axis in the left medium. Therefore image formed under given condition shall always be virtual.



7. Ans (A,B)



For mirror $\frac{1}{f} = -\frac{1}{90} - \frac{1}{180} = -\frac{1}{60}$ For lens $\frac{1}{f} = \frac{1}{30} + \frac{1}{30} + \frac{1}{60} = \frac{1}{20}$

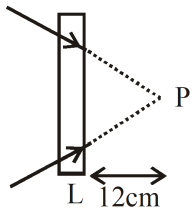
8. Ans (A,B,C,D)

$\frac{m_1}{m_2} = 4 \Rightarrow m_1 m_2 = 1 \Rightarrow m_1 = 2, m_2 = \frac{1}{2}$

$x + y = 96; \frac{y}{x} = 2$, $y = 2x$, $3x = 96$, $x = 32$

distance between two positive = $x - y = 32$

9. Ans (B,D)



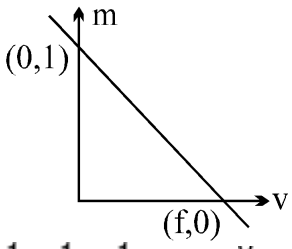
(A) If L is convex & focal length 20

$f = -20 \quad u = +12 \quad \Rightarrow \quad v = +7.5$

(C) If L is concave lense

than $f = +16 \quad u = +12 \quad \Rightarrow v = +48$

10. Ans (A,B,C)



$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow 1 - \frac{v}{u} = \frac{v}{f}$

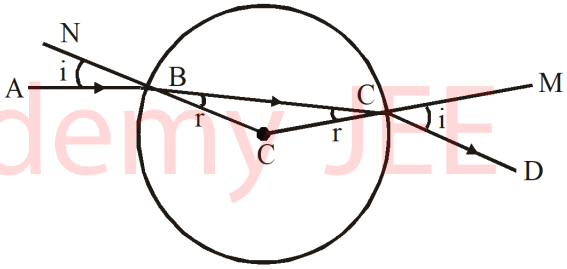
or $m = -\left(\frac{1}{f}\right)v + 1$

comparing with $y = mx + c$

A, B & C are correct

12. Ans (B,C)

The diagram shows the path of a ray AB incident at B on the sphere at angle i. The path ABCD shows the refracted ray BC, inside the sphere with angle r of refraction and the emergent ray CD with angle of emergence i.



Obviously $\angle OBC = \angle OCB = r$. Hence $\angle ABN = \angle DCM = i$ for all values of i.

Also, if $\angle ABN = 90°$, grazing incidence at B, then $\angle DCM = 90°$, grazing emergence at C.

But AB and CD will not be parallel.

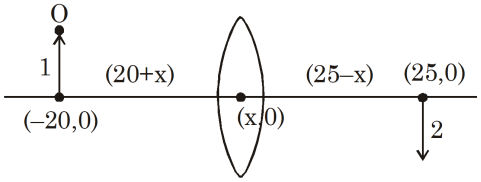
13. Ans (A,B,C)

Light will emerge from prism if $A < \theta_C$

Light will not emerge from prism if $A > 2\theta_C$ $\therefore$ $\theta_C = 45°$

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14. Ans (B,C)



Let's say the lens is located at (x, 0), then following is the representations of object and image. Assuming x > 0.

$$\Rightarrow \frac{1}{(25-x)}+\frac{1}{(20+x)}=\frac{1}{f} \quad \text{... (i)}$$

$$\Rightarrow \frac{25-x}{20+x}=\frac{2}{1}=2 \quad \text{... (ii)}$$

By solving above two equations we get the answer.

15. Ans (A,C,D)

The cavity will behave like a thick parallel slab and we know that it does not bring any deviations to light, it just laterally displaces the light and that does not matter for our case.

So independent of what is filled in the cavity the minimum angle of deviation will not change.

$\Rightarrow$ We know that, $u=\dfrac{\sin\left(\frac{A+\delta_{min}}{2}\right)}{\sin(A/2)}$

$$\Rightarrow \sqrt{2}=\frac{\sin\left(\frac{60+\delta_{min}}{2}\right)}{\sin\left(\frac{60}{2}\right)}$$

$\Rightarrow$ $\delta_{min} = 30°$ So, (A,C,D)

17. Ans (A,B,C)

$y_1 = f\,\delta_1 = 20 \times \dfrac{\pi}{100}$ cm = $\pi/5$ cm ; $y_2 = f\,\delta_2 = 20 \times \dfrac{3\pi}{50}$ cm = 1.2π cm

PHYSICS

SECTION-II

2. Ans (6)

[Number of images = 2, distance between them = 0.003 m]

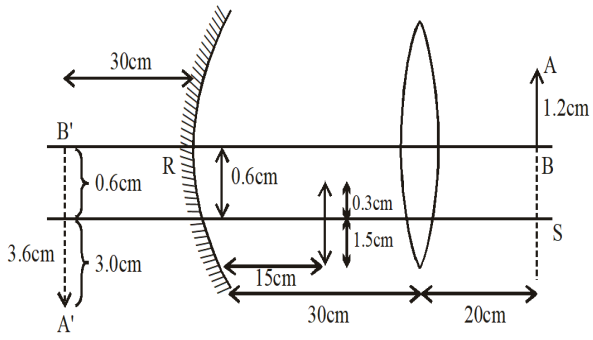
3. Ans (20)

Ans. 20 cm behind the mirror

Parallel ray on equi-concave lens will make the image on the left side of the lens at it's focal length. This will be considered an object for the plane mirror. This object will be at the distance of |d + f| from the plane mirror on the left so image will be |d + f| on the right side of mirror.

4. Ans (15)

Ans. 15 cm right of the mirror, magnification = 1.5; (−15 cm, −1.5 cm; − 15 cm, 0.3 cm)



For lens,

u = −20 cm, f = +15 cm

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f} \Rightarrow v_1=\frac{uf}{u+f}$$ = 60 cm left of lens

This A′B′ will act as object for mirror.

So, $u_2 = +30$ cm

$f_2 = -30$ cm

$$\frac{1}{v_2}+\frac{1}{u_2}=\frac{1}{f_2} \Rightarrow v_2=\frac{u_2 f_2}{u_2-f_2}$$ = −15 cm right of mirror

PHYSICS

SECTION-III

1. Ans (6)

Image of O will be formed at O itself.

Image of A will form, $\dfrac{1}{v}-\dfrac{3/2}{-5}=\dfrac{1-3/2}{-10} \Rightarrow \dfrac{1}{v}=\dfrac{1}{20}-\dfrac{3}{10}=\dfrac{-5}{20}$ $\therefore$ v = −4

$\therefore$ Distance between A & O = (10−4) = 6 cm

2. Ans (4)

$\dfrac{1}{F_{lens}}=(1.5-1)\left[\dfrac{1}{-40}-\dfrac{1}{-20}\right]=\dfrac{1}{80}$; $F_\ell = 80$ cm

$F_m = -\dfrac{20}{2} = -10$ cm; $\dfrac{1}{F_{eq}}=\dfrac{1}{f_m}-\dfrac{2}{f_\ell}=\dfrac{1}{-10}-\dfrac{2}{80}$; $f_{eq} = -8$ cm

Hence object should be placed at x = 16 cm, i.e. at the centre of curvature.

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3. **Ans (2)**

Refraction from a spherical surface is given by,

$$\frac{\mu_2}{v}=\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$$

For refraction from the first surface,

$$\frac{7}{4v_1}-\frac{1}{-24}=\frac{\frac{3}{4}}{6}$$

For refraction from second surface,

$$\frac{4}{3v_2}-\frac{7}{4v_1}=\frac{\frac{3}{4}-\frac{7}{4}}{\infty}$$

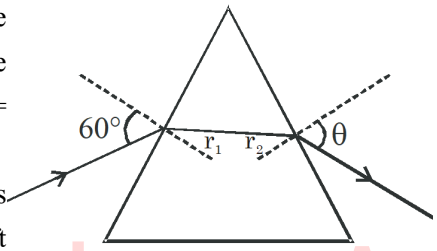
Adding the two equations gives,

$v_2 = 16$ cm

Thus distance from the bottom = depth $- v_2 = 18$ cm $- 16$ cm $= 2$ cm

4. **Ans (2)**

From the geometry we have $r_1 + r_2 = 60°$



Using Snells law at the left side we have $\frac{\sin 60°}{\sin r_1}=n$

or $\sin r_1=\frac{\sqrt{3}}{2n}$

Using Snells law at the right end

or $\frac{\sin r_2}{\sin\theta}=\frac{1}{n}$

$\sin\theta = n \sin r_2$

$$= n\sin(60°-r_1)=n\left(\frac{\sqrt{3}}{2}\cos r_1-\frac{1}{2}\sin r_1\right)$$

$$\text{or} = n\left(\frac{\sqrt{3}}{2}\sqrt{1-\sin^2 r_1}-\frac{1}{2}\sin r_1\right)$$

$$\text{or} = n\left(\frac{\sqrt{3}}{2}\sqrt{1-\frac{3}{4n^2}}-\frac{1}{2}\frac{\sqrt{3}}{2n}\right)$$

Solving we get

$$\sin\theta=\frac{\sqrt{3}}{4}\sqrt{4n^2-3}-\frac{\sqrt{3}}{4}$$

Differentiating

$$\cos\theta d\theta=\frac{\sqrt{3}}{4}\times\frac{1}{2\sqrt{4n^2-3}}\times 8ndn$$

or $$\frac{d\theta}{dn}=\frac{\sqrt{3}n}{\sqrt{4n^2-3}}\times\frac{1}{\cos\theta}$$

$$=\frac{\sqrt{3}\times\sqrt{3}}{\sqrt{4\times3-3}}\times\frac{1}{1/2}=2$$

PHYSICS

SECTION-IV

1. **Ans (A->QS,B->PS,C->QS,D->S)**

At minimum deviation $i_1 = i_2$, FG||BC

At maximum deviation $i_1 = 90°$ or $i_2 = 90°$

For $i_1 = 0$, TIR will not take place at AC

2. **Ans (A->S,B->P,C->R,D->Q)**

For (A) : $\frac{1}{f}=(\mu-1)\left(\frac{1}{p_1}=\frac{1}{p_2}\right)=(1.5-1)\left(\frac{1}{20}+\frac{1}{20}\right)=\frac{1}{20}\Rightarrow$ f= 20cm

For (B) : $\frac{1}{f}=\left(\frac{1.5}{4/3}-1\right)\left(\frac{1}{20}+\frac{1}{20}\right)=\frac{1}{80}\Rightarrow$ f = 80 cm

For (C) : $\frac{1.5}{v_1}-\frac{4/3}{\infty}=\left(1.5-\frac{4}{3}\right)\left(\frac{1}{20}\right)$ & $\frac{1}{f}-\frac{1.5}{v_1}(1-1.5)\left(\frac{1}{-20}\right)\Rightarrow$ f = 30 cm

For (D) : $\frac{1.5}{v_1}-\frac{1}{\infty}=(1.5-1)\left(\frac{1}{20}\right)$ & $\frac{4/3}{f}-\frac{1.5}{v_1}=\left(\frac{4}{3}-1.5\right)\left(-\frac{1}{20}\right)\Rightarrow$ f = 40 cm

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GUIDED REVISION

JEE (Main + Advanced) 2025
ENTHUSIAST & LEADER COURSE
PHASE : ALL

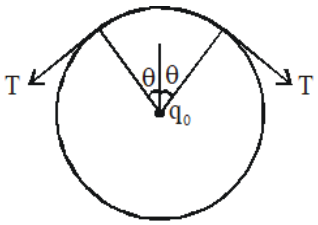
SOLUTION

GR_Electrostatics & Gravitation-1

PHYSICS

SECTION-I (i)

1. Ans (D)



θ is small, then

$2T \sin\theta = kq_0\left(\frac{Q}{2\pi R}\right)\frac{(R2\theta)}{R^2}$

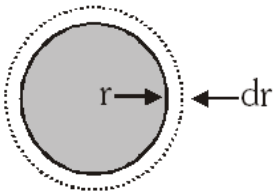
Putting $\sin\theta = \theta$ we get

$T = \frac{1}{8\pi^2\epsilon_0}\frac{Qq_0}{R^2}$

Now $Y = \frac{TR}{A\Delta R}$

$\therefore \Delta R = \frac{TR}{AY} = \left(\frac{1}{8\pi^2\epsilon_0}\frac{Qq_0}{R}\right)\left(\frac{1}{AY}\right)$

2. Ans (A)



Charge in layer of width dr

$dq = 2\pi r dr\sigma$

Work to bring dq = dU=Vdq

$dU = 4\sigma r \cdot 2\pi r dr\sigma = 8\pi\sigma^2 r^2 dr$

$U = 8\pi\sigma^2\int_0^R r^2 dr = \frac{8\pi\sigma^2}{3}R^3$

3. Ans (A)

$mv = 2m\,(v')$

$\frac{KQ(-Q)}{5R} = \frac{KQ(-Q)}{2R} + \frac{1}{2}mv^2 + \frac{1}{2}(2m)(v')^2 \Rightarrow v = \sqrt{\frac{2KQ^2}{5mR}}$

4. Ans (B)

Charge on sphere $Q = \int_0^R \rho_0\left(4\pi r^2 dr\right) = \pi\rho_0 R^4$

By definition of potential $V_{2R} - V_\infty = \frac{kQ}{2R} \Rightarrow$

$V_\infty = -\frac{kQ}{2R} = -\frac{Q}{8\pi\epsilon_0 R}$

5. Ans (D)

$\vec{E} = -\frac{dv}{dx}\hat{i} - \frac{dv}{dy}\hat{j} = -2x\hat{i} + 2y\hat{j}$

$\tan\theta = -\frac{dy}{dx} = -\frac{y}{x} \Rightarrow \frac{dy}{y} = -\frac{dx}{x}$

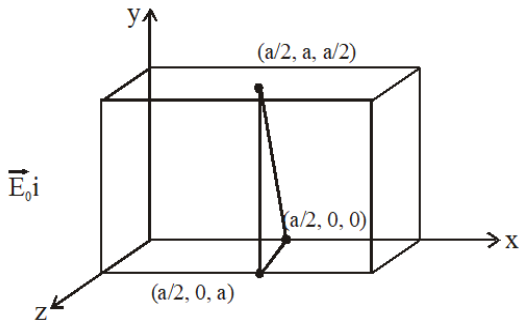
$\ell ny = -\ell nx + \ell nC \quad \ell n(xy) = \ell nC$

$xy = C$

shape will be rectangular hyperbola

6. Ans (B)

The flux of y-component of $\vec{E}$ is zero, through this loop

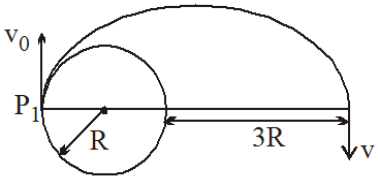


$\therefore \phi = E_0 \times \frac{1}{2} \times a \times a$

$\phi = \frac{E_0 a^2}{2}$

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7. Ans (A)



Let speed at max height is v

Since, it is at max height, v is perpendicular to r

⇒ By conservation of angular momentum

$mv_0R = mv(4R)$

$v = \frac{v_0}{4}$

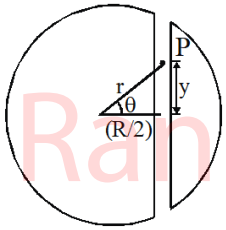
⇒ By conservation of mechanical energy

$$\frac{1}{2}mv_0^2 - \frac{GMm}{R} = \frac{1}{2}m\left(\frac{v_0}{4}\right)^2 - \frac{GMm}{4R}$$

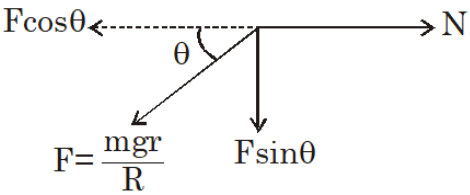
$$v_0 = \sqrt{\frac{8}{5}\frac{GM}{R}}$$

$$g = \frac{GM}{R^2}, v_0 = \sqrt{\frac{8}{5}gR}$$

8. Ans (D)



Gravity at point $P = \left(\frac{gr}{R}\right)$



$F\cos\theta - N = 0$

$F\sin\theta = ma$

$$\frac{mgr}{R}\left(\frac{y}{r}\right) = ma$$

$$a = \left(\frac{g}{R}\right)y = \frac{g}{R}\sqrt{r^2 - \left(\frac{R}{2}\right)^2}$$

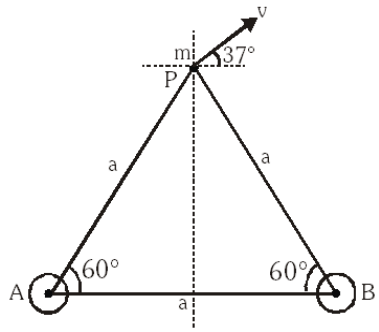
$a \longrightarrow f(r)$

9. Ans (C)

$$v_e = \sqrt{2}\sqrt{\frac{GM}{r_0}}$$

$$\therefore \Delta v = v_e - v_0 = \left(\sqrt{2}-1\right)\sqrt{\frac{GM}{r_0}}$$

10. Ans (D)



Total mechanical energy = 0

$$-2\frac{GM}{a}m + \frac{1}{2}mv_e^2 = 0$$

$$v_e = 2\sqrt{\frac{GM}{a}}$$

PHYSICS

SECTION-I (ii)

11. Ans (A,B,C,D)

$\vec{E} = (10-5x)\,i$

m = 5kg

$\vec{F} = q\vec{E}$

$\vec{F} = (10-5x)\,i$

(a) $F \propto -x$

so motion of charge particle is oscillatory

(b) F = 0 at x = 2

oscillation about x = 2

so maximum displacement is 4

$F = 10 - 5x$

$a = \frac{10-5x}{2}$

$a = 2 - x$

$a = -(x-2)$

so $\omega^2 = 1$

$\omega = 1$

(c) $v_{max} = A\omega$

A = 2

$\omega = 1$

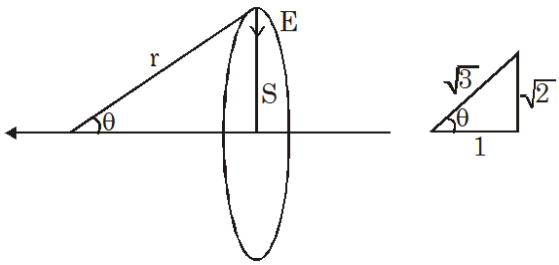
$v_{max} = 2$

(d) velocity at mean position

at x = 2

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12. **Ans (B,D)**



For dipole $\tan\alpha = \frac{1\tan\theta}{2}$ where α is angle of electric field with position vector

$\tan\theta = \sqrt{2}$

$\frac{S}{r} = \sin\theta$

$qE = \frac{mv^2}{R}$

$\frac{qkp}{mr^3}\sqrt{1+3\cos^2\theta} = \frac{v^2}{s}$

$r = \frac{s}{\sin\theta}$, $\cos\theta = \frac{1}{\sqrt{3}}$; since $= \frac{\sqrt{2}}{\sin\theta}$

$\frac{kpq}{ms^2}\frac{2\sqrt{2}}{3\sqrt{3}}\left(\sqrt{2}\right) = \frac{v^2}{s}$

$\frac{pq}{\pi\epsilon_0 m3\sqrt{3}} = v^2 \Rightarrow \frac{1}{s}\sqrt{\frac{pq}{\pi\epsilon_0 m3\sqrt{3}}}$

14. **Ans (A,B,D)**

Finally potential on all sphere will be equal

$\frac{KQ_1}{R} = \frac{KQ_2}{2R} = \frac{KQ_3}{3R}$ and

$Q_1 + Q_2 + Q_3 = (Q) + (-2Q) = -Q$

$\Rightarrow Q_1 = \frac{-Q}{6}, Q_2 = \frac{-Q}{3}, Q_3 = \frac{-Q}{2}$

15. **Ans (A,B,C)**

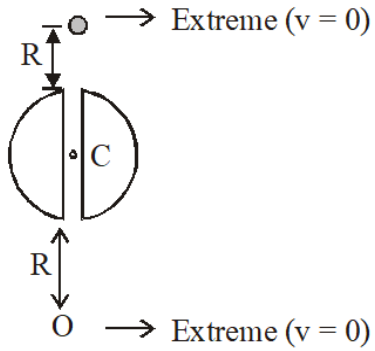
For (A) : By COME total mechanical energy of the two objects

For (B) : By using reduced mass concept

$\frac{1}{2}\mu v_{rel}^2 - \frac{G(4m)(m)}{r} = 0$ where $\mu = \frac{(m)(4m)}{m+4m} = \frac{4}{5}m \Rightarrow v_{rel} = \sqrt{\frac{10Gm}{r}}$

For (C) total KE = $-$PE = $\frac{G(m)(4m)}{r} = \frac{4Gm^2}{r}$

16. **Ans (A,C,D)**



Let r is distance from center,

If r > R $F = \frac{GMm}{r^2}$ (Not SHM)

r < R $F = \frac{GMmr}{R^3}$ (SHM)

$\Rightarrow$ But motion is periodic to find velocity at center,

Apply COME,

$\frac{1}{2}m\left(0^2\right) - \frac{GMm}{2R} = \frac{1}{2}mv_C^2 - \frac{3}{2}\frac{GMm}{R}$

$v_C = \sqrt{\frac{2GM}{R}}$

17. **Ans (A,C)**

$V_{CM} = 0, Mv_1 + 2Mv_2 = 0$

$\frac{1}{2}Mv_1^2 + \frac{1}{2}2Mv_2^2 = \frac{GM(2M)}{3R} - 0$

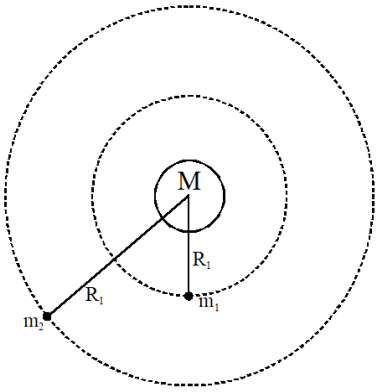
$\Rightarrow$ relative velocity at t = T is $\frac{1}{3}\sqrt{\frac{2GM}{R}}$

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PHYSICS

SECTION-I (iii)

19. Ans (B)



$\left.\begin{matrix}\frac{m_1}{m_2}=2\\ \frac{R_1}{R_2}=\frac{1}{4}\end{matrix}\right\}$ given

$$\frac{GMm_1}{R_1^2}=\frac{m_1v_1^2}{R_1}$$

$$v_1^2=\frac{GM}{R_1},\ v_2^2=\frac{GM}{R_2}$$

$$\frac{v_1^2}{v_2^2}=\frac{R_2}{R_1}=4$$

(P) $\frac{v_1}{v_2}=2$

(Q) $L = mvR$

$$\frac{L_1}{L_2}=\frac{m_1v_1R_1}{m_2v_2R_2}=2\times2\times\frac{1}{4}=1$$

(R) $K=\frac{1}{2}mv^2$

$$\frac{K_1}{K_2}=\frac{m_1v_1^2}{m_2v_2^2}=2\times(2)^2=8$$

(S) $T = 2\pi R/V$

$$\frac{T_1}{T_2}=\frac{R_1}{v_1}\times\frac{v_2}{R_2}=\frac{R_1}{R_2}\times\frac{v_2}{v_1}=\frac{1}{4}\times\frac{1}{2}=\frac{1}{8}$$

PHYSICS

SECTION-II

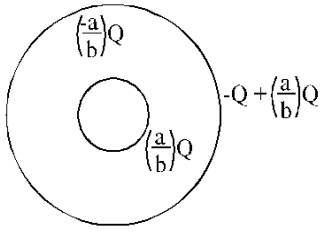
1. Ans (81)

$$\frac{\frac{2kQq}{L}}{R}=\frac{2\times9\times10^9\times3\times10^{-3}\times5\times10^{-6}}{R}\times\frac{3}{5}=81$$

2. Ans (180)

When outer surface is grounded charge '−Q' resides on the inner surface of sphere 'B'

Now sphere A is connected to earth potential on its surface becomes zero.



Let the charge on the surface A becomes q

$$\frac{kq}{a}-\frac{kQ}{b}=0\Rightarrow q=\frac{a}{b}Q$$

Consider the figure. In this position energy stored

$$E_1=\frac{1}{8\pi\varepsilon_0 a}\left[\frac{a}{b}Q\right]^2+\frac{Q^2}{8\pi\varepsilon_0 b}+\frac{1}{4\pi\varepsilon_0 b}\left[\frac{a}{b}Q\right](-Q)$$

when 'S_3' is closed, total charge will appear on the outer surface of shell 'B'. In this position energy stored $E_2=\frac{1}{8\pi\varepsilon_0 b}\left(\frac{a}{b}-1\right)^2Q^2$ Heat produced = $E_1-E_2=\frac{Q^2a(b-a)}{8\pi\varepsilon_0 b^3}=180$

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PHYSICS

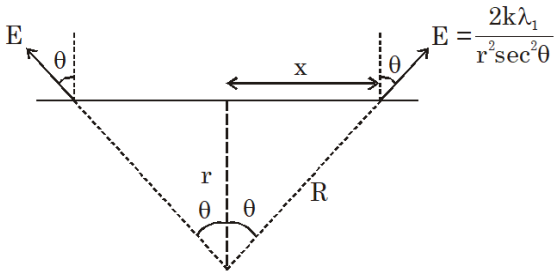
SECTION-III

1. Ans (0)

Electric field due to 1 at 2

$E = \frac{2k\lambda_1}{r}$ and it is radially away from z-axis.

So



$r \sec\theta = R$

$r \tan\theta = x$

$dx = r \sec^2\theta \, d\theta$

So component of force along y-axis added while along x force is zero.

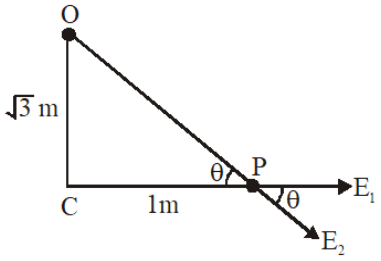
So force = $dF = (\lambda_2 dx) E\cos\theta = \left(\frac{2k\lambda_1}{r\sec\theta}\right)\cos\theta$ $(\lambda_2 \, r \sec^2\theta \, d\theta)$

$= 2k\lambda_1\lambda_2$

$$F_{net} = \int_{-\pi/2}^{\pi/2} 2k\lambda_1\lambda_2 d\theta = \frac{\lambda_1\lambda_2}{2\varepsilon_0}$$

So n = 0

2. Ans (9)

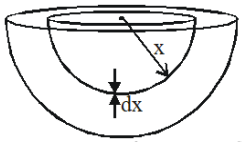


$$E_r = \vec{E}_1 + \vec{E}_2$$

$$E_r = \sqrt{E_1^2 + E_2^2 + 2E_1E_2\cos\theta}$$

where $E_1 = \frac{kQ}{12}$, $E_2 = \frac{kQ}{(\sqrt{3})^2}$ & $\theta = 60°$

3. Ans (3)



$$V = \int_a^b \frac{K\left(2\pi x^2 dx\right)\rho}{x}$$

$$V = 2\pi k\rho\left[\frac{b^2 - a^2}{2}\right] = \pi\rho k\,(b^2 - a^2)$$

b = 2a

$v = 3\pi k\rho b^2$

4. Ans (1)

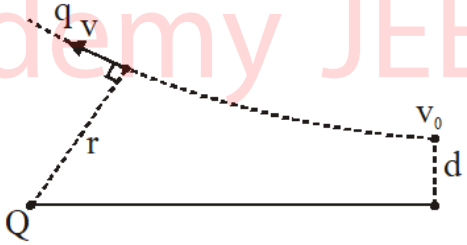
Potential due to one dipole

$V = \frac{kPi\cdot\left(\cos 30\hat{i} + \sin 30\hat{j}\right)}{\left(\frac{a}{\sqrt{3}}\right)^2}$; so total potential

= 3V

5. Ans (2)

Due to repulsion forces the point of q will be a line curved away from Q.



at least distance, r1 we can conserve angular momentum & energy of system.

$mv_0 d = mvr$

$$\& \; 0 + \frac{1}{2}mv_0^2 = \frac{KqQ}{r} + \frac{1}{2}mv^2$$

$$\frac{1}{2}mv_0^2 = \frac{KqQ}{r} + \frac{1}{2}m\left(\frac{v_0 d}{r}\right)^2$$

$$\frac{1}{2}mv_0^2\left(1 - \frac{d^2}{r^2}\right) = \frac{KqQ}{r}$$

$$\frac{1}{2}mv_0^2\left(r^2 - d^2\right) = KqQr$$

$$\left(\frac{1}{2}mv_0^2\right)r^2 - (KqQ)\,r - \frac{1}{2}mv_0^2 d^2 = 0$$

$$r = \frac{KqQ + \sqrt{(KqQ)^2 + m^2v_0^4d^2}}{mv_0^2} = 2m$$

KqQ = 9

$mv_0^2 = 12$

d = 1

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6. **Ans (6)**

$V_{(0,0,0)} = 0$ and $V_{(0,0,0.1)} = \frac{a(2)}{100} = \frac{a}{50}$

$W_{elect} = q(V_1 - V_2) \Rightarrow 50 \times 10^{-6} = 4 \times 10^{-6} \times \frac{a}{50}$

$\Rightarrow a = 625$

6250 volts $= 625\,(x^2) + 625\,(y^2) + 625\,(2)\,(2)^2$

$\Rightarrow 6 = x^2 + y^2$ Thus $R = \sqrt{6}$

7. **Ans (7)**

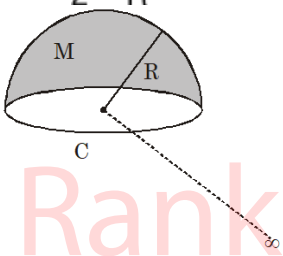
$\vec{E} = \frac{a\hat{r}}{r^2}$; $V = \frac{-a}{r}$; $\Delta V = \frac{a}{r_2} - \frac{a}{r_1} = \frac{a}{5} - \frac{a}{7} = \frac{2a}{35} = 2 \times \frac{122.5}{35} = 7$ V

$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$|\vec{r}| = r = \sqrt{x^2 + y^2 + z^2}$

8. **Ans (3)**

$A = -\frac{3}{2}\frac{GmM}{R}$



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$W_g = \Delta U_g$

$= -m\,(\Delta V_g)$

$= -m\left(V_{gf} - V_{gi}\right)$

$= -m\,(V_\infty - V_C)$

$W_g = -m\left(0 - \left(\frac{-3GM}{2R}\right)\right)$

$= -\frac{3GmM}{2R}$

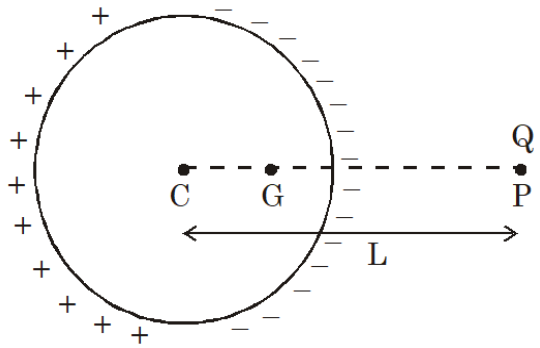
9. **Ans (6)**

$\frac{L_{total}}{L_B} = \frac{m_1 r_1^2}{m_2 r_2^2} + 1$

PHYSICS

SECTION-I (i)

1. Ans (A)



Inside metallic ball, potential is same everywhere.

$V_G = V_C$

V_C = V due to Q + V due to induced charge

$= \frac{KQ}{L} + 0$

$V_G = V_C = \frac{KQ}{L}$

2. Ans (B)

Use gauss theorem $\phi = \frac{q_{enclosed}}{\epsilon_0}$ = Number of electric field lines

3. Ans (B)

$\rho = \alpha r^2$ [where α is a constant]

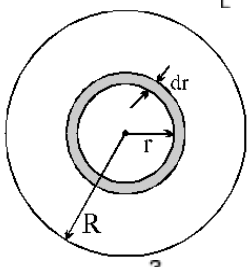
Charge in the shell (element)

$dq = r(4\pi r^2 dr) = \alpha(4\pi)r^4 dr$

$\therefore$ Charge enclosed in sphere of radius r, $q = 4\pi\alpha \int_0^r r^4 dr = \frac{4\pi\alpha r^5}{5}$

By Gauss's theorem,

at r = R/2 $(E_{r=R/2})\left[4\pi\left(\frac{R}{2}\right)^2\right] = \frac{4\pi\alpha(R/2)^5}{5\varepsilon_0}$



$E_{r=R/2} = \frac{\alpha R^3}{40\varepsilon_0}$

Total charge enclosed, $Q = \frac{4\pi\alpha R^5}{5}$

$\therefore E_{r=2R} = \frac{4\pi\alpha R^5}{5\varepsilon_0} \times \frac{1}{4\pi(2R)^2} = \frac{\alpha R^3}{20\varepsilon_0}$

$\frac{E_{r=2R}}{E_{r=R/2}} = 2$

4. Ans (C)

$V_A = V_B$

$r_A \geq r_B$

so, charge density at A > charge density at B

$E_A > E_B$

flux = 0

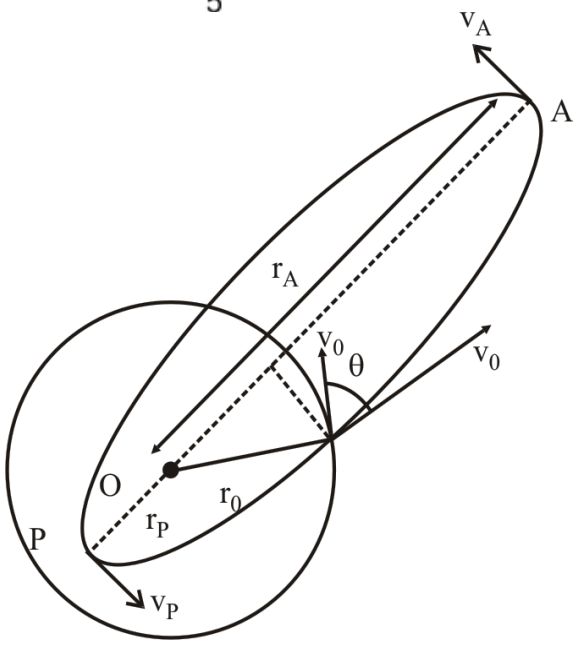
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5. Ans (B)

Using conservation of angular momentum about O.

$mV_Pr_P = mV_Ar_A = mV_0r_0\cos\theta$

$V_Ar_A = V_Pr_P = \frac{3v_0r_0}{5}$



using conservation of energy

$$\frac{1}{2}mV_A^2 + \frac{-GMm}{r_A} = \frac{-GMm}{r_0} + \frac{1}{2}mV_0^2$$

$\Rightarrow$

$$\frac{9V_0^2r_0^2}{50r_A^2} - \frac{V_0^2r_0}{r_A} + \frac{V_0^2}{2} = 0 \quad \left[\text{Let}\frac{r_0}{r_A} = x\right]$$

$\Rightarrow 9x^2 - 50x + 25 = 0 \Rightarrow x = 5$ or $(5/9)$

$\Rightarrow r_A = \frac{9r_0}{5}$; $r_P = \frac{r_0}{5}$

$\Rightarrow \frac{V_P}{V_A} = \frac{r_A}{r_P} = 9$

PHYSICS

SECTION-I (ii)

7. Ans (A,C)

$\uparrow \frac{\sigma_1}{2\epsilon_0}$

$+ + + + + + + + +$

$\downarrow \frac{\sigma_1}{2\epsilon_0}$

$$E_1 = \frac{8.85 \times 10^{-9}}{2 \times 8.85 \times 10^{-12}} = 500$$

$\uparrow \frac{\sigma_2}{2\epsilon_0}$

$+ + + + + + + + +$

$\downarrow \frac{\sigma_2}{2\epsilon_0}$

$$E_2 = \frac{17.7 \times 10^{-9}}{2 \times 8.86 \times 10^{-12}} = 1000$$

$$E_{net} = \sqrt{\left(\frac{10^3}{2}\right)^2 + \left(10^3\right)^2 + 2 \times \frac{10^3}{2} \times 10^3 \cos 60}$$

$$\sqrt{\frac{7}{4}} \times 1000 = 500\sqrt{7}$$

8. Ans (A,B,C)

From gauss theorem, $\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$

$$\left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right) \cdot \left(x\hat{i} + y\hat{j} + z\hat{k}\right) = \frac{\rho}{\epsilon_0}$$

that gives uniform density.

10. Ans (A,B)

$\phi = \vec{E}.\vec{A}$

ϕ_{in} = –ve and ϕ_{out} = +ve

$Q_{net} = \frac{q_{in}}{\varepsilon_0}$

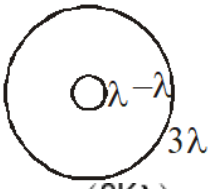
$\phi_1 = 0, \phi_2 = E_0a^3, \phi_3 = E_0a^3$;

$\phi_4 = 0, \phi_5 = 0, \phi_6 = 0$

$Q_{net} = \phi_1 + \phi_2 + \phi_3 + \phi_4 + \phi_5 + \phi_6 = 2E_0a^3 \Rightarrow 2E_0a^3 = \frac{q_{in}}{\epsilon_0} \Rightarrow q_{in} = 2\varepsilon_0E_0a^3$

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11. **Ans (A,C,D)**



$\vec{E} = 2\left(\frac{2K\lambda}{r}\right)\hat{r}(r < R)$

$\vec{E} = 3\left(\frac{2K\lambda}{r}\right)\hat{r}(r > R)$

$\phi_{ABCD} = \phi_{GFEH} + \phi_{BEFD} + \phi_{CGFD} = \frac{q}{4\epsilon_0}$

13. **Ans (A,D)**

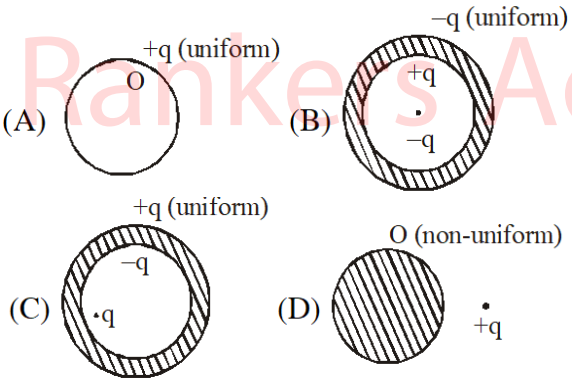
Due to surface tension bubble try to contract.

Due to surface charge bubble try to expand.

So excess pressure inside the bubble $= \frac{4T}{R} - \frac{\sigma^2}{2\varepsilon_0}$

and electrostatic pressure $= \frac{\sigma^2}{2\varepsilon_0}$

14. **Ans (A,C)**



15. **Ans (A,D)**

(A) $V_{centre} = \frac{KQ}{a} + \frac{K2\pi\lambda b}{b} = \frac{KQ}{a} + \frac{\lambda}{2\epsilon_0}$

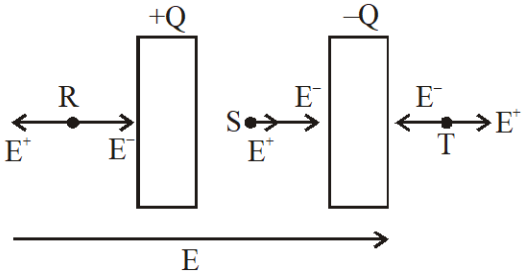
(B) $V_{centre} = 0$ for eathing $\frac{KQ'}{a} + \frac{K2\pi\lambda b}{b} = 0$

$Q' = -2\pi\lambda.a$

(C) $\vec{E}_{inside\ sphere} = \vec{0}$ always (property of conductor)

(D) Conductor without cavity is always equipotential.

16. **Ans (A,D)**



17. **Ans (A,D)**

ω in both is same

so $T_1 = T_2$

and by $v_{max} = A\omega$ $V_1 > V_2$

18. **Ans (A,B,C,D)**

$V = \sqrt{\frac{GM}{r}}$

$\frac{V_A}{V_B} = \sqrt{\frac{r_B}{r_A}} = 2$

$E = \frac{-GMm}{2r}$ $\frac{E_A}{E_B} = \frac{m_A}{m_B} \times \frac{r_B}{r_A} = 12$

$u_A = \sqrt{\frac{2GM}{r}}$ $\frac{u_A}{u_B} = 2$

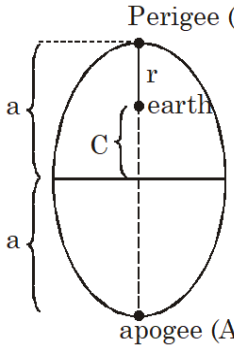
$T = \frac{2\pi}{\sqrt{GM}}(r)^{3/2}$

$\frac{T_A}{T_B} = \left(\frac{r_A}{r_B}\right)^{3/2} = \frac{1}{8}$

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19. Ans (B,C)

Initially parabolic path means escape velocity & hence (TE = 0)



$r_{min} = (a - c) = \frac{a}{2}$

$r_{max} = (a + c) = \frac{3a}{2}$

$e = \frac{c}{a} = \frac{1}{2} \Rightarrow c = \frac{a}{2}$

$r = R + h = R + \frac{R}{2} = \frac{3R}{2}$

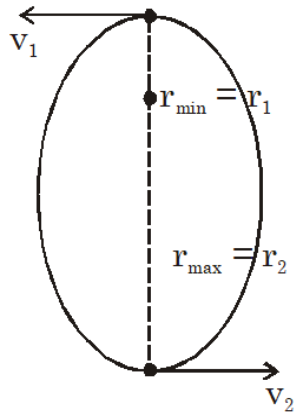
If $r_{min} = \frac{3R}{2} \Rightarrow a = 3R$

and if $r_{max} = \frac{3R}{2} \Rightarrow a = R$

If a = 3R, then, $C = \frac{a}{2} = \frac{3R}{2}$

$r_{min} = (a - c) = \frac{3R}{2} = r_1$

and $r_{max} = (a + c) = \frac{9R}{2} = r_2$



L = constant

$mv_1r_1 = mv_2r_2$

$v_1\left(\frac{3R}{2}\right) = v_2\left(\frac{9R}{2}\right)$

$v_1 = 3v_2$

TE = conserved

$\frac{1}{2}mv_1^2 - \frac{GmM}{r_1} = \frac{1}{2}mv_2^2 - \frac{GmM}{r_2}$

$\frac{v_1^2}{2} - \frac{GM}{\left(\frac{3R}{2}\right)} = \frac{\left(\frac{v_1}{3}\right)^2}{2} - \frac{GM}{\left(\frac{9R}{2}\right)}$

$\frac{v_1^2}{2}\left(1 - \frac{1}{9}\right) = \frac{GM}{R}\left(\frac{2}{3} - \frac{2}{9}\right)$

$v_1^2\left(\frac{4}{9}\right) = \frac{GM}{R}\left(\frac{4}{9}\right)$

$v_1^2 = \frac{GM}{R}$

$TE = \frac{1}{2}mv_1^2 - \frac{GMm}{r_1}$

$= \frac{1}{2}m\frac{GM}{R} - \frac{GMm}{\left(\frac{3R}{2}\right)}$

$TE_{(finally)} = \frac{GmM}{R}\left(\frac{1}{2} - \frac{2}{3}\right) = \left[-\frac{GmM}{R}\left(\frac{1}{6}\right)\right]$

So, change in TE required = $\frac{GmM}{6R}$

Similarly solving for (a = R)

then we will get

$TE_{final} = -\frac{GmM}{2R}$

So, change in $TE = \frac{GmM}{2R}$

Alternate Method

For perihilion

$a(1 - e) = R + h = \frac{3R}{2}$

$\Rightarrow a = 3R$

Also, for elliptical path

$T.E. = -\frac{GMm}{2a} = \frac{-GMm}{6R}$

$\Rightarrow$ change in $TE = \frac{GMm}{6R}$

For aphelion :

$a(1 + e) = R + h = \frac{3R}{2}$

$\Rightarrow a = R$

& $T.E. = -\frac{GMm}{2a} = -\frac{GMm}{2R}$

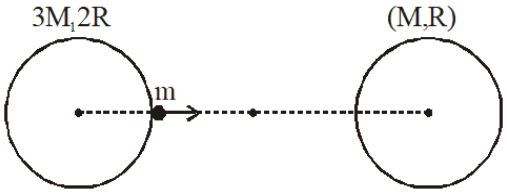
$\Rightarrow$ change in $TE = \frac{GMm}{2R}$

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20. Ans (A,B)

(a) 2R

(b) $3\sqrt{3}\left(\sqrt{3}-1\right)R$



(a) speed will maximum at projection point

(b) To just reach the other planet small body must be able to reach the point where gravitation field due to both planet is zero. At this point speed will be minimum and equal to zero.

$$\frac{G\times 3M}{x^2}=\frac{GM}{(6R-x)^2}\Rightarrow \sqrt{3}(6R-x)=x$$

$$\Rightarrow \left(1+\sqrt{3}\right)x=6\sqrt{3}R$$

$$x=\frac{6\sqrt{3}\left(\sqrt{3}-1\right)R}{2}$$

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PHYSICS

SECTION-II

3. Ans (2.00)

(a) The COM of the system divides the distance between the stars in the inverse ratio of thin masses. If d_1 & d_2 are the distances of M & m from the COM. (COM = Centre of mass)

$$d_1=\frac{d}{M+m}\times m \ \& \ d_2=\frac{d}{M+m}\times M$$

The stars will rotate in circles of radii d_1 & d_2 about their COM. The same force of attraction provides the necessary contripetal force for their circular motion.

$$\therefore \frac{GmM}{d^2}=M\omega_1^2 d_1=m\omega_2^2 d_2$$

$$\text{or } \omega_1^2=\frac{Gm}{d^2 d_1}=\frac{Gm}{d^2}\times\frac{M+m}{d\times m}=\frac{G(M+m)}{d^3}$$

$$\text{and } \omega_2^2=\frac{G(M)}{d^2 d_2}=\frac{GM}{d^2}\times\left(\frac{M+m}{d\times M}\right)=\frac{G(M+m)}{d^3}$$

$$\Rightarrow \omega_1=\omega_2=\sqrt{\frac{G(M+m)}{d^3}}$$

(b) From the fact that the moment of momentum is also angular momentum.

$$\frac{I_M}{I_m}=\frac{(Mv_1)d_1}{(Mv_2)d_2}=\frac{M}{m}\frac{d_1^2}{d_2^2}=\frac{M}{m}\times\frac{m^2}{M^2}=\frac{m}{M}$$

(c) $$\frac{K_M}{K_m}=\frac{\frac{1}{2}Mv_1^2}{\frac{1}{2}Mv_2^2}=\frac{M}{m}\times\frac{\omega_1^2 d_1^2}{\omega_2^2 d_2^2}=\frac{M}{m}\frac{d_1^2}{d_2^2}$$

$$\Rightarrow \frac{K_M}{K_m}=\frac{M}{m}\times\frac{m^2}{M^2}=\frac{m}{M}$$

PHYSICS

SECTION-III

1. **Ans (4)**

In non uniform electric field force on dipole

$$F = \vec{p}.\frac{d\vec{\in}}{dr}$$

$$\vec{E} = 4xy^2\hat{i} + 4x^2y\hat{j}$$

$$\frac{d\in}{dx} = 4y^2\hat{i} + 8xy\hat{j}$$

$$\frac{d\in}{dy} = 8y\hat{i} + 4x^2\hat{j}$$

$$F_x = \vec{p}\cdot\frac{d\vec{\in}}{dx} = 4py^2$$

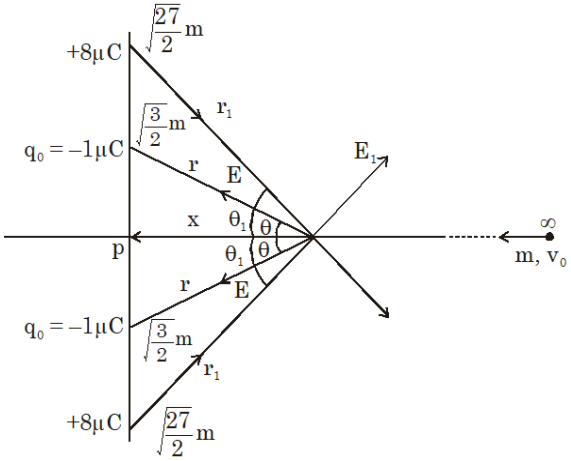
$$F_y = \vec{p}\cdot\frac{d\vec{\in}}{dy} = 8xyp$$

$$F_{Net} = \sqrt{F_x^2 + F_y^2} = \sqrt{16p^2y^4 + 64p^2x^2y^2}$$

$$= 4py\left(y^2 + 4x^2\right)^{\frac{1}{2}}$$

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2. **Ans (3)**



P is origin. We can see from diagram electric field at P is zero but we have to check if it is zero at any other point x-axis we have to give particle just as much velocity to reach to point where electric field is zero. (Called neutral point)

Let field is zero at θ.

$(E_\theta)_{Net} = 0$

$2E\cos\theta = 2E_0\cos\theta_1$

$$2 \times \frac{kq_0}{r^2} \times \frac{x}{r} = 2 \times \frac{k8q_0}{r_1^2} \times \frac{x}{r_1}$$

$$\frac{1}{r^3} = \frac{8}{r_1^3} \Rightarrow \frac{r_1}{r} = 2$$

For Q $\frac{r_1}{r} = +2$

$r_1 = 2r$ again r1 = 4

$$\sqrt{x^2 + \frac{27}{2}} = 2\sqrt{x^2 + \frac{3}{2}} \qquad r = 2$$

$$x^2 + \frac{27}{2} = 4x^2 + \frac{12}{2}$$

$$3x^2 = \frac{15}{2} \Rightarrow x = \sqrt{\frac{5}{2}}$$

To reach at origin particle has to reach at Q the it will be attracted towards origin by electric fields of $-1\mu C$

Apply work energy theorem from ∞ to Q.

$-q\Delta V = k_f - k_i$

$$-q\left(v_Q - v_\infty\right) = 0 - \frac{1}{2}mv_0^2$$

$$-1 \times 10^{-6}\left[\left(\frac{2 \times 8 \times 10^{-6} \times 9 \times 10^9}{r_1} - \frac{2 \times 10^{-6} \times 9 \times 10^9}{r}\right) - 0\right]$$

$$= -\frac{1}{2}mv_0^2$$

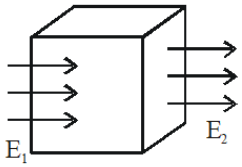
$$-10^6 \times 2 \times 10^{-6} \times 9 \times 109\left(\frac{8}{4} - \frac{1}{2}\right) = -\frac{1}{2}mv_0^2$$

Solving we get $v_0 = 3$ m/s

3. Ans (3)

Electric field is increasing uniformly according to $|E| = 1 + x$

total flux $= +E_2A - E_1A$

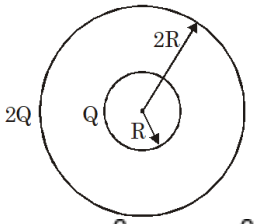


$= [(1 + x_2) - (1 + x_1)]A$

$= (x_2 - x_1)A = (10\text{ cm})\,(100\text{ cm}^2) = 10^{-3}\text{ m}^3$

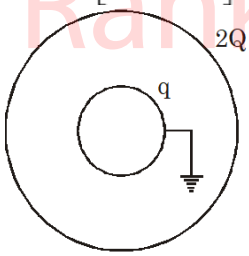
$Q_{enc} = 10^{-3}\,\epsilon_0 = 8.85 \times 10^{-15}\text{ C}$

4. Ans (5)



$$U_1 = \frac{kQ^2}{2R} + \frac{k[2Q]^2}{2\times 2R} + \left[\frac{kQ}{2R}\right]\times 2Q$$

$$= \frac{kQ^2}{R}\left[\frac{1}{2} + 1 + 1\right] = \frac{5}{2}\frac{kQ^2}{R}$$



$$V_A = 0 = \frac{kq}{R} + \frac{k[2Q]}{2R} = 0$$

$q = -Q$

$$U_2 = \frac{k(Q)^2}{2R} + \frac{k[2Q]^2}{2\times 2R} - \left(\frac{kQ}{2R}\right)\times 2Q$$

$$= \frac{kQ^2}{R}\left[\frac{1}{2} + 1 - 1\right] = \frac{kQ^2}{2R}$$

$$\frac{U_1}{U_2} = 5$$

5. Ans (2)

$$\text{Force} = \left[\frac{\sigma^2}{2\varepsilon_0}\right]\times A$$

$$= \left[\frac{Q}{A}\right]^2 \times \frac{A}{2\varepsilon_0} = \frac{Q^2}{2A\varepsilon_0} = \frac{Q^2}{2a^2\varepsilon_0}$$

N = 2

PHYSICS

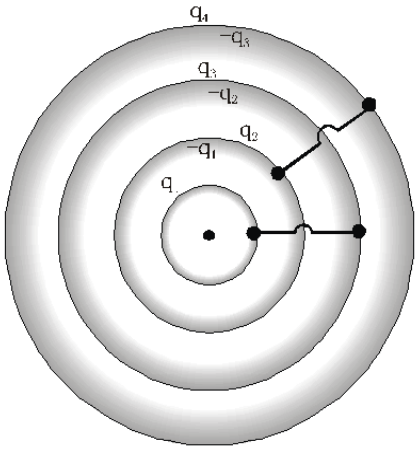
SECTION-IV

1. Ans (A->R,B->P,C->S,D->Q)

Let the charge distribution be as shown in the figure below. From Gauss's theorem, we know that facing surfaces of the conductor acquire equal and opposite charges.

$\Rightarrow q_1 + q_3 - q_2 = +4Q$, and .....(i)

Now, $q_2 - q_1 + q_4 - q_3 = -6Q$ .....(ii)



$$V_1 = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{R} + \frac{q_2 - q_1}{2R} + \frac{q_3 - q_2}{3R} + \frac{q_4 - q_3}{4R}\right]$$

$$V_2 = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{2R} + \frac{q_2 - q_1}{2R} + \frac{q_3 - q_2}{3R} + \frac{q_4 - q_3}{4R}\right]$$

$$V_3 = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{3R} + \frac{q_2 - q_1}{3R} + \frac{q_3 - q_2}{3R} + \frac{q_4 - q_3}{4R}\right]$$

and

$$V_4 = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{4R} + \frac{q_2 - q_1}{4R} + \frac{q_3 - q_2}{4R} + \frac{q_4 - q_3}{4R}\right]$$

From $V_1 = V_3$, $q_1 = -\frac{q_2}{3}$; From $V_2 = V_4$, $q_2 = -\frac{q_3}{2}$

On solving eq. (i), $q_1 = \frac{2Q}{5}, q_2 = -\frac{6Q}{5}$ and $q_3 = \frac{12Q}{5}$

Substituting these values in Eq.(ii), $q_4 = -2Q$

For (A) Charge on the inner surface of 3rd conductor $= -q_2 = \frac{6Q}{5}$

For (B) Charge on 4th conductor $= q_4 - q_3 = -2Q - \frac{12Q}{5} = \frac{-22Q}{5}$

For (C) Potential of conductor 1, $V_1 = \frac{-3Q}{40\pi\varepsilon_0 R} = V_3$

For (D) Potential of conductor 2, $V_2 = -\frac{Q}{8\pi\varepsilon_0 R} = V_4$

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2. **Ans (A->QR,B->RS,C->PQ,D->PS)**

Since cavity is spherical charge induces uniformly on its inner surface but an outer surface charge distribution is non-uniform.

Due to point charge electric field is radially outwards.

Near conducting surface net electric field is always perpendicular to the conductor.

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PHYSICS

SECTION-I (i)

1. Ans (A)

$J = \sigma E \qquad \sigma = \frac{1}{\rho}$

$J = \frac{E}{\rho} \Rightarrow E = \rho J \Rightarrow E \propto \rho$

$E_1 : E_2 : E_3$

$\rho_1 : \rho_2 : \rho_3$

$4\rho : 2\rho : 6\rho$

$2 : 1 : 3$

2. Ans (D)

$$T - T_0 = \frac{I^2R}{2\pi r\ell k}\left(1 - e^{-\frac{2kt}{rCd}}\right)$$

$$\frac{dQ}{dt} = I^2R - kA(T - T_0)$$

$$ms\frac{dT}{dt} = I^2R - k(2\pi rl)(T - T_0)$$

$$d\pi r^2 \ell C\frac{dT}{dt} = I^2R - 2\pi r\ell k(T - T_0)$$

$$\int_{T_0}^{T}\frac{dT}{I^2R - 2\pi r\ell k(T - T_0)} = \int_0^t \frac{dt}{Cd\pi r^2\ell}$$

$$-\frac{1}{2\pi r\ell k}\ell n\left(\frac{I^2R - 2\pi r\ell k(T - T_0)}{I^2R}\right) = \frac{t}{Cd\pi r^2\ell}$$

$$\ell n\left(\frac{I^2R - 2\pi r\ell k(T - T_0)}{I^2R}\right) = -\frac{2kt}{rCd}$$

$$1 - \frac{2\pi r\ell k}{I^2R}(T - T_0) = e^{-\frac{2kt}{rCd}}$$

$$\left[T - T_0 = \frac{I^2R}{2\pi r\ell k}\left(1 - e^{-\frac{2kt}{rCd}}\right)\right]$$

PHYSICS

SECTION-I (ii)

3. Ans (B,C,D)

Current through the wire is same

$i_A = i_B = i_C$ $\because i = neAV_d$ and $J = \frac{i}{A}$, $J = \sigma E$

4. Ans (A,C)

$$I = \frac{dV}{dR} = \frac{Edr}{\frac{dr}{\sigma 4\pi r^2}} = kE^2 4\pi r^2 \quad \text{..... (i)}$$

$dV = E.dr$

from (1) $E^2 = \frac{I}{4\pi kr^2}$

$$\Rightarrow dV = \int_a^b \sqrt{\frac{I}{4\pi k}}\frac{1}{r}dr = \sqrt{\frac{I}{4\pi k}}\ell n\left(\frac{b}{a}\right)$$

5. Ans (B,C)

Heat produced = $\int I^2Rdt = 10\,[\,64 \times 1 + 16 \times 1 + 4 \times 1 + 1 \times 1] = 850$ J

Total charge = $\int Idt = [8 + 4 + 2 + 1] \times 1 = 15C$

6. Ans (B,D)

$$i = \frac{10}{2 + 8} = 1A$$

$$\int \left(\vec{E}_C + \vec{E}_{N.C}\right).\vec{d\ell} = iR$$

Across 8Ω $E_{N.C} = 0$

$$\int \vec{E}_C.\vec{d\ell} = 8 \text{ J/C}$$

(1) $\int \vec{E}_C.\vec{d\ell} = 8$ J

Since $\int_{loop} \vec{E}_C.\vec{d\ell} = 0$

So work done by conservative electric field while an unit charge goes from Q to P is -8 joule.

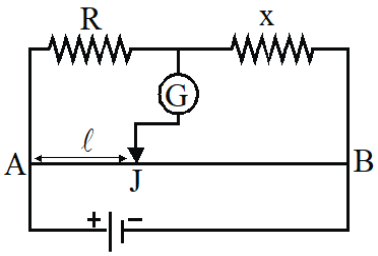
Now across the battery, we have

(1) $\int \left(\vec{E}_{N.C}\right).\vec{d\ell} = \varepsilon = 10$joule

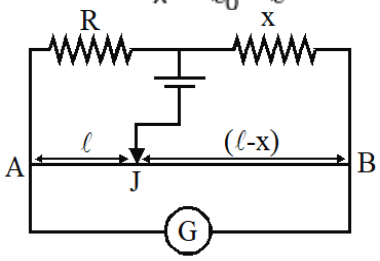
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7. **Ans (A,C)**

(i)



$$\frac{R}{x} = \frac{\ell}{\ell_0 - \ell} \quad \text{... (i)}$$



Here $V_A = V_B$ ⇒ current through galvanometer is zero.

$\frac{R}{\ell} = \frac{x}{\ell_0 - \ell}$ [it is also a balance Wheatstone bridge]

(iii) Ratio $\frac{R}{x} = \frac{\ell}{\ell_0 - \ell}$ is independent of cross-section area and resistivity

(iv) $\frac{\ell}{\ell_0 - \ell} = \frac{R}{x} \Rightarrow \frac{2x}{2R} \neq \frac{\ell}{\ell_0 - \ell}$

8. **Ans (C,D)**

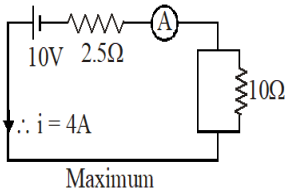
$i = \frac{3}{30} = \frac{1}{10}$

$v_{AB} = \frac{1}{10} \times 20 = 2v$ = maximum voltage read by potentiometer

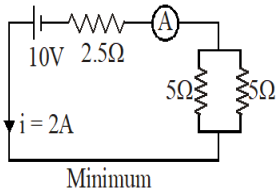
also voltage per mm = $\frac{2}{20 \times 100} = 10^{-4}v = 0.1mv$

⇒ accuracy of potentiometer is 0.1 mv

9. **Ans (B,C,D)**

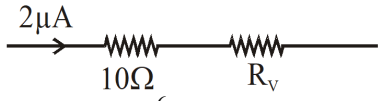


Maximum



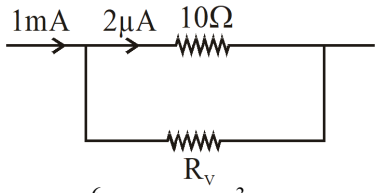
Minimum

10. **Ans (A,C)**

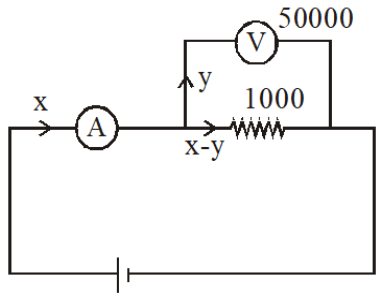


$0.1 = 2 \times 10^{-6} (10 + R_V)$

$\therefore R_V = 49990\ \Omega$



$2 \times 10^{-6} \times 10 = 10^{-3} R_A \therefore R_A = 0.02\ \Omega$



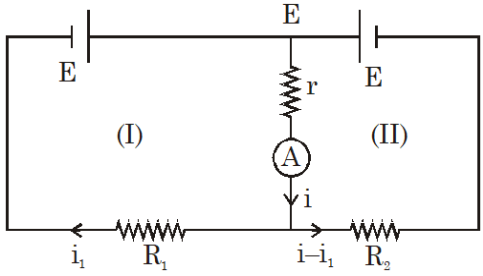
$y\ 50000 = (x - y)\ 1000$

$\therefore 51y = x$

Reading = $\frac{y\,50000}{x} \simeq 980$

11. **Ans (A,D)**

$\frac{\varepsilon}{r + R_0(f - f^2)}$; for I_{max}, f = 0, 1; I_{min}, f = 1/2



$R_1 = fR_0 \quad R_2 = (1 - f)R_0$

Ist Loop

$E - ir - fi_1R_0 = 0$

IInd Loop

$E - ir - (i - i_1) R_0 (1 - f) = 0$

$fi_1R_0 = (i - i_1) R_0 (1 - f)$

$i_1 = i (1 - f)$

$\therefore E - ir - f(1 - f)\, iR_0 = 0$

$$i = \frac{E}{r + R_0(f - f^2)}$$

For f = 0,1 I will be maximum

For f = 1/2 I will be minimum

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12. Ans (A,B,C)

(i) $r_A = \frac{1 \times 99}{100} = 0.99A$

$3 = \frac{12}{r + 2 + 0.99}$

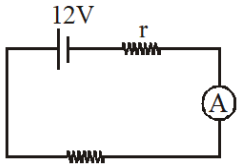
$\Rightarrow r + 2.99 = 4$

$r = 1.01\ \Omega$

(ii) $r_v = 99 + 101 = 200\ \Omega$

$i = \frac{12}{1.01 + \frac{2\times200}{200}} = 4A$

$V = 8V$

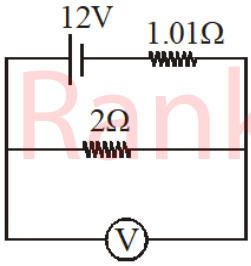


Range of voltmeter $= \frac{5}{4} \times 8 = 10V$

Range of ammeter = 5A

(ii) $1(5 - i_g) = 99\ i_g$

$5 = 100\ i_g$



$i_g = 0.05$ A

PHYSICS

SECTION-I (iii)

14. Ans (C)

(i) Total charge = Area under the i-t graph.

$= \frac{1}{2} i_0 t_0$

(ii) $\frac{i}{i_0} + \frac{t}{t_0} = 1 \Rightarrow i = i_0\left(1 - \frac{t}{t_0}\right)$

(iii) $H = \int_{t_0} i^2 R\, dt$

$= \int_0 i_0^2\left(1 - \frac{t}{t_0}\right)^2 \cdot R dt$

15. Ans (C)

$i = \frac{\varepsilon}{R_g + R}$

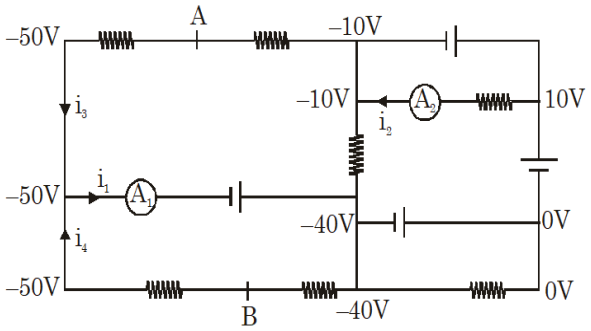
16. Ans (C)

$10 \times 10^{-3} = \frac{1}{20 + R} \Rightarrow R = 80\Omega$

17. Ans (A)

$I = \frac{1}{20 + 80 + 100} = 5$ mA

18. Ans (B)



V_2 = 40 V when S_4 opened and V_2 = 45 V otherwise

V_1 = 50 V when S_3 opened and V_1 = 30 V otherwise

A_2 = 0A when S_1 opened and A_2 = 4A otherwise

A_1 = 5A when S_3 and S_4 are closed (no effect of S_1 and S_2)

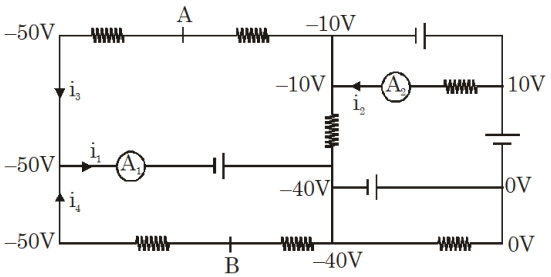
A_1 = 4A when S_4 opened

A_1 = 1A when S_3 opened

	V_1	V_2	A_1	A_2	$\lvert A_1 - A_2 \rvert$
S_1opened	30	45	5	0	5
S_2opened	30	45	5	4	1
S_3opened	50	45	1	4	3
S_4opened	30	40	4	4	0

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19. **Ans (C)**



V_2 = 40 V when S_4 opened and V_2 = 45 V otherwise

V_1 = 50 V when S_3 opened and V_1 = 30 V otherwise

A_2 = 0A when S_1 opened and A_2 = 4A otherwise

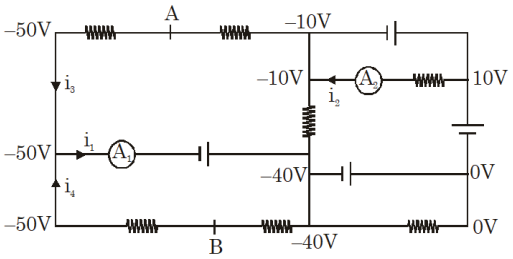
A_1 = 5A when S_3 and S_4 are closed (no effect of S_1 and S_2)

A_1 = 4A when S_4 opened

A_1 = 1A when S_3 opened

	V_1	V_2	A_1	A_2	$\lvert A_1 - A_2 \rvert$
S_1opened	30	45	5	0	5
S_2opened	30	45	5	4	1
S_3opened	50	45	1	4	3
S_4opened	30	40	4	4	0

20. **Ans (D)**



V_2 = 40 V when S_4 opened and V_2 = 45 V otherwise

V_1 = 50 V when S_3 opened and V_1 = 30 V otherwise

A_2 = 0A when S_1 opened and A_2 = 4A otherwise

A_1 = 5A when S_3 and S_4 are closed (no effect of S_1 and S_2)

A_1 = 4A when S_4 opened

A_1 = 1A when S_3 opened

	V_1	V_2	A_1	A_2	$\lvert A_1 - A_2 \rvert$
S_1opened	30	45	5	0	5
S_2opened	30	45	5	4	1
S_3opened	50	45	1	4	3
S_4opened	30	40	4	4	0

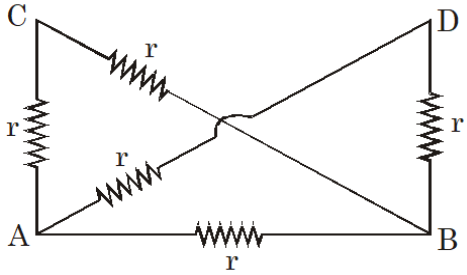
PHYSICS

SECTION-I (iv)

21. **Ans (A,D)**

By symmetry

$V_C = V_D$



$R_{AB} = \frac{r}{2}$

PHYSICS

SECTION-II

1. **Ans (20)**

$$\left(\frac{V}{R+r}\right)^2 R = \left(\frac{V}{r+nR}\right)^2 \times nR$$

$\Rightarrow r^2 + n^2R^2 = nR^2 + nr^2$

$(n^2 - n)\,R^2 = (n-1)r^2$

$nR^2 = r^2$

$r = R$

$r = 20\ \Omega$

2. **Ans (60)**

In parallel

$$H = \frac{V^2}{R/2}t_1 = \frac{2V^2}{R}t_1$$

In series

$$H = \frac{V^2}{2R}t_2$$

$$2t_1 = \frac{t_2}{2}$$

$t_2 = 4t_1 = 60$ min

3. **Ans (3)**

$$\frac{1}{R} = \frac{\pi r^2}{3l}(2\sigma_2 + \sigma_1)$$

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4. **Ans (40)**

In case–1

'i' is very small $\Rightarrow R_V$ is very high

In case - 2

V across ammeter is V, small $\Rightarrow R_A$ is very low

$\Rightarrow$ from case - 1, we can say that

$V_{Cell} = 100$ volt.

From case 2, we can say

$25 \times 10^{-3} = V_{cell} - i_r$

$= 100 - 2.5\ r$

$r = 40\ \Omega$

5. **Ans (35)**

Total resistance $= \int_0^L \lambda x\,dx = \frac{\lambda}{2}L^2$

For balance condition

$$\frac{R_1}{R_2} = \frac{(\lambda/2)l_1^2}{(\lambda/2)(L^2 - l_1^2)}$$

Similarly, $\frac{R_2}{R_1} = \frac{l_2^2}{L^2 - l_2^2}$

$\therefore \frac{R_1}{R_2} = \frac{l_1^2}{L^2 - l_1^2} = \frac{L^2 - l_2^2}{l_2^2} = n$

Solving, $l_1 = L\sqrt{\frac{n}{n+1}}$ and $l_2 = L\sqrt{\frac{1}{n+1}}$

$\therefore \Delta = l_1 - l_2 = \frac{L}{\sqrt{n+1}}(\sqrt{n} - 1)$

Putting L = 100 cm and n = 3

$\Delta = \frac{100}{2}(\sqrt{3} - 1) = 35$ cm

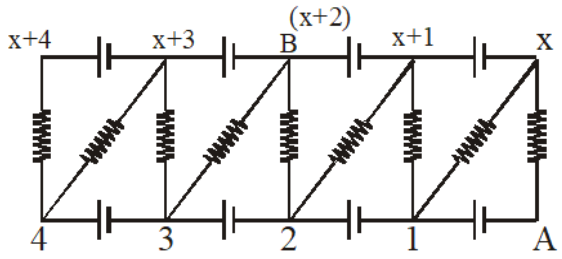
6. **Ans (22.00)**

$-\frac{22}{9}V$

$\frac{x+4-4}{1} + \frac{(x+3-4)}{1} + \frac{x}{1} + \frac{(x+2-3)}{1} + \frac{x}{1} + \frac{x+1-2}{1} + \frac{x}{1} + \frac{x-1}{1} + \frac{x}{1} = 0$

$5x + 4(x - 1) = 0$

$9x - 4 = 0$



$x = \frac{4}{9}V$

$V_A - V_B = -(x + 2 - 0) = -\frac{22}{9}V$

PHYSICS

SECTION-III

1. **Ans (2)**

Let x be the resistance between a & b then, x || r.

The total resistance of circuit $= \frac{rx}{r+x} + (R - x)$

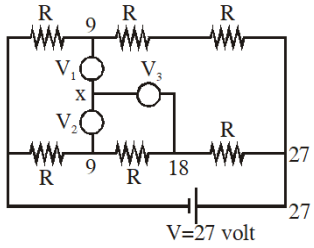
So, current through ammeter $I_A = \frac{I_0 r}{r+x} = \frac{V_0 r}{rx + (R-x)(r+x)}$

for its minimum value,

$\left(\frac{dI_A}{dx}\right) = 0 \Rightarrow$ we get $x = \frac{R}{2}$

So, $(I_A)_{min} = \frac{V_0 r}{R\left(r + \frac{R}{4}\right)} = \frac{(12)(2)}{4(2+1)} = 2A$

2. **Ans (6)**



$\frac{x-9}{R} + \frac{x-9}{R} + \frac{x-18}{R} = 0 \Rightarrow x = 12 \therefore V_3 = 6$ volt

3. **Ans (2)**

$P_i = \left(\frac{V}{5+20}\right)^2 \times 5$

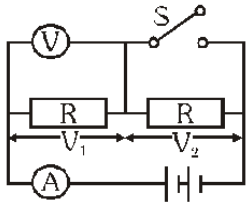
$P_f = \left[\left(\frac{\frac{V}{5R}}{5+R} + 20\right) \times \frac{5R}{5+R}\right]^2 \frac{1}{5}$

$\Rightarrow P_f = \frac{P_i}{9}$

PHYSICS

SECTION-IV

1. **Ans (A->P,B->P,C->Q,D->QS)**



After closing the switch net resistance decreases therefore there will be increases in the current.

Initially $V = V_1 + V_2$ after closing the switch V_2 becomes zero hence $V = V_1$.

After short circuiting current in the resistance becomes zero therefore power become zero.

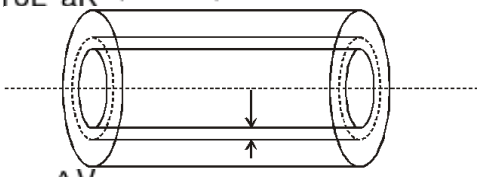
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PHYSICS

SECTION-V

1. **Ans ( )**

(a) $\frac{\Delta V}{L}, \frac{\Delta V r^2}{La}$; (b) $\frac{(\Delta V)^2(2\pi)}{La}\left(\frac{R^4}{4}\right)$, (c) $T_S = T_A + \frac{2\pi(\Delta V)^2}{La\lambda}\left(\frac{R^4}{4}\right)$, (d) $T = T_s + \frac{(\Delta V)^2}{16L^2aK}(R^4 - r^4)$



$$E = \frac{\Delta V}{L}$$

$J = \sigma E$

$$= \frac{r^2}{a}\frac{\Delta V}{L}$$

$$P = i^2R = i^2\left(\frac{\rho L}{A}\right)$$

$$P = \int\left(\frac{\Delta V r^2}{L\ a}\right)^2 (2\pi r dr)^2.\left(\frac{a\ \ L}{r^2\ 2\pi r dr}\right)$$

$$P = \frac{(\Delta V)^2}{La}.2\pi\frac{R^4}{4}$$

$P = \lambda(T - T_S)$

$$\frac{(\Delta V)^2}{La}.2\pi\frac{R^4}{4} = \lambda(T - T_S)$$

$$T = T_S + \frac{2\pi(\Delta V)^2}{La\lambda}.\frac{R^4}{4}$$

Also at a distance r from central axis

$$\frac{(\Delta V)^2}{La}.\frac{2\pi r^4}{4} = k(2\pi rL)\frac{dT}{dr}$$

$$\int dT = \frac{(\Delta V)^2}{4L^2ak}\int_r^R r^3 dr$$

$$T = T_s + \frac{(\Delta V)^2}{16L^2aK}(R^4 - r^4)$$

2. **Ans ( )**

First case $V_1 > V_2 > V_3$; Second case $V_1 = V_2 = V_3$

$I = \frac{6}{R_{eq}}$, $V_{1k\Omega} = I \times 1$, $V_{2k\Omega} = 6 - V_{1k\Omega}$

(a) $R_{eq} = 1 + \frac{2 \times 30}{2 + 30} = \frac{23}{8}k\Omega$

$$I = \frac{6}{\left(\frac{23}{8}\right)} = \frac{48}{23}mA,\ V_{1k\Omega} = \frac{48}{23}V$$

$$V_{2k\Omega} = 6 - \frac{48}{23} = \frac{90}{23}V = 3.91\ V$$

(b) $R_{eq} = 1 + \frac{2 \times 1}{2 + 1} = \frac{5}{3}k\Omega$

$$I = \frac{6}{\left(\frac{5}{3}\right)} = \frac{18}{5}mA,\ V_{1\,k\Omega} = \frac{18}{5}V$$

$$V_{2\,k\Omega} = 6 - \frac{18}{5} = \frac{12}{5}V = 2.4\ V$$

(c) $R_{eq} = 1 + \frac{1}{\frac{1}{1} + \frac{1}{2} + \frac{1}{30}} = \frac{38}{23}k\Omega$

$$I = \frac{6}{\left(\frac{38}{23}\right)} = \frac{138}{38}mA,\ V_{1\,k\Omega} = \frac{138}{38}V$$

$$V_{2kW} = 6 - \frac{138}{38} = \frac{90}{38}V = 2.37\ V$$

$[V_1 > V_2 > V_3]$

In second case

$V_A - V_B = 6\ V$

$V_1 = V_2 = V_3$

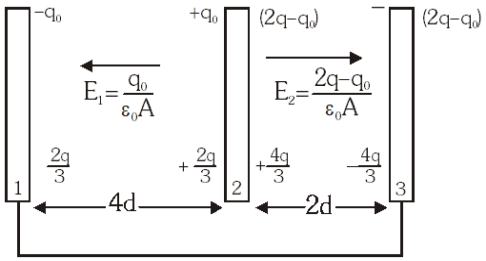
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SOLUTION

PHYSICS

SECTION-I (i)

1. Ans (B)



$E_1 4d = E_2\, 2d$

$$\frac{q_0}{\epsilon_0 A}4d = \frac{(2q-q_0)}{\epsilon_0 A}2d \Rightarrow q_0 = \frac{2q}{3}$$

$$V_{23} = \frac{4q}{3\epsilon_0 A} \times 2d = \frac{8qd}{3\epsilon_0 A} \ldots\ (i)$$

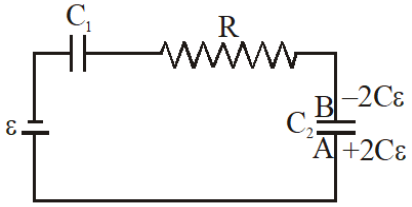
2. Ans (C)

Apply KVL

$$\varepsilon = \frac{q}{C} - iR + \frac{RC\varepsilon - q}{C} = 0$$

$$\frac{3C\varepsilon - 2q}{C} = iR$$

$$i = \frac{dq}{dt}$$



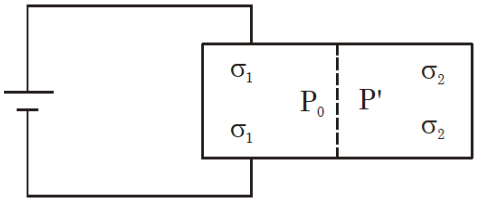
$$\frac{dq}{dt} = \frac{3C\varepsilon - 2q}{C}$$

$$\int_0^q \frac{dq}{3C\varepsilon - 2q} = \frac{1}{RC}$$

$$q = \frac{3C\varepsilon}{2}(1 - e^{-1/RC})$$

On plate B, Charge is $-2C\varepsilon + q = \frac{-C\varepsilon}{2}(3 + 3e^{-1/RC})$

3. Ans (C)



$$E = \frac{V}{d}$$

$\sigma_1 \neq \sigma_2$

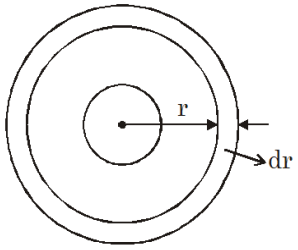
But

$$E = \frac{V}{d}$$

So electric field at P and P' is equal.

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4. Ans (B)



$$\text{Resistance} = R = \int_R^\infty \frac{\rho dr}{4\pi r^2}$$

$$R_{es} = \frac{\rho}{4\pi}\int_R^\infty \frac{dr}{r^2}$$

$$R_{es} = \frac{\rho}{4\pi}\int_R^\infty r^{-2}dr$$

$$R_{es} = \frac{\rho}{4\pi} - \left[\left(\frac{1}{r}\right)\right]_R^\infty = \frac{-\rho}{4\pi}\left[\frac{1}{\infty} - \frac{1}{R}\right]$$

$$R_{es} = \frac{\rho}{4\pi R}$$

For discharging circuit

$$Q = Q_0 e^{-\frac{t}{R_{es}C}}$$

$$\frac{Q_0}{2} = Q_0 e^{-\frac{t}{R_{es}C}}$$

$$\frac{1}{2} = e^{-\frac{t}{R_{es}C}}$$

$$-\ell n2 = -\frac{t}{R_{es}C}$$

$$t = R_{es}C\ell n2$$

$$t = \frac{\rho}{4\pi R} \times k4\pi\varepsilon_0 R\ell n2$$

$$t = \rho k\varepsilon_0 \ell n2$$

5. Ans (A)

$$E \times \frac{1}{k} \Rightarrow \Delta V \times \frac{1}{k}$$

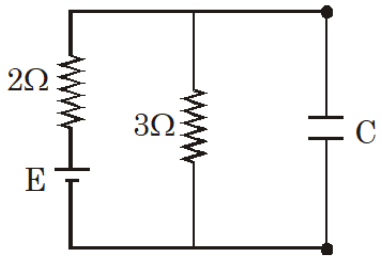
$$\Rightarrow E_3 < E_2 < E_1 \text{ and } \Delta V_3 < \Delta V_2 < \Delta V_1$$

6. Ans (A)

we know $q' = -q\left(1 - \frac{1}{k}\right)$,

Therefore, $|q'| < |q|$

7. Ans (A)

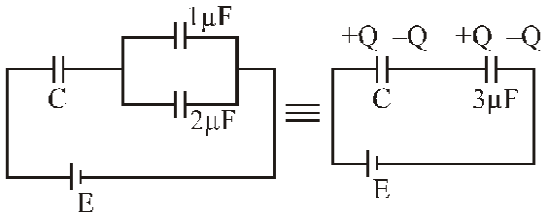


$q_1 = CE$, when switch is closed.

$$E_{eq} = \frac{3E}{5}$$

$$q_2 = CE \times \frac{3}{5}$$

8. Ans (D)



$$C_{eq} = \frac{3C}{3+C} \quad Q = Ceq.E$$

$$Q_2 = \frac{2CE}{(C+3)} = \frac{3CE}{(C+3)}$$

$$= 2E\left[1 - \frac{3}{C+3}\right]$$

$$Q_{2/\text{when } C = 1\mu F} = 2E\left[\frac{1}{4}\right] = \frac{E}{2}$$

$$Q_{2/\text{when } C = 3\mu F} = 2E\left[\frac{1}{2}\right] = E$$

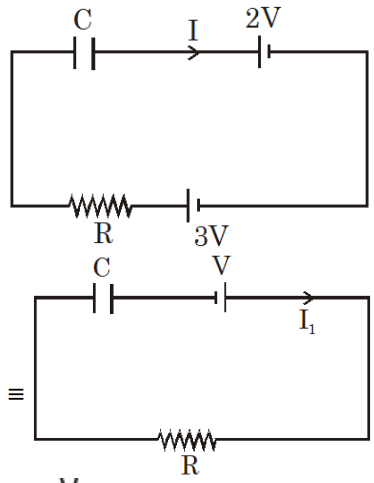
$$\frac{dQ}{dC} = -\frac{2E \cdot 3}{(C+3)}(-1) = \frac{6E}{(C+3)^2} > 0$$

$$\frac{d^2Q}{dC^2} = \frac{6E(-2)}{(C+3)^3} = -\frac{12E}{(C+3)^2} < 0$$

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9. Ans (A)

$\frac{V}{R}\left(e^{\frac{-t}{RC}}-1\right)$



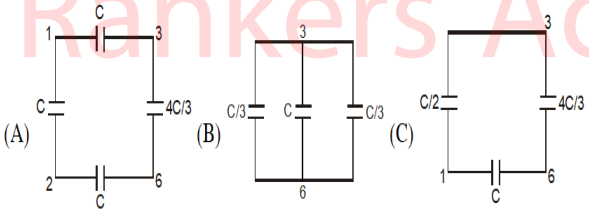
$I_1 = \frac{V}{R}\left(1-e^{-\frac{t}{RC}}\right)$

$\because I = -I_1 = \frac{V}{R}\left(e^{-\frac{t}{RC}}-1\right)$

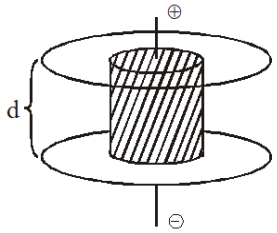
PHYSICS

SECTION-I (ii)

10. Ans (A,B,C)



12. Ans (A,C)



$C = C_{di} + C_{empty}$

$= \frac{k\in_0\left(\pi R^2\right)}{d} + \frac{\in_0\pi\left(3R^2\right)}{d}$

$C = \frac{(k+3)\in_0\pi R^2}{d}$

14. Ans (A,B,C,D)

$V_0 = I_0 R = 10 \times 10 = 100$ volts (since, $I_0 = 10$ amp from figure)

Hence (A) is correct

Also: $I = I_0 e^{-t/RC}$

Taking log; $\ln\left(\frac{I_0}{I}\right) = \frac{t}{RC} \Rightarrow C = \frac{t}{R\ln(I_0/I)}$

At; t = 2 sec, I = 2.5 A

$C = \frac{2}{10\ \ln\left(\frac{10}{25}\right)}$

$C = \frac{2}{10\ \ln 4} = \frac{2}{10 \times 2\ \ln 2} = \frac{1}{10\ \ln 2}$ F Hence (B) is correct.

$C = \frac{1}{10\ \ln 2}$.

Heat produced $= \frac{1}{2}CV^2 = \frac{1}{2}\left(\frac{1}{10\ \ln 2}\right)(100^2)$

$= \frac{500}{\ln 2}$ joules.

Hence, (C) is correct

Thermal power in the resistor will decrease with a time constant $\frac{1}{2\ \ln 2}$ second.

Hence, (D) is correct.

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17. **Ans (A,B,C,D)**

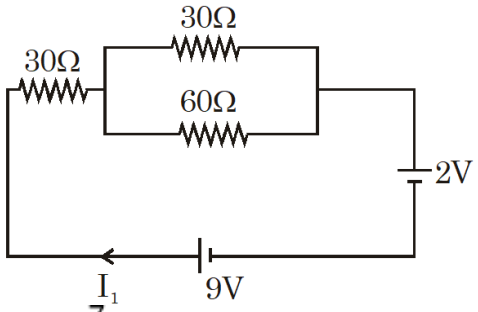
(a) $\frac{7}{50}A$ or $\frac{11}{50}A$, (b) $Q_1 = 9\mu C$, $Q_2 = 0$

(a) Potential difference across 'C_1' just before switch is closed.

$$\Delta V = \frac{q}{C} = \frac{2}{1} = 2V$$

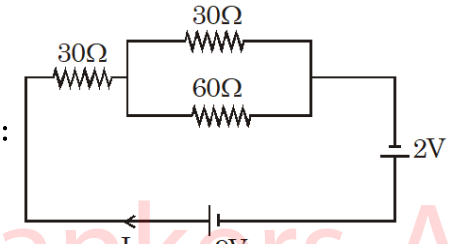
$\therefore$ Just after switch is closed

Case-I :



$$I_1 = \frac{7}{50}A$$

Case-II :



$$I_2 = \frac{9}{50}A$$

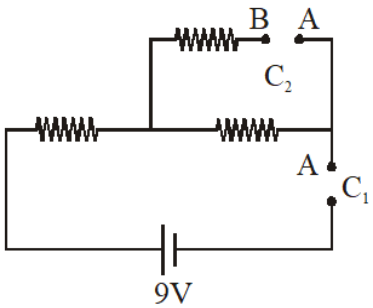
(b) In steady state there will be no current in the circuit

P.d across $C_2 = 0$

So charge = 0

P.d across $C_1 = 9V$

$Q = 9\ \mu C$



PHYSICS

SECTION-I (iii)

18. **Ans (B)**

$$\left(i - \frac{dq}{dt}\right)R = \frac{q}{C}$$

$$\Rightarrow \frac{dq}{i - \frac{q}{RC}} = dt$$

$$\& \ q = i\,RC\left(1 - e^{-\frac{t}{RC}}\right)$$

19. **Ans (C)**

$$\frac{d}{dt}\left(\frac{q^2}{2C}\right) = \left(i - \frac{dq}{dt}\right)^2 R$$

$$\Rightarrow \frac{q}{C}\frac{dq}{dt} = \left(i - \frac{dq}{dt}\right) \times R\left(i - \frac{dq}{dt}\right)$$

$$\Rightarrow \frac{2dq}{dt} = i \Rightarrow t = RC\,\ell n2$$

20. **Ans (A)**

When switch is closed for a very long time capacitor will get fully charged & charge on capacitor will be $q = CV$

Energy stored in capacitor $\in_C = \frac{1}{2}CV^2$ ...(i)

Work done by battery $(\omega) = Vq = VCV = CV^2$

dissipated across resistance $\in_D$ = (work done by battery) – (energy store)

$$\in_D = CV^2 - \frac{1}{2}CV^2 = \frac{1}{2}CV^2 \text{ ...(ii)}$$

from (i) & (ii)

$$\in_D = \in_C$$

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21. **Ans (A)**

For process (1)

Charge on capacitor = $\frac{CV_0}{3}$

energy stored in capacitor = $\frac{1}{2}C\frac{V_0^2}{9} = \frac{CV_0^2}{18}$

work done by battery = $\frac{CV_0}{3} \times \frac{V}{3} = \frac{CV_0^2}{9}$

Heat loss = $\frac{CV_0^2}{9} - \frac{CV_0^2}{18} = \frac{CV_0^2}{18}$

For process (2)

Charge on capacitor = $\frac{2CV_0}{3}$

Extra charge flow through battery = $\frac{CV_0}{3}$

Work done by battery : $\frac{CV_0}{3}.\frac{2V_0}{3} = \frac{2CV_0^2}{9}$

Final energy store in capacitor : $\frac{1}{2}C\left(\frac{2V_0}{3}\right)^2 = \frac{4CV_0^2}{18}$

energy store in process 2 : $\frac{4CV_0^2}{18} - \frac{CV_0^2}{18} = \frac{3CV_0^2}{18}$

Heat loss in process (2) = work done by battery in process (2) − energy store in capacitor process (2)

$= \frac{2CV_0^2}{9} - \frac{3CV_0^2}{18} = \frac{CV_0^2}{18}$

For process (3)

Charge on capacitor = CV_0

extra charge flow through battery : $CV_0 - \frac{2CV_0}{3} = \frac{CV_0}{3}$

work done by battery in this process : $\left(\frac{CV_0}{3}\right)(V_0) = \frac{CV_0^2}{3}$

find energy store in capacitor : $\frac{1}{2}CV_0^2$

energy stored in this process : $\frac{1}{2}CV_0^2 - \frac{4CV_0^2}{18} = \frac{5CV_0^2}{18}$

heat loss in process (3) : $\frac{CV_0^2}{3} - \frac{5CV_0^2}{18} = \frac{CV_0^2}{18}$

Now total heat loss (E_D) : $\frac{CV_0^2}{18} + \frac{CV_0^2}{18} + \frac{CV_0^2}{18} = \frac{CV_0^2}{6}$

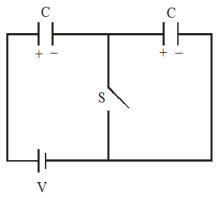
final energy store in capacitor : $\frac{1}{2}CV_0^2$

so we can say that $E_D = \frac{1}{3}\left(\frac{1}{2}CV_0^2\right)$

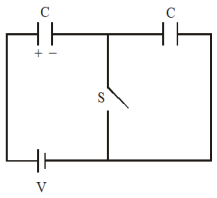
PHYSICS

SECTION-II

1. **Ans (4.50)**



initial



final

$U_i = \frac{1}{2}C.\left(\frac{V}{2}\right)^2.2$ $\quad$ $U_f = \frac{1}{2}CV^2$

$= \frac{CV^2}{4}$

$W_{Cell} = V.\Delta q = V.\left(CV - \frac{CV}{2}\right) = \frac{CV^2}{2}$

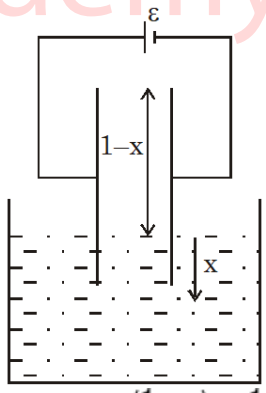
Now $U_i + W = U_f + H$

$\therefore H = \frac{CV^2}{4} = \frac{2 \times 9}{4} = 4.50$

2. **Ans (4.8)**

$\frac{\sqrt{4.8}}{2\sqrt{9.1}} \times 10^8$ m/s

3. **Ans (8.85)**



$C_{eq} = \frac{\epsilon_0(1-x) \times 1}{d} + \frac{k\epsilon_0(x \times 1)}{d}$

$= \frac{\epsilon_0}{d}[1 - x + kx]$

$= \frac{\epsilon_0}{d}[1 + (k-1)x]$

$q = C_{eq}\varepsilon$

$\frac{dq}{dt} = \frac{dC_{eq}}{dt}\varepsilon$

$\Rightarrow \frac{dq}{dt} = \frac{\epsilon_0}{d}(k-1)\varepsilon$

$\Rightarrow I = \frac{\left(8.85 \times 10^{-12}\right)(11-1) \times 500}{0.01}$

$I = 4.425 \times 10^{-9}$ A

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4. **Ans (3.00)**

Initial charge on capacitor = CE

Initial potential energy of capacitor = $\frac{CE^2}{2}$

Now C = $\frac{\varepsilon_0 A}{d}$

And new capacitance : C' = $\frac{\varepsilon_0 A}{d/2}$, in series with $\frac{\varepsilon_0 kA}{d/2}$

$\Rightarrow$ C' = $\frac{2\varepsilon_0 A}{d}$ series $\frac{4\varepsilon_0 A}{d}$

$= \frac{\varepsilon_0 A}{d}\left(\frac{2\times 4}{2+4}\right) = \frac{4\varepsilon_0 A}{3d} = \frac{4}{3}C$ $\Rightarrow$ New charge $= \frac{4}{3}CE$

New energy = $\frac{1}{2}\times\frac{4}{3}CE^2 = \frac{2}{3}CE^2$

Now $W_{but} = \Delta H + \Delta U$

$\Rightarrow E\left(\frac{4}{3}CE - CE\right) = \Delta H + \frac{2}{3}CE^2 - \frac{1}{2}CE^2$

$\Rightarrow \Delta H = \frac{1}{6}CE^2$

PHYSICS

SECTION-III

1. **Ans (4)**

$\left(\frac{Q^2}{2\epsilon_0 A}\right)\times x = \frac{1}{2}Kx^2 \Rightarrow x = \frac{Q^2}{\epsilon_0 AK}$

2. **Ans (7)**

$F = -\frac{dU}{dx}$ & $U = \frac{Q^2}{2C}$

where, $C = \frac{\epsilon_0 \ell(\ell - x)}{d} + \frac{\epsilon_r \epsilon_0 \ell(x)}{d}$

PHYSICS

SECTION-IV

1. **Ans (A->QRST,B->PS,C->RT,D->PRT)**

(A) Motion is along force then work done is positive

(B) P–iniitial $\frac{E_0}{K}$, final E_0

S–initial $\frac{v_0}{d}$ decreases so E increase

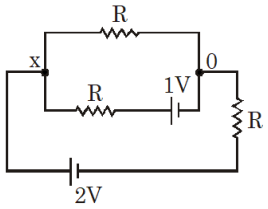
(C) Charge remain same capacitance increases $\frac{Q^2}{2C}$

(D) F = $\frac{Q}{2A\epsilon_0}Q$

where battery is not connected

2. **Ans (A->QRT,B->PS,C->RS,D->PS)**

For A

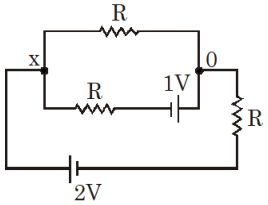


By nodal analysis

x + x – 1 + x – 2 = 0

x = 1 V, $I_2 = 0$, $I_1 = 1A$

For B



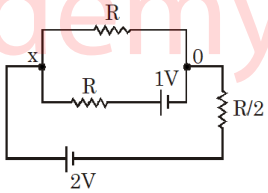
By nodal analysis

x + x + 1 + x – 2 = 0

x = $\frac{1}{3}$ V,

$I_2 = \frac{4}{3}A$, $I_1 = \frac{5}{3}A$

For C



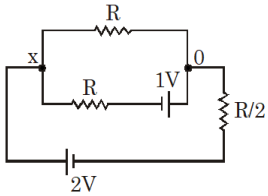
By nodal analysis

x + x – 1 + 2(x – 2) = 0

x = $\frac{5}{4}$V,

$I_2 = \frac{1}{4}A$, $I_1 = \frac{6}{4}A$

For D



By nodal analysis

x + x + 1 + 2(x – 2) = 0

x = $\frac{3}{4}$,

$I_2 = \frac{7}{4}A$, $I_1 = \frac{10}{4}A$

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3. **Ans (A->PQR,B->QST,C->ST,D->PR)**

For (A) : Field remains same as charge is same. Hence energy stored increases. (P) & (Q) as field is same and (R).

For (B) : Since distance between plate and charge remains same and field decreases, (S & T)

$F = \frac{Q^2}{2\epsilon_0 A}$, charge is same hence force remain same (Q).

For (C) : Since connected to battery P.D. remains same and as distance increases the electric field decreases. (T) Also as capacitance decreases the charge also decreases decreasing the force of interaction and energy inside decrease (S).

For (D) : Capacitance increases and P.D. remaining same, thus charge and energy increases (P) and force of interaction increases. Field remains same as distance and P.D. remains same (R) and energy increases.

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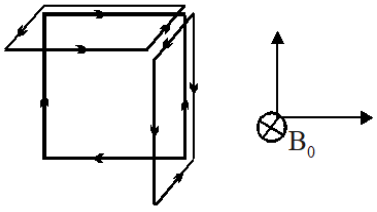
SOLUTION

PHYSICS

SECTION-I (i)

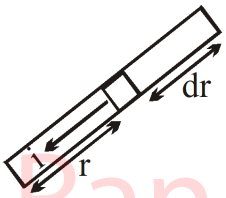
1. Ans (D)

Imagine 3 loops.



$B = \sqrt{B_0^2 + B_0^2 + B_0^2} = \sqrt{3}B_0$ towards F

2. Ans (A)



Torque due to magnetic force $|d\vec{\tau}| = \int_0^L (idrB)r =$ $\frac{iL^2B}{2}$

In equilibrium $\frac{iL^2B}{2} = (kx) \times L\sin 30^\circ$ or x = $\frac{5iLB}{8k}$

3. Ans (C)

Along y $v_y^2 = 2 \times \frac{qE}{m} \times y$

Net speed = $\sqrt{v^2 + v_y^2}$

4. Ans (A)

$B_E - B_P = B_1$ $\quad B_E + B_P = B_2$

$B_E = \frac{B_1 + B_2}{2}$

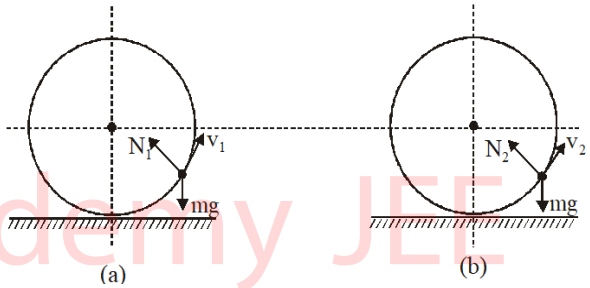
$B_P = \frac{B_2 - B_1}{2} = \frac{\mu_0 J}{2}$

So answer is = $J\left(\frac{B_1 + B_2}{2}\right)$

5. Ans (D)

Current sensitivity = $\frac{NAB}{C}$

7. Ans (A)



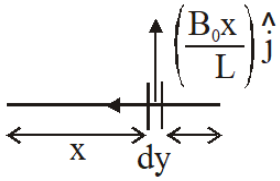
F.B.D in two cases is as shown in case (b) normal will be less and incase (b) work done by magnetic field is zero so height in two cases will be same.]

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PHYSICS

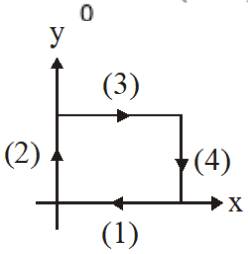
SECTION-I (ii)

8. Ans (A,B,C,D)



Wire segment (1)

$$(d\vec{f}) = \int_0^L i\,dx\left(\frac{B_0x}{2}\right) = \frac{iB_0L}{2};\ d\vec{f} = \frac{-iB_0L}{2}\hat{k}$$



Similary for $\vec{f}_2 = -\frac{-iB_0L}{2}\hat{k}$; $\vec{f}_3 = \frac{iB_0L}{2}\hat{k}$; $\vec{f}_4 = \frac{iB_0L}{2}\hat{k}$ $\vec{f}_n = 0$

If coil is constrained to rotate about y-axis $|\vec{\tau}| = i\left(\frac{B_0L}{2}\right)L = \frac{iAB_0}{2}$

$\Rightarrow$ Similarly for torque about torque has same magnitude

9. Ans (C,D)

Current = $\frac{(4q)\omega}{2\pi}$ for each face from which axis is passing

$$B = 2\times\frac{\mu_0 iR^2}{2\left(R^2+x^2\right)^{\frac{3}{2}}} = \frac{2\times\mu_0(4q)\omega\left(\frac{a}{\sqrt{2}}\right)^2}{2\cdot 2\pi\left[\left(\frac{a}{\sqrt{2}}\right)^2+\left(\frac{a}{2}\right)^2\right]^{\frac{3}{2}}} = \frac{8\mu_0q\omega}{3\sqrt{3}\pi a}$$

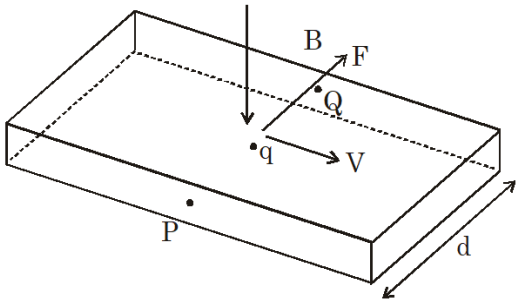
Reversing any of the four charges mass the current either zero or in opposite sense at two surface.

10. Ans (A,D)

(A) $\frac{\mu_0K}{2}$ hence K = σV

(D) F = $q\vec{u}\times\vec{B}$ hence upwards.

11. Ans (A,B)



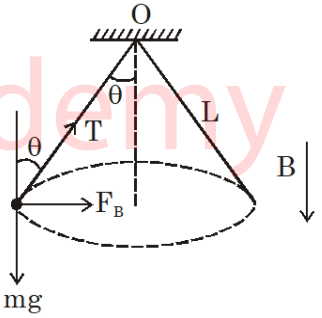
$\therefore$ Charges separate between P & Q

hence potential difference develops between P & Q.

$\Delta V = Ed = v_dBd$

$\therefore$ if 'B' increases, ΔV increases.

12. Ans (A,D)



$T\cos\theta = mg$ ... (i)

$T\sin\theta + F_B = mw^2(L\sin\theta)$ .... (ii)

From equation (i) and (ii)

$$\left(\frac{mg}{\cos\theta}\right)\sin\theta + q(\omega L\sin\theta)B = m\omega^2L\sin\theta$$

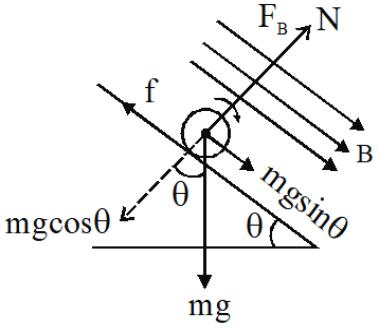
$$\Rightarrow B = \frac{m\omega}{q} - \frac{mg\sin\theta}{\cos\theta\cdot(\omega L\sin\theta)}$$

$$B = \frac{1}{\beta}\left[\omega - \frac{g}{\omega L\cos\theta}\right],\left(\because\beta = \frac{q}{m}\right)$$

Since torque due to force mg and F_B is not zero about 'O'. Hence angular momentum is not conserved about point of suspension 'O'.

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13. **Ans (B,C,D)**



Using $\vec{F}_B = i\left(d\vec{\ell} \times \vec{B}\right)$, we can find direction of F_B, as shown in figure.

for angular velocity 'ω' to be constant,

$\tau_{cm} = 0$

$\Rightarrow$ $f = 0$

$\Rightarrow$ $N = 0$

$\Rightarrow$ $F_B = mg\cos\theta$

Due to component of mg, i.e., $mg\sin\theta$ translational velocity will change.

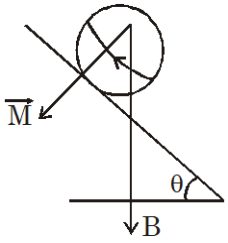
15. **Ans (A,C,D)**

$I\vec{\alpha} = \vec{\tau}$

$\vec{\tau} = \vec{\mu} \times \vec{B}$

$\vec{\mu} = \pi R^2 \times I\left(-\hat{k}\right)$ & $I = \dfrac{MR^2}{2}$

16. **Ans (B,C)**

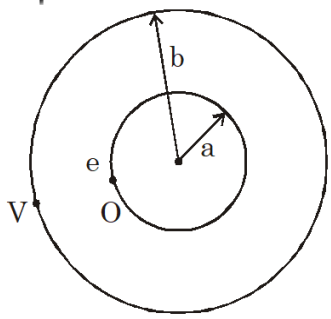


$MB\sin\theta = mg\sin\theta \times R$

$B = \dfrac{mg}{\pi iR}$

17. **Ans (A,C)**

(i) $\sqrt{\dfrac{2eV_0}{m_e}}$, (ii) $\dfrac{2b\,m_e\,u_0}{e(b^2-a^2)}$

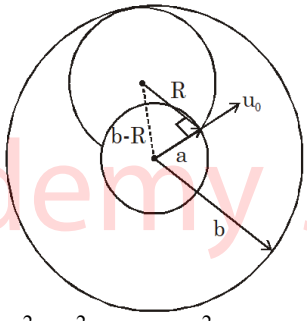


(i) From work-energy theorem between inner and outer cylinder.

$eV = \dfrac{1}{2}m_e v^2$

$\Rightarrow v = \sqrt{\dfrac{2eV}{m_e}}$

(b) From figure



$R^2 + a^2 = (b - R)^2$

$\Rightarrow R^2 + a^2 = b^2 + R^2 - 2bR$

$\Rightarrow R = \dfrac{b^2 - a^2}{2b}$ ... (i)

Also, $R = \dfrac{m_e u_0}{eB_C}$ ... (ii)

From (i) and (ii)

$\dfrac{b^2 - a^2}{2b} = \dfrac{m_e u_0}{eB_C} \Rightarrow B_C = \dfrac{2m_e b u_0}{e\left(b^2 - a^2\right)}$

PHYSICS

SECTION-I (iii)

18. **Ans (D)**

Applying Gauss law,

$\dfrac{-ne}{\epsilon_0}\pi r^2\ell = E2\pi rl \Rightarrow \vec{E} = \dfrac{-ner}{2\epsilon_0}\hat{r}$

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19. **Ans (B)**

Applying Ampere's law : $\oint \vec{B}.d\vec{l} = \mu_0 I$,

$B2\pi r = \mu_0 ne\pi r^2 v \Rightarrow B = \frac{\mu_0 ner}{2} v$

20. **Ans (B)**

Net force on electron =

$q\vec{E} + q\vec{v} \times \vec{B} = \frac{ne^2 r}{2\epsilon_0}\left[1 - \mu_0 \epsilon_0 v^2\right]\hat{r}$

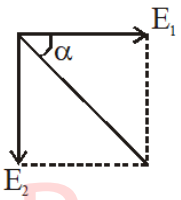
21. **Ans (D)**

$qv_dB = qE$

$E = v_dB = \left(\frac{i}{nqA}\right)B = \frac{iB}{nq(YZ)}$

$\Delta V_H = E \cdot Y = \frac{iB}{nq(Z)}$

22. **Ans (A)**



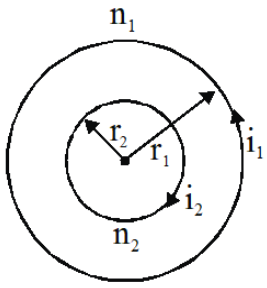
$\tan\alpha = \frac{E_2}{E_1} = \frac{E_2}{\left(\frac{J}{\sigma}\right)} = \frac{\left(\frac{iB}{nqA}\right)}{\frac{J}{\sigma}} = \frac{B\sigma}{nq}$

PHYSICS

SECTION-II

1. **Ans (216.00)**

Energy stored permit length = $\frac{B^2}{2\mu_0} \times \pi\left(r_1^2 - r_2^2\right)$



2. **Ans (160.00)**

$r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{eB}$

$0.75 = \frac{\sqrt{2 \times 16 \times 10^{-27} \times 8 \times 10^6 \times 1.6 \times 10^{-19}}}{1.6 \times 10^{-19} \times B}$

$\Rightarrow B = \frac{4}{7.5}$ so $300B = 160$ T

PHYSICS

SECTION-III

1. **Ans (1)**

dBQ_0

$IdB = m\frac{dv}{dt}$

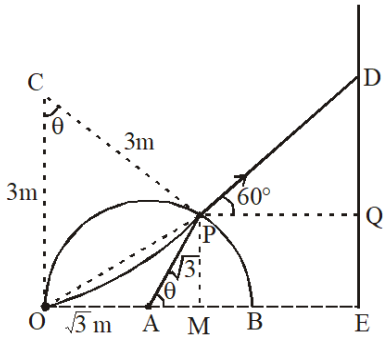
$dB(Idt) = mdv$

$\Rightarrow dB\int_0^{Q_0} Idt = m\int_0^{v} dv$

$\Rightarrow dBQ_0 = mv$

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2. Ans (3)



Considering the upper half of the area

Radius of paticle is circular motion $R = \frac{mv}{qB} = \frac{\left(2\times10^{-3}\right)(0.3)}{\left(1\times10^{-3}\right)\times0.2} = 3$ m

Radius of circular arc, r = $\sqrt{3}$ m

from ΔCOP and ΔOAP

$OP = 2\times3\times\sin\frac{\theta}{2} = 2\times\sqrt{3}\sin\left(\frac{180-\theta}{2}\right)$

$\Rightarrow \tan\left(\frac{\theta}{2}\right) = \frac{1}{\sqrt{3}} \Rightarrow \theta = 60^\circ$

From, ΔAPM,

$PM = \sqrt{3}\sin60^\circ = \frac{3}{2} = \theta E$

$AM = \sqrt{3}\cos60^\circ = \frac{\sqrt{3}}{2}$

From diagram,

$PQ = ME = \sqrt{3} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$

From ΔPQD,

$\tan60^\circ = \frac{DQ}{PQ} \Rightarrow DQ = \sqrt{3}\times\frac{3\sqrt{3}}{2} = \frac{9}{2}$m

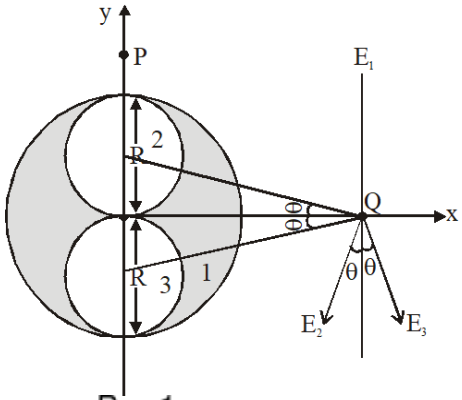
$\therefore DE = QE + DQ = \frac{3}{2} + \frac{9}{2} = 6$m

Similarlly we can get, from lower half

$\therefore$ Distance between the points where charges will strike after deflection = 6 + 6 = 12m = 4x

$\Rightarrow$ x = 3

3. Ans (9)



$\tan\theta = \frac{R}{2R} = \frac{1}{2}$

Assume hollow cylinders as current density ± J and find E_1, E_2 & E_3 individually finally add all of them

4. Ans (5)

$\oint B\cdot d\vec{\ell} = \mu_0 i_{net}$

$B\times2\pi r = \mu_0\int_a^r\left(J_0\frac{r}{a}\right)2\pi r^2dr$

5. Ans (1)

Consider unit length of cylinder charge = $\sigma 2\pi R\times1$

time = $2\pi/\omega$

So $ni = \frac{2\pi R\sigma\omega}{2\pi} = \sigma R\omega$

Now $B = \mu_0 ni$

$B = \mu_0\sigma R\omega$

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SOLUTION

PHYSICS

SECTION-I(i)

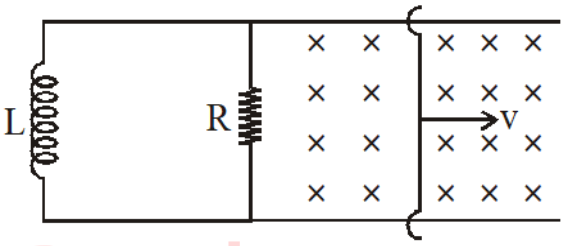
1. Ans (C)

$$30 - L\frac{di}{dt} = i(20 + 5t)$$

$$\because \frac{di}{dt} = -ve \Rightarrow 30 - L\frac{di}{dt} > 30$$

$\Rightarrow$ at t = 2sec ; i > 1 amp

2. Ans (B)



Work done by external goes in magnetic energy + heat dissipiated.

$$BLv = L\frac{di}{dt} \Rightarrow i = B\ell vt = Li$$

$$\therefore \text{ Energy in inductor} = \frac{1}{2}Li^2 = \frac{1}{2}L\frac{B^2v^2t^2\ell^2}{L^2} = 16 \text{ J}$$

$$\text{Energy dissipiated in heat} = i^2Rt = \left(\frac{B^2\ell^2v^2}{R^2}\right)Rt = 16 \text{ J}$$

3. Ans (C)

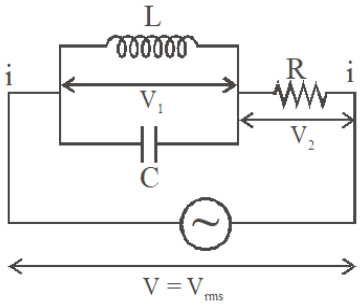
Current through inductor remain same just after switch is opened

$$\Rightarrow i = \frac{E}{r}$$

$$\text{Total heat generated } H = \frac{1}{2}Li^2 = \frac{1}{2}L\left(\frac{E}{r}\right)^2$$

$$\text{Heat generated in solenoid} = \frac{r}{r+R}[H] = \frac{E^2L}{2r(r+R)}$$

4. Ans (B)



$$V_1^2 + V_2^2 = V^2$$

$$\frac{V_1}{X_L} - \frac{V_1}{X_C} = i$$

$$V_1\left(\frac{1}{\omega L} - \omega C\right) = i$$

$$V_1 = \frac{\omega L\, i}{1 - \omega^2 LC}$$

$$V_2 = iR$$

$$\frac{\omega^2L^2}{(1-\omega^2LC)^2} + R^2 = \frac{V^2}{i^2}$$

$$i = \frac{V}{\sqrt{R^2 + \frac{\omega^2L^2}{(1-\omega^2LC)^2}}}$$

5. Ans (B)

$$I = \sqrt{\left(\frac{V}{R}\right)^2 + \left(\frac{V}{X_L} - \frac{V}{X_C}\right)^2}$$

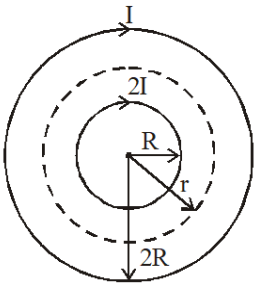
6. Ans (D)

$$\frac{f_{max}}{\Delta f_{\text{half of max power}}} = \text{Quality factor}$$

$$\Rightarrow Q \approx \frac{7}{5} = 1.4$$

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8. **Ans (B)**



$I = Rt$

$B = \mu_0 nRt$

$\phi = \pi R^2 \times 2B + \pi r^2 \times B$

$= (2R^2 + r^2)\,\pi\mu_0 nRt$

$$E(2\pi r) + \frac{d\phi}{dt} = \left(2R^2 + r^2\right)\pi\mu_0 nR$$

$$E = \left(\frac{2R^2 + r^2}{r}\right)\frac{\mu_0 nR}{2}$$

for charged particle

$$\frac{mv^2}{r} = qvB$$

$v = a_t\, t$

$$\boxed{v = \left(\frac{qE}{m}\right)t}$$

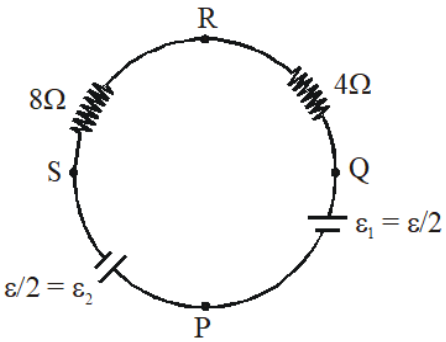
$$\frac{m}{r}\left[\frac{\left(2R^2 + r^2\right)}{r}\frac{\mu_0 nR}{2}\cdot\frac{q}{m}t\right] = q\mu_0 nRt$$

$$\frac{2R^2 + r^2}{2r^2} = 1 \quad \left(r = \sqrt{2}\,R\right)$$

PHYSICS

SECTION-I(ii)

9. **Ans (A,C)**



Electrical equivalent :

$$i = \frac{\in_1 + \in_2}{4 + 8} \Rightarrow \frac{\in}{12} = 4$$

$\in = 48$ V

$\Delta V_{PR} = \in_1 - i(4) = 24 - 4(4) = 8$V

10. **Ans (A,B,D)**

$$M = \frac{\phi_2}{i_1},\ \phi_2 = BA$$

where $B = \frac{\mu_0 i_1}{2\pi x}$

$A = 4 \times 10^{-6}\ m^2$

$$M = \frac{\mu_0}{2\pi x}\cdot A = \frac{2 \times 10^{-7}}{1} \times 4 \times 10^{-6} = 8 \times 10^{-13}$$ H

$i_1 = i_0 e^{-t/RC}$

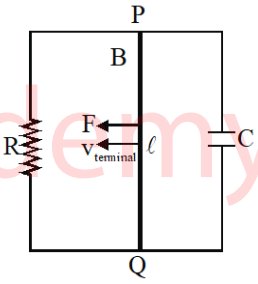
$$e_2 = \frac{-d\phi_2}{dt} \& i_2 = \frac{e_2}{(8 \times 0.5)}$$

$$\Rightarrow \quad i_2 = \frac{\frac{M}{RC}\cdot\frac{q}{RC}e^{-t/RC}}{4} = \frac{8 \times 10^{-13} \times 20 \times 10^{-6}}{\left(10 \times 10^{-6}\right)^2 \times 4} e^{-\frac{2RC\ell n2}{RC}}$$

$= 10^{-8}$ A

11. **Ans (B,C)**

In steady state



$$B\ell V_{terminal} = \frac{q_{max}}{C}$$

also $iB\ell = F$ & $i = \frac{B\ell v_{terminal}}{R}$

$$\Rightarrow \frac{B\ell v_T}{R}\cdot B\ell = F \Rightarrow v_T = \frac{FR}{B^2\ell^2} \text{ and } q_{max} = B\ell C\left(\frac{FR}{B^2\ell^2}\right)$$

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12. **Ans (B,D)**

$i = \frac{E}{R}\left(1 - e^{-Rt/L}\right) \Rightarrow \frac{di}{dt} = \frac{E}{L}e^{-Rt/L}$ = Slope in i-t graph

Circuit-1

$\frac{di}{dt} = \frac{E_1}{L_1}e^0 = \tan 30^\circ \Rightarrow \frac{E_1}{L_1} = \frac{1}{\sqrt{3}}$

Also $i_0 = \frac{E_1}{R_1}$

$\Rightarrow \frac{i_0}{1/\sqrt{3}} = \frac{E_1/R_1}{E_1/L_1} \Rightarrow \frac{L_1}{R_1} = \sqrt{3}i_0 = \tau_1$

Circuit-2

$\frac{di}{dt} = \frac{E_2}{L_2}e^0 = \tan 60^\circ \Rightarrow \frac{E_2}{L_2} = \sqrt{3}$

Also $2i_0 = \frac{E_2}{R_2}$

$\Rightarrow \frac{2i_0}{\sqrt{3}} = \frac{E_2/R_2}{E_2/L_2} \Rightarrow \frac{L_2}{R_2} = \frac{2i_0}{\sqrt{3}} = \tau_2$

$\Rightarrow \frac{\tau_1}{\tau_2} = \frac{3}{2}$

Also $\left(\frac{di}{dt}\right)_2 = 3\left(\frac{di}{dt}\right)_1$

13. **Ans (A,B,C)**

$i_R = \frac{V}{R}$

$\phi = \frac{\pi}{4} \Rightarrow i_L = i_R$

$\Rightarrow \omega_L = R$

$i = \sqrt{\left(\frac{V}{R}\right)^2 + \left(\frac{V}{R}\right)^2}$

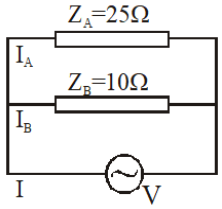
$= \frac{\sqrt{2}100}{100} = \sqrt{2}A$

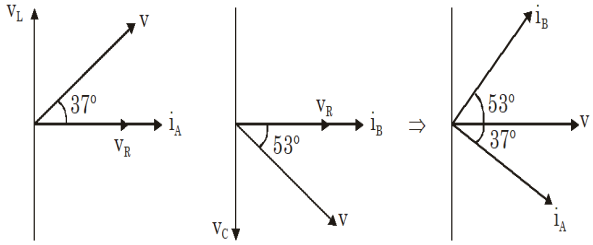
PHYSICS

SECTION-I(iii)

15. **Ans (C)**

Parallel circuit :





Circuit A Circuit B

$i = \sqrt{i_A^2 + i_B^2} \Rightarrow \left(\sqrt{29}\right)^2 = \left(\frac{v}{25}\right)^2 + \left(\frac{v}{10}\right)^2 \Rightarrow \frac{29v^2}{2500} = 29 \Rightarrow v = 50$

$i_A = \frac{50}{25} = 2A$ and $i_B = \frac{50}{10} = 5A$

16. **Ans (B)**

Power $= I_A^2R_A + I_B^2R_B = (2)^2(20) + (5)^2(6)$

$= 80 + 300 = 380$ W

17. **Ans (A)**

$EMF = \pi b^2\frac{dB}{dt} \Rightarrow K\pi b^2 = E(2\pi b)$

$\Rightarrow F = qE = \frac{qKb}{2}$

$\tau = Fb = \frac{qKb^2}{2} = (mb^2)\alpha$

$\Rightarrow \alpha = \frac{qK}{2m}$

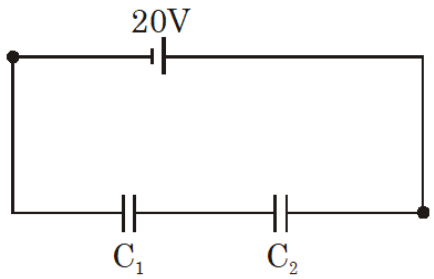
18. **Ans (B)**

$v = (\alpha b)t = \frac{qKb}{2m}t$

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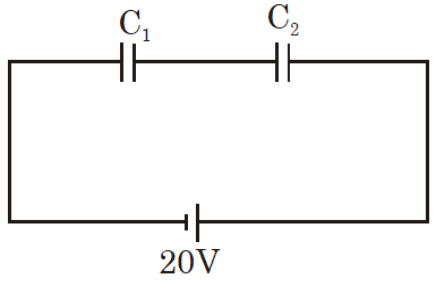
19. Ans (C)

When k_1 closed & k_2 open



After very long time P.D

across both capacitor = 20 volt



$\frac{V_1}{V_2} = \frac{C_2}{C_1} = \frac{3}{2}$

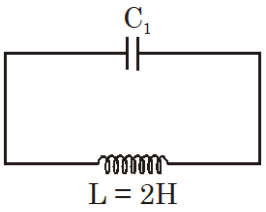
$Q = CV \Rightarrow V \propto \frac{1}{C}$

$V_1 = \frac{3}{5} \times 20 = \frac{60}{5} = 12$ volt

$V_2 = 8$ volt

$Q = C_2V_2 = 3 \times 8 = 24\ \mu C$

20. Ans (C)



$Q = 24\ \mu C$

$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2 \times 2 \times 10^{-6}}}$

$= \frac{1000}{2} = 500$

$T = \frac{2\pi}{\omega} = \frac{2\pi}{500}$

at t = 0, charge will be maximum across capacitor

$\xrightarrow[\text{again max.}]{} t = \frac{T}{2} \Rightarrow \frac{\pi}{500}$ sec

21. Ans (A)

Maximum energy in inductor

$\Rightarrow \frac{1}{2}Li^2 = \frac{Q^2}{2C} \Rightarrow \frac{\left(24 \times 10^{-6}\right)^2}{2 \times \left(2 \times 10^{-6}\right)}$

$\frac{24 \times 24 \times 10^{-12}}{4 \times 10^{-6}} = 0.144$ mJ

22. Ans (A)

$\Rightarrow Z = \sqrt{\left(\frac{1}{\omega C}\right)^2 + R^2}$

$\Rightarrow i_{max} = \frac{V_{in}}{Z}$

$\Rightarrow v_{out} = i \times R$

$v_{out} = \frac{V_{in}}{Z} \times R$

$\frac{v_{in}}{v_{out}} = \frac{Z}{R} = \frac{\sqrt{\left(\frac{1}{\omega C}\right)^2 + R^2}}{R}$

23. Ans (D)

$x_C = \frac{1}{\omega C} = x_C = \infty \Rightarrow$ all potential drop will be across capacitor

$v_{out} = 0 \Rightarrow \frac{\Delta v_{out}}{\Delta v_{in}} = 0$

24. Ans (A)

$Z = \sqrt{\left(\frac{1}{\omega C}\right)^2 + R^2}$

$i_{max} = \frac{V_{in}}{Z}$

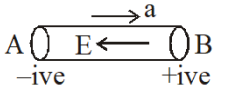
$\Rightarrow v_{out} = ix_C$

$v_{out} = \frac{V_{in}}{Z} \times \left(\frac{1}{\omega C}\right)$

$\frac{v_{out}}{v_{in}} \frac{\frac{1}{\omega C}}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$

25. Ans (A,C)

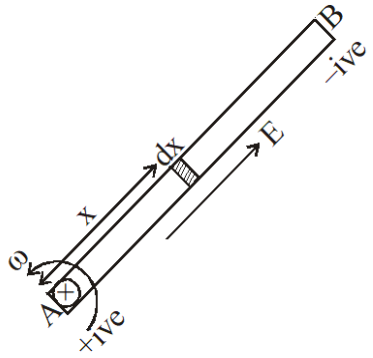
Due to pseudo force, electrons move from B to A.



$\Rightarrow v_B > v_A$

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26. Ans (A,C)



Due to centrifugal force electrons are moving from A to B.

At equilibrium, Electric force due to induce charges become equal to centrifugal force $m\omega^2 x = eE$

$$\int_0^{\ell} m\omega^2 x dx = \int_0^{\ell} eE dx = e\int_0^{\ell} E dx$$

$$\Rightarrow \frac{m\omega^2\ell^2}{2} = ev \text{ (v = voltage across rod)}$$

$$\Rightarrow v = iR = \frac{m\omega^2\ell^2}{2e} \Rightarrow \frac{e}{m} = \frac{\omega^2\ell^2}{2iR}$$

PHYSICS

SECTION-III

1. Ans (2)

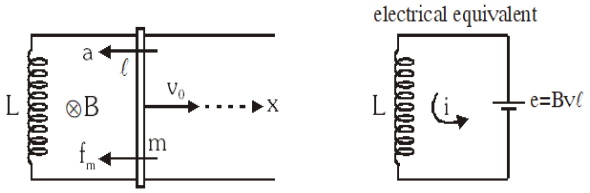
Magnetic field is non zero only in the region between the two solenoids , where $B = \mu_0 n_2 i_2$

$\therefore$ energy stored per unit volume $= \frac{B^2}{2\mu_0} = \frac{\mu_0 n_2^2 i_2^2}{2}$

The energy per unit length. = energy per unit volume × area of cross section were $B \neq 0$

$= \frac{\mu_0 n_2^2 i_2^2}{2}[\pi(r_2^2 - r_1^2)] = \frac{\mu_0 n_1^2 i_1^2}{2}[\pi(r_2^2 - r_1^2)]$, since $n_1 i_1 = n_2 i_2$

2. Ans (5)



According to Len's law, magnetic force, will oppose the motion of conducting rod.

$F_m = i\ \ell\ B$; $ma = i\ell B$ and $\left(a = -\frac{dv}{dt}\right)$ so,

$$m = \left(-\frac{L}{B\ell}\frac{d^2i}{dt^2}\right) = i\ell B \text{ or } \frac{d^2i}{dt^2} = -\left(\frac{B^2\ell^2}{mL}\right)i$$

and $\varepsilon = L\frac{di}{dt}$; $Bv\ell = L\frac{di}{dt}$; $(B\ell)\left(\frac{dv}{dt}\right) = L\frac{d^2i}{dt^2}$

This is a second order differential equation; So, $i = i_0\sin(\omega t)$

where $\frac{1}{2}mv_0^2 = \frac{1}{2}Li_0^2 \Rightarrow \left[i_0 = v_0\sqrt{\frac{m}{L}}\right]$ and $\left[\omega = \sqrt{\frac{B^2\ell^2}{mL}}\right]$

We have, $Bv\ell = L\frac{di}{dt}$ or B (vdt) ℓ = L(di) or $B\ell(dx) = L(di)$

or $x = \frac{Li}{B\ell} = \frac{Li_0\sin\omega t}{B\ell} = \frac{Lv_0\sqrt{\frac{m}{L}}}{B\ell}\sin\omega t \Rightarrow x = \frac{v_0}{\omega}\sin\omega t$

So amplitude $= \frac{v_0}{\omega} = \frac{v_0}{B\ell}\sqrt{mL} = 5cm$

3. Ans (5)

$$V = \frac{Br^2\omega}{2}$$

$$I = \frac{V}{R} = \frac{Br^2\omega}{2R}$$

$$\frac{B^2r^3\omega}{2R}\cdot\frac{r}{2} - mg\rho = 0$$

$$\omega = \frac{4mg\rho R}{B^2r^4} = 100s^{-1}$$

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SOLUTION

PHYSICS

SECTION-I(i)

9. Ans (B)

Equation of demped oscillation

$x = \left[Ae^{-\frac{b}{2}mt}\right] \sin\omega t$

$I = \left[I_0 e^{-\frac{R}{2L}t}\right] \sin\omega t$

$I_0 e^{-(R/2L)t} = \frac{I_0}{e}$ For maximum

$\frac{R}{2L}t = 1 \quad \phi = \omega t$

$t = \frac{2L}{R} \quad \phi = \frac{2L\omega}{R} (X_L = \omega L)$

$(2\pi)n = \frac{2 \times 200}{20} = n = \frac{10}{\pi}$

10. Ans (D)

$\omega' = \sqrt{\frac{K}{m} - \frac{b^2}{4m^2}}$ and $\omega' = \frac{2\pi}{T}$

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11. Ans (D)

This is the case of Damped oscillation.

12. Ans (C)

Considering gravity, buoyant force and viscouse force oscillation will be damped.

$\Omega_1^2 = \omega_2^2 - C$, ω_2 is natural frequency.

$\Rightarrow T_1 > T_2$ C is a constant.

15. Ans (B)

$T_0 = 2\pi\sqrt{\frac{m}{k}}$

$= \frac{2\pi}{\sqrt{10}}$

$A = A_0 e^{-t/\gamma}$

$\therefore$ for $A = \frac{A_0}{e}$, $t = \gamma$

$t = \gamma = \frac{2m}{b} = \frac{2m}{\frac{B^2\ell^2}{R}} = 10^4 s$

$\therefore$ No of oscillation $\frac{t}{T_0} = \frac{10^4}{2\pi/\sqrt{10}} \approx 5000.$

SOLUTION

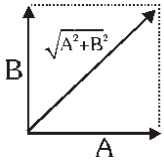
PHYSICS

SECTION-I(i)

1. Ans (C)

It is combination of two wave of phase difference $\frac{\pi}{2}$.

$\therefore$ Resultant amplitude = $\sqrt{A^2 + B^2}$



Compare with the wave y (x, t) = D sin(kx−ωt +ϕ)

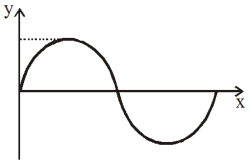
$\therefore D^2 = A^2 + B^2$

2. Ans (C)

A_i positive

$A_r = \frac{V_2 - V_1}{V_2 - V_1} A_i$ $V_2 < V_1$ = negative

3. Ans (D)



Wave moving in +ve x direction

At t = 0

At t = 0, x = 0, y = 0

ϕ = 0

y = A sin(kx − ωt)

Wave moving in −ve x direction

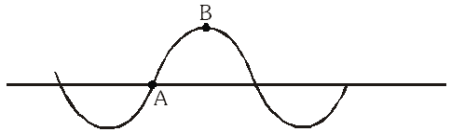
y = A sin (ωt + kx + ϕ)

at t = 0, x = 0, y = 0

ϕ = 0

y = 0 A sin(ωt + kx)

4. Ans (B)

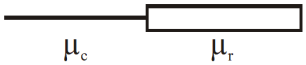


Both KE and PE are maximum at y = 0 and minimum at y = A.

V = 0 at point B so KE = 0.

Slope zero at B so PE = 0.

5. Ans (C)



A = 1

A_t – Amplitude of transmitted wave.

$A_t = \left[\frac{2\sqrt{\mu_\ell}}{\sqrt{\mu_\ell} + \sqrt{\mu_r}}\right]$

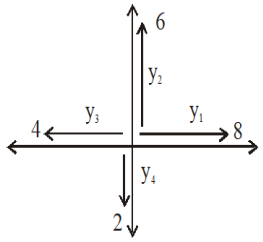
6. Ans (A)

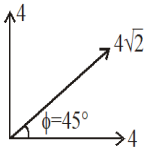
$y_1 = 8\sin(\omega t + kx)$

$y_2 = 6\sin(\omega t + kx + \pi/2)$

$y_3 = 4\sin(\omega t + kx + \pi)$

$y_4 = 2\sin(\omega t + kx + 3\pi/2)$





$y_5 = 4\sqrt{2}\sin(\omega t - kx + \pi/4)$

$y_1 + y_2 + y_3 + y_4$

$y' = 4\sqrt{2}\sin(\omega t + kx + \pi/4)$

$y_{net} = y' + y_5$

$= 4\sqrt{2}(\sin(\omega t - kx + \pi/4) + \sin(\omega t + kx + \pi/4)$

$= 4\sqrt{2}[2\sin(\omega t + \pi/4)\cos kx]$

$= 8\sqrt{2}\sin\left(\omega t + \frac{\pi}{4}\right)\cos kx$

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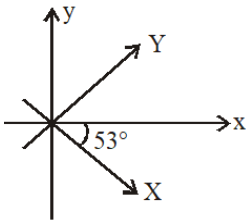
7. **Ans (B)**

T.E. = KE + P.E. = $\int \left[\frac{1}{2}(\mu dx)v_P^2 + \frac{1}{2}Tdx\left(\frac{dy}{dx}\right)^2\right] = m^2Ta$

8. **Ans (C)**

In new cartesian coordinate system X-Y, wave equation can be written as

$\phi = \phi_n \sin(2t - x + \frac{\pi}{3})$



9. **Ans (B)**

The given equation can be written in the form y = 2 × 2 $\sin\left(\frac{2\pi x}{30}\right)\cos 2\pi\frac{(1440t)}{30}$

a = 2 cm, λ =30 cm, v = 1440 cm/sec; y_{max} = 4 sin $\frac{2\pi x}{30}$

For x =5, $y_{max} = 4\sin\frac{2\pi \times 5}{30} = \frac{4\sqrt{3}}{2} = 2\sqrt{3}$ cm

PHYSICS

SECTION-I(ii)

10. **Ans (A,B,C,D)**

y = A sin (kx − ωt + ϕ)

at t = 0, x = 0, y = 5 mm

at $t = \frac{1}{24}$, x = 0, y = 0

also λ = 2m, A = 10 mm

Solve to get answer.

12. **Ans (C,D)**

Time taken by point P from one extreme to other is $\frac{T}{2}$

So, $\frac{T}{2} = 0.1s$

In time $t = \frac{T}{4}$, energy flows from node to antinodes.

Velocity = $\sqrt{\frac{T}{\mu}} = 10m/s$, frequency is 5 Hz

$\frac{nv}{2\ell} = f \Rightarrow n = 2$

So, the string oscillates with two loops

and x = 1 is a node.

Also x = 0.5 is antinode and at t = 0.05

rate of energy flow through P is zero

14. **Ans (A,B,C,D)**

At node cos(10πx) = 0 & at antinode cos(10πx) = 1

& ω = 50 π, k = 10 π & v = $\frac{\omega}{k}$, k = $\frac{2\pi}{\lambda}$

15. **Ans (A,C)**

To produce node at x = 0

$y_1 + y_2 = 0$ x = 0

$\Rightarrow A\sin\left(-ut + \frac{\pi}{3}\right) + \frac{1}{2} = 0$

$\Rightarrow \frac{1}{2} = -A\sin\left(-4t + \frac{\pi}{3}\right)$

Phase difference between two wave force fig are π

$\frac{1}{2}$ may be = $A\sin\left(-2x - 4t + \frac{\pi}{3} \pm \pi\right)$

16. **Ans (A,C)**

Let $\lambda_1 = \lambda$

$\lambda_2 = 2\lambda$

Let of wave become 2 time's have frequency will remain same.

18. **Ans (A,B,D)**

PE and KE both maximum at y = 0. [Simultaneously for two particles in phase]

Amplitude and frequency same for all particles.

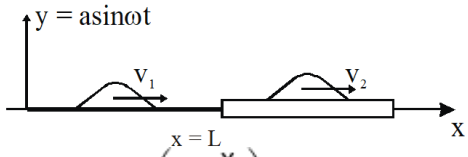
There will be phase diffrence in motion of all particles.

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PHYSICS

SECTION-I(iii)

19. Ans (A)



$$Y_i = a\sin\omega\left(t-\frac{x}{v_1}\right)$$

Transmitted at x = L

$$Y_{(x=L)} = a\sin\omega\left(t-\frac{L}{v_1}\right)$$

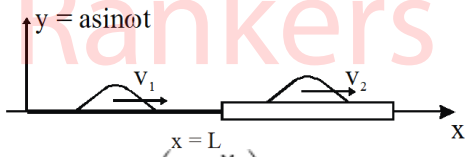
$$Y_t = a_t\sin\omega\left[t-\frac{L}{v_1}-\left(\frac{x-\ell}{v_2}\right)\right]$$

Reflected at x = L

$$y_{x=2L} = a\sin\omega\left(t-\frac{\ell}{v_1}\right)$$

$$y_{Reff} = a_r\sin\omega\left[t-\frac{\ell}{v_1}+\left(\frac{x-\ell}{v_1}\right)+\pi\right]$$

20. Ans (C)



$$Y_i = a\sin\omega\left(t-\frac{x}{v_1}\right)$$

Transmitted at x = L

$$Y_{(x=L)} = a\sin\omega\left(t-\frac{L}{v_1}\right)$$

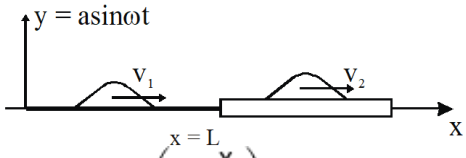
$$Y_t = a_t\sin\omega\left[t-\frac{L}{v_1}-\left(\frac{x-\ell}{v_2}\right)\right]$$

Reflected at x = L

$$y_{x=2L} = a\sin\omega\left(t-\frac{\ell}{v_1}\right)$$

$$y_{Reff} = a_r\sin\omega\left[t-\frac{\ell}{v_1}+\left(\frac{x-\ell}{v_1}\right)+\pi\right]$$

21. Ans (B)



$$Y_i = a\sin\omega\left(t-\frac{x}{v_1}\right)$$

Transmitted at x = L

$$Y_{(x=L)} = a\sin\omega\left(t-\frac{L}{v_1}\right)$$

$$Y_t = a_t\sin\omega\left[t-\frac{L}{v_1}-\left(\frac{x-\ell}{v_2}\right)\right]$$

Reflected at x = L

$$y_{x=2L} = a\sin\omega\left(t-\frac{\ell}{v_1}\right)$$

$$y_{Reff} = a_r\sin\omega\left[t-\frac{\ell}{v_1}+\left(\frac{x-\ell}{v_1}\right)+\pi\right]$$

22. Ans (D)



A = 3 cm

f = 100 Hz

$$2m = \frac{3\lambda}{2}$$

$$\lambda = \frac{4}{3}m$$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi\times 3}{4} = \frac{3\pi}{2}$$

23. Ans (B)

$$E = (\text{No of loop's})\times\left(\frac{1}{8}\omega^2A_S^2\lambda\right)$$

24. Ans (C)

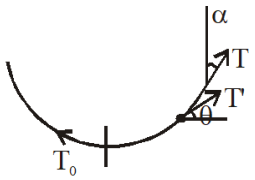
$$V = f\times\lambda \qquad \lambda = \frac{4}{3}$$

$$\lambda = 100\times\frac{4}{3} = \frac{400}{3}$$

$$\text{time} = \frac{\lambda}{4V} = \frac{4}{4\times 3}\times\frac{3}{400} = \frac{1}{400}$$

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25. Ans (D)



$T\cos\alpha = \frac{Mg}{2}$

$T\sin\alpha = T_0$

$T'\cos\alpha = T_0$

$T'\sin\theta = \frac{M}{L}xg$

$\tan\theta = \frac{dy}{dx} = \frac{2x}{L}\cot\alpha$

$y = \frac{x^2}{L}\cot\alpha$

26. Ans (A)

$T\cos\alpha = \frac{Mg}{2}$

$T\sin\alpha = T0$

$\tan\alpha = \frac{2T_0}{Mg}$

$v = \sqrt{\frac{T_0}{m}\ell} = \sqrt{\frac{Mg\ell\tan\alpha}{2m}}$

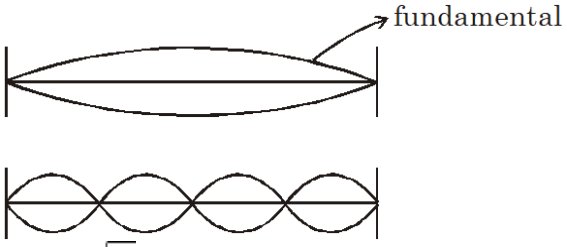
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PHYSICS

SECTION-II

1. Ans (17.5)

17.5°



$f = \frac{n}{2\ell}\sqrt{\frac{T}{\mu}}$

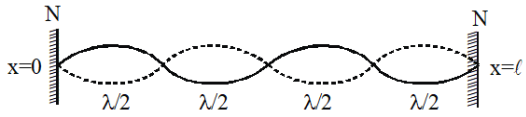
$f^2 = \frac{4}{\ell^2} \times \frac{Y\alpha\Delta\theta A}{\rho A}$

$f^2 = \frac{4}{\ell^2} \times \frac{Y\alpha\Delta\theta}{\rho}$

$\Delta\theta = \frac{f^2\ell^2\rho}{4Y\alpha}$

$\Delta q = 2.5°C$

2. Ans (9.00)



We have taken x = 0 as node so that

Amplitude = (A_{max} sink x)

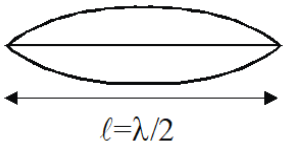
$K = \frac{2\pi}{\lambda}$ where

$\lambda = \frac{\ell}{2} \Rightarrow A = A_{max}\sin\left[\frac{2\pi}{\frac{\ell}{2}} \times \frac{\ell}{3}\right] = A\sin\frac{4\pi}{3} = \frac{\sqrt{3}A_{max}}{2} = 3$

3. Ans (375.00)

$\rho = \frac{m}{A\ell} \Rightarrow \ell = \frac{m}{\rho A}$ and $\mu = \frac{m}{\ell} \Rightarrow v = \sqrt{\frac{T}{\mu}}$

fundamental frequency



$f = \frac{v}{\lambda}$

SOLUTION

PHYSICS

SECTION-I(i)

2. Ans (B)

In rod, $\frac{\lambda}{4} = \ell \Rightarrow \lambda = 4\ell \Rightarrow v_r = \lambda\nu = 4\ell\nu$

In air column $v_a = 2\Delta\ell\nu$;

$\frac{v_r}{v_a} = \frac{2\ell}{\Delta\ell} \Rightarrow v_r = \frac{2 \times 1 \times 330}{0.6} = 1100 m/s$

3. Ans (B)

Let n_1 be the frequency of the wire having tension $T + \Delta T$ and n_2 be the frequency of the wire having tension T, then $\frac{n_1}{n_2} = \sqrt{\frac{T+\Delta T}{T}} = \frac{500+5}{500} = \frac{505}{500}$

$\Rightarrow \frac{T+\Delta T}{T} = \left(1 + \frac{5}{500}\right)^2 \approx 1 + \frac{2}{100} \approx 1.02 \Rightarrow \frac{\Delta T}{T} = 0.02$

OR

For stretched wire frequency $n \propto \sqrt{T} \Rightarrow \frac{\Delta n}{n} = \frac{1}{2}\frac{\Delta T}{T}$

$\Rightarrow \frac{\Delta T}{T} = 2\left(\frac{\Delta n}{n}\right) = 2\left(\frac{5}{500}\right) = 0.02$

4. Ans (C)

The second wire resonates best with the first wire when the two wires have exactly the same frequency.

$f = \frac{n}{2\ell}\sqrt{\frac{T}{\rho A}} = \frac{n}{2\ell}\sqrt{\frac{T}{\rho\pi r^2}} \Rightarrow n \propto r$

5. Ans (D)

At closed end pressure antinode and displacement nodes are formed so at closed end variation in pressure is maximum.

6. Ans (A)

Loudness $\beta = 10 \log_{10}\frac{I}{I_0}$ $\therefore \beta_2 - \beta_1 = 10\log_{10}\frac{I_2}{I_1}$ $\because I = \frac{P}{4\pi r^2} \therefore \frac{I_2}{I_1} = \frac{r_1^2}{r_2^2}$

$\therefore (\beta + 20) - \beta = 10 \log_{10}\frac{r^2}{r_2^2} = 20 \log_{10}\frac{r}{r_2} \Rightarrow \frac{r}{r_2} = 10 \Rightarrow r2 = 0.1r$

$\therefore$ shift = r – 0.1 r = 0.9 r.

7. Ans (C)

Appearent frequency $f = \left(\frac{v}{v - u\cos\theta}\right) f_0$

Angle of approaches θ,

$\theta_C > \theta_B > \theta_A$

$\therefore f_1 > f_2 > f_3$

8. Ans (C)

Wave originating at t =0 from S_1 reaches point A at t = 1.

Wave originating at $t = \frac{1}{2}$ from S_2 reaches point A at t = 1.

So phase difference in these waves = $\frac{\pi}{2} + \frac{\pi}{6}$

$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\phi} = 5$

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9. Ans (C)

$\Delta\phi = 2n\pi \Rightarrow \frac{\pi}{2} + \frac{2\pi}{\lambda} d \sin\theta = 2n\pi; \frac{2\pi}{\lambda} d \sin\theta$

$= \left(2n - \frac{1}{2}\right)\pi$

$\sin\theta = \left(2n - \frac{1}{2}\right)\frac{\lambda}{2d} = \frac{1}{2} \times \frac{\lambda}{2 \times 3\lambda} = \frac{1}{12} \Rightarrow$

$\frac{y}{\sqrt{(100\lambda)^2}} = \frac{1}{12}$

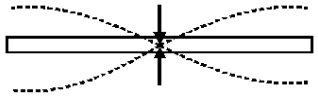
$144y^2 = (100\lambda)^2 ; y \approx \frac{100\lambda}{12} = \frac{25\lambda}{3}$

PHYSICS

SECTION-I(ii)

12. Ans (A,C)

$L = \frac{\lambda}{2} \Rightarrow \lambda = 2L$



$\Rightarrow f = \frac{C}{2L}$

2nd OT $\Rightarrow L = \frac{5\lambda}{2}$



$\Rightarrow \lambda = \frac{2L}{5}; f = \frac{5C}{2L}$

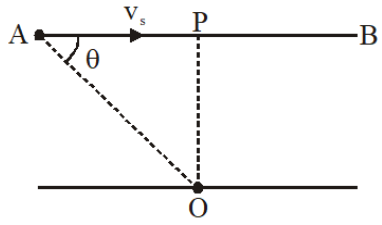
f is $\frac{C}{2L}, \frac{3C}{2L}, \frac{5C}{2L}$ etc.

13. Ans (C,D)

The graph shows the situation shown in figure below. The observed frequency will initially be more than the natural frequency. When the source is at P, observed frequency is equal to its natural frequency i.e., 2000 Hz.

For region AP : $f = f_0\left(\frac{v}{v - v_s\cos\theta}\right)$

For PB : $f = f_0\left(\frac{v}{v + v_s\cos\theta}\right)$



Minimum value of f will be:

$f_{min} = f_0\left(\frac{v}{v + v_s}\right)$ when $\cos\theta = 1$

or $1800 = 2000\left(\frac{300}{300 + v_s}\right)$

Solving this we get, $v_s = 33.33$ m/s

and maximum value of f can be

$f_{max} = f_0\left(\frac{v}{v - v_s}\right)$ when $\cos\theta = 1$

or $f_{max} = 2000\left(\frac{300}{300 - 33.33}\right) = 2250$ Hz

14. Ans (A,C)

For sound, air is denser medium than water.

15. Ans (B,D)

From open end compression will be reflected as refraction and from closed end these is no phase change so compression will be reflected back as compression.

16. Ans (A,B,C)

For open organ pipe $f = \frac{nv}{2L}$ & for closed $f = \frac{(2n+1)}{4L}v$

$\frac{v}{2(24+22)} = \frac{3v}{4(38+e)} \Rightarrow e = 1$ cm

Fundamental of open pipe $= \frac{340}{2(26\text{cm})} = 654$ Hz

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17. **Ans (A,B,D)**

(A) $y_1 = A \sin 2\pi ft \quad y_2 = y_1 + y_2$

$y_2 = A\sin 2\pi f_2 t$

$f = \frac{f_1 - f_2}{2}$

for pressure wave it behave like open end.

PHYSICS

SECTION-I(iii)

18. **Ans (A)**

In first case, no relative motion between source and observer so no doppler effect. Only due to distance, last soldiers heard sound after some time.

19. **Ans (B)**

time interval between strike left foot and next strike right foot $= \frac{60\,\text{sec}}{120} = \frac{1}{2}\,\text{sec.}$

time interval between first and last soldier is $= n\left(\frac{1}{2}\,\text{sec.}\right)$

n is integer

distance between first and last soldier S = vt

$S = (320)\,(n \times \frac{1}{2}) = 160\text{m}$

S = 160, 320, 480,.....

20. **Ans (D)**

In part b, soldiers recieve sound of frequency $\frac{121}{60}\text{Hz}$

$\Rightarrow f' = \left(\frac{v + v_0}{v}\right) f$

$\Rightarrow \frac{121}{60} = \left(\frac{320 + v_0}{320}\right)\frac{120}{60} \Rightarrow 320 \times 121 = 320 \times 120 + 120\,v$

$\Rightarrow v_0 = \frac{320}{120} = 2.66\ \text{m/s}$

21. **Ans (B)**

$f' = \left(\frac{v - v_0}{v}\right) f = \left(\frac{320 - \frac{320}{120}}{320}\right) \cdot 120 = 119$ steps/min.

22. **Ans (A)**

$f = \frac{C - V_R}{C + V_R} f_0 \,;\, \frac{\Delta f}{f_0} = \frac{2V_R}{C + V_R} \approx \frac{2V_R}{C}$

$\Rightarrow C = \frac{2 \times 0.1}{600} \times 5 \times 10^6 \approx 1700\ \text{m/s}$

23. **Ans (D)**

$f' = \frac{C - V_R}{C + V_R} f_0 \,;\, \frac{\Delta f}{f_0} = \frac{2V_R}{C + V_R}$

velocity is 4 times ($A_1v_1 = A_2v_2$) $\Rightarrow \Delta f = 4\Delta f_0 =$ 2400 Hz

24. **Ans (A)**

$\frac{1}{2}\frac{dm}{dt}(v_2^2 - v_1^2) = \frac{1}{2} \times 1.5 \times 10^3 \times 0.1 \times 10^{-4} \times 0.1\,[(4 \times 0.1)^2 - (0.1)^2]$

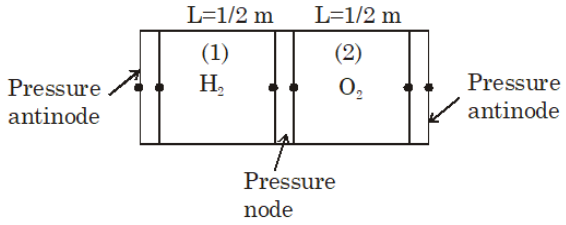
$= 0.75 \times 10^{-5}\,[15] = 11.25 \times 10^{-5} = 1.125 \times 10^{-4}$ W

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PHYSICS

SECTION-II (ii)

3. **Ans (1650)**



$n_1 \times \frac{v_1}{4L} = n_2 \times \frac{v_2}{4L}$

$\frac{n_1}{n_2} = \frac{300}{1100} = \frac{3}{11}$

$f_1 = \frac{3 \times 1200}{4 \times \frac{1}{2}} = 1650\text{Hz}$

PHYSICS

SECTION-III

1. **Ans (2)**

$f_{\text{reflected}} = \frac{(300 - 150)}{150} \times 75\text{Hz}$

$\therefore \lambda_{\text{reflected}} = 2\ \text{m}$

2. Ans (4)

$f = \frac{v}{2(\ell + 2e)}$ where e = end correction = 0.6 r

$\therefore f = \frac{v}{2(\ell + 2 \times 0.6r)} = \frac{v}{2(\ell + 1.2r)} \therefore \frac{\Delta f}{f} = \frac{\Delta v}{v}$

$- \frac{\Delta(\ell + 1.2r)}{\ell + 1.2r} = \frac{\Delta v}{v} - \frac{\Delta\ell + 1.2\Delta r}{\ell + 1.2r}$

here $\frac{\Delta v}{v} = 0$ (given) $\frac{\Delta f}{f} \times 100 = - \frac{\Delta\ell + 1.2\Delta r}{\ell + 1.2r} \times 100$

for maximum % error : $\Delta\ell = 0.1$, $\Delta r = 0.05$

$\left(\frac{\Delta f}{f} \times 100\right)_{max} = \frac{0.1 + 1.2 \times 0.05}{94 + 1.2 \times 5} \times 100 =$ 0.16%

3. Ans (2)

$I = \frac{P^2}{2\rho V}$, $v = \sqrt{\frac{B}{\rho}} = \sqrt{\frac{4 \times 10^4}{1}} = 200$ m/s

Also $20 = 10 \log \frac{I}{I_0} \Rightarrow I = 10^{-19}$ W/m^2

$10^{-10} = \frac{P^2}{2 \times \rho \times v} \Rightarrow P = 2 \times 10^{-4}$ N/m^2

4. Ans (4)

Energy flow rate reaching surface of the ball = $\pi R^2 \frac{(\Delta p_m)^2}{2\rho\nu\lambda}$

5. Ans (2)

Let be ℓ is wavelength of sound in air

$\ell_1 = \frac{\lambda}{4}$ and $\ell_2 = \frac{3\lambda}{4}$

$\ell = 2(\ell_2 - \ell_1)$

$v = f\lambda = 165 \times 2 (150 - 50) \times 10^{-2} = 330$ m/s

$\frac{dv}{v} = \frac{|d\ell_2| + |d\ell_1|}{\ell_2 - \ell_1}$

$dv = \frac{0.5 + 0.1}{150 - 50} \times 330 = 1.98$m/s

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SOLUTION

PHYSICS

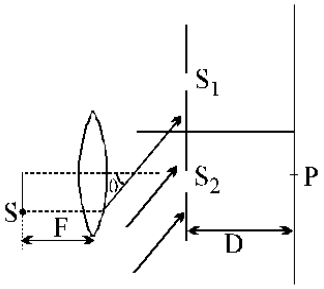
SECTION-I (i)

1. Ans (A)

$\tan\theta = \theta = \sin\theta = \frac{d/2}{F}$

suppose P is max.

then $d\sin\theta = S_1P - S_2P = \frac{dy}{D}$



$\therefore \frac{dD}{2F} = y$ Ans.]

2. Ans (A)

Phase difference between waves through S_1 and S_2 is $2 \times (\lambda/6) \times (2\pi/\lambda) = 2\pi/3$ and that through S_1 and S_3 is 2. $2\lambda/3.2\pi/\lambda = 8\pi/3$

$y = A\sin\theta + A\sin(\theta - 2\pi/3) + A\sin(\theta - 8\pi/3)$

$\therefore$ Resultant amplitude $= \sqrt{3}A$

As, Intensity $\propto$ (Amplitude)^2, new intensity $= 3I$

3. Ans (A)

On comparing optical path we get

$$t\mu_0\left(1 - \frac{Z^2}{Z_0}\right) + \sqrt{Z^2 + F^2} = t\mu_0 + F$$

$$\Rightarrow \sqrt{Z^2 + F^2} = \frac{\mu_0 t Z^2}{Z_0^2} + F$$

$$\Rightarrow Z^2 + F^2 = \mu_0^2 t^2\left(\frac{Z^2}{Z_0^2}\right)^2 + F^2 + 2\mu_0 t\frac{Z^2}{Z_0^2}F$$

since t is very small

$$\Rightarrow Z^2 = 2\mu_0 t\frac{Z^2}{Z_0^2}F \quad \Rightarrow F = Z_0^2/(2\mu_0 t)$$

7. Ans (B)

For an EM wave, direction of wave propagation is given by $\vec{E} \times \vec{B}$ & also $\vec{E}$ & $\vec{B}$ are mutually pependicular & in phase.

8. Ans (B)

$$I = \frac{1}{2}C\epsilon_0 E_0^2$$

$$= \frac{1}{2} \times 3 \times 10^8 \times 8.85 \times 10^{-12} \times 1296 = 1.72 \text{ W/m}^2$$

9. Ans (C)

Average energy density of electric field

$$= \frac{1}{2}\varepsilon_0 E_{rms}^2 \frac{1}{2}\varepsilon_0 E_{rms}^2$$

$$= \frac{1}{2}\varepsilon_0\left[\frac{E_0}{\sqrt{2}}\right]^2 = \frac{1}{4}\varepsilon_0 E_0^2 = \frac{1}{4} \times 8.8 \times 10^{-12} \times 4^2$$

$= 35.2 \times 10^{-12}$ J/m^3

10. Ans (B)

$J_c = \sigma E_0 \sin(\omega t - kx)$

$$i_d = \epsilon\epsilon_0\frac{d\phi_E}{dt} = \epsilon\epsilon_0 A\frac{dE}{dt}$$

$= \epsilon\epsilon_0 \times A E_0\omega\cos(\omega t - kx)$

$Jd = \epsilon\epsilon_0 E_0\omega$

$$\frac{J_c}{J_d} = \frac{\sigma}{\epsilon\epsilon_0\omega}$$

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11. **Ans (B)**

$\vec{E} = 10\hat{j}\cos(6x + 8z)$ =

$10\hat{j}\cos\left[\left(6\hat{i} + 8\hat{k}\right).\left(x\hat{i} + z\hat{k}\right)\right]$

$\therefore\ \vec{k} = 6\hat{i} + 8\hat{k}$: This is the direction of 'c'.

$\vec{c} \times \vec{E}$ is direction of $\vec{B}$: $\left(6\hat{i} + 8\hat{k}\right) \times \hat{j} = 6\hat{k} - 8\hat{i}$

$\therefore \vec{B} = \left(\frac{6\hat{k} - 8\hat{i}}{c}\right)\cos[6x + 8z - 10ct]$ as

$B = \frac{E}{c}$

12. **Ans (B)**

$I = \frac{P}{4\pi r^2} = \frac{E_{rms}^2}{c\mu_0}$

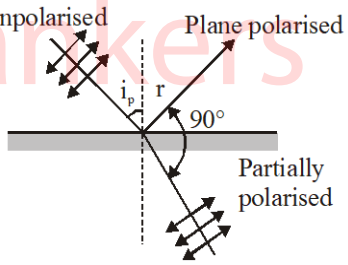
$E_{rms} = \sqrt{\frac{250 \times 3 \times 10^8 \times 4\pi \times 10^{-7}}{4\pi(0.9)^2}} = 96 V/m$

13. **Ans (B)**

Polarization by scattering.

14. **Ans (C)**

From Brewster's law reflected ray is perpendicular to refracted ray.



The reflected ray os obtained is plane polarized having is electric vector in the plane of incidence.

15. **Ans (A)**

Resolving power is given by the receprocal of the minimum separation between two points seen as distinct $= \frac{D}{1.22\lambda L}$

16. **Ans (B)**

$A = A_0\left(\frac{\sin\beta}{\beta}\right)$

or $\frac{A}{A_0} = \frac{\sin\beta}{\beta}$

Now $\beta = \frac{\pi a \sin\theta}{\lambda} = \frac{\pi}{2}$

$\therefore \frac{A}{A_0} = \frac{\sin(\pi/2)}{(\pi/2)} = \frac{2}{\pi}$

17. **Ans (A)**

50

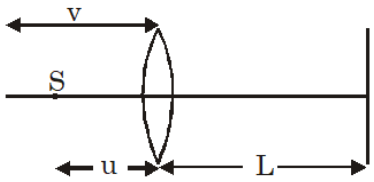
PHYSICS

SECTION-I (ii)

18. **Ans (A,D)**

$\Delta\phi = (2n+1)\ \lambda/2$

19. **Ans (B,C)**



$v = \frac{fu}{f-u}$ on the left of lens

$d = 2\left(\frac{t}{2}\right)\left(\frac{v}{u} - 1\right) = t\left(\frac{u}{f-u}\right)$

$D = L + \frac{uf}{f-u}$

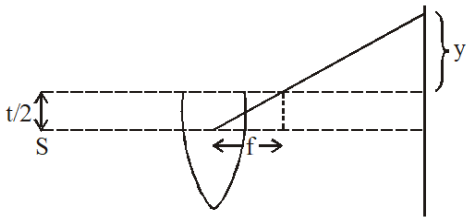
Fringe width

$B = \frac{\lambda\left(L + \frac{uf}{f-u}\right)}{t\left(\frac{u}{f-u}\right)} = \frac{\lambda}{t}\left[f + \frac{L(f-u)}{u}\right]$

$\simeq \frac{\lambda f}{t} = \frac{6 \times 10^{-7} \times 0.4}{10^{-3}}$

= 0.24 mm

$\frac{y}{L-f} = \frac{t}{2f} \Rightarrow y = \frac{t(L-f)}{2f}$



Length of interference pattern

$\frac{t(L-f)}{f} + t$

$= t\left(\frac{L}{f} - 1 + 1\right)$

$= t\frac{L}{f}$

$= \frac{(10-3) \times 50}{40}$

$= \frac{5}{4 \times 10^3} m = \frac{5}{40} cm = \frac{1}{8} cm$

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20. Ans (A,B,C)

$d = 2a(\mu - 1)\alpha$

$\beta = \frac{D\lambda}{d} \Rightarrow \frac{D\lambda}{2a(\mu-1)\alpha} = \beta$

(A) a is decreased so $\beta \uparrow$

(B) $\beta' = \frac{D(\lambda/\mu_m)}{2a\alpha\left(\frac{\mu}{\mu_m}-1\right)} = \frac{D\lambda}{2a\alpha}\frac{1}{(\mu-\mu_m)}$

Now, $(\mu - \mu_m) < (\mu - 1)$

so, $\beta \uparrow$

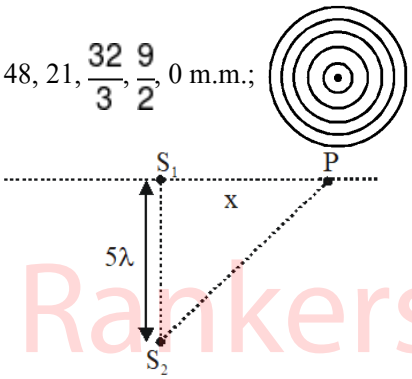
(C) $\alpha \downarrow \Rightarrow \beta \uparrow$

21. Ans (A,B,C)

$\beta = \frac{\mu\lambda}{2\theta}$

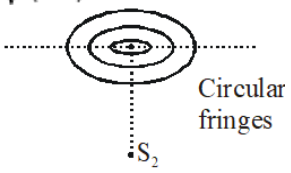
22. Ans (A,B,C)

48, 21, $\frac{32}{3}$, $\frac{9}{2}$, 0 m.m.;



$S_2P - S_1P = n\lambda$

$\sqrt{(5\lambda)^2 + x^2} - x = n\lambda$



23. Ans (A,B)

$OA = y = \frac{3D\lambda}{d}$

$\frac{1}{y} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{10} - \frac{1}{15} = \frac{1}{30}$

v = 30 cm

D = 1.30 − 0.30

D = 1.00 m

$d_0 = m = \frac{v}{u} = \frac{30}{15} = 2$

$d_0 = 2 \times 0.5$ mm = 0.5 mm

$d = 0.5 + 2d_0 = 1.5$ mm

$OA = \frac{3 \times 1 \times 5 \times 10^{-7}}{1.5 \times 10^{-3}}$

$OA = 10 \times 10^{-4}$

OA = 1mm

27. Ans (A,B,C,D)

$\frac{2\pi}{\lambda^2} I_0 \ln\left(\frac{a}{s}\right) \sin(2\pi vt)\hat{k},$

$+\left(\frac{a}{2}\right)^2\left(\frac{2\pi}{\lambda}\right)^2 I_0 \sin 2\pi vt$

29. Ans (A,C)

(a) ; (b)

30. Ans (A,C)

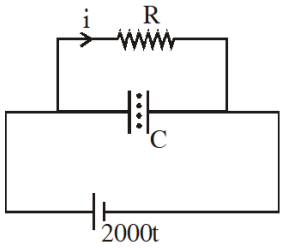
(a) $I_d = \varepsilon_0\frac{d\phi_E}{dt} = \frac{\varepsilon_0 Ad\left(\frac{V}{d}\right)}{dt} = \frac{\varepsilon_0 A}{d}\frac{dq}{dt}$

$= C\frac{dv}{dt} = 5 \times 10^{-6} \times 2000 = 0.01A$

(b) iR = 2000 t

$0.01 \times 4 \times 10^5 = 2000\ t$

$\therefore$ t = 2s



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PHYSICS

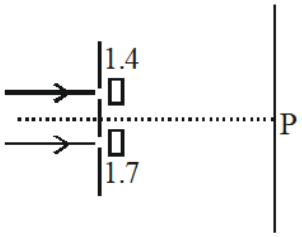
SECTION-I (iii)

35. Ans (A)

PHYSICS

SECTION-II

1. Ans (9.3)



$$\frac{3}{4} = \cos^2\frac{\Delta\phi}{2}$$

$$\Delta\phi = 2n\pi + \frac{\pi}{3} \quad \left(n + \frac{1}{2}\right)\lambda$$

5th max & 6th min

phase diff : 10π & 11π

$\frac{2\pi}{\lambda}(\mu_2 - \mu_1)t = \Delta\phi = 10\pi + \frac{\pi}{3}$ is acceptable

$$t = \frac{31\pi}{3}\frac{x\lambda}{x2\pi \times (\mu - \mu_1)} = \frac{31}{3} \times 5400 \times 10^{-10}$$

2. Ans (1.25)

$\Delta X = 2\ \mu t$

For destructive interference

$2\mu t = \lambda/2$

$$\mu = \frac{\lambda}{4t} = 1.25$$

PHYSICS

SECTION-III

1. Ans (2)

$$\frac{\lambda^2 v}{4\pi^2 \varepsilon_0 a^2}\hat{k}$$

2. Ans (1)

1 mJ

PHYSICS

SECTION-IV

1. Ans (A->R,B->T,C->QR,D->PS)

Path difference remains same on a circle for case D

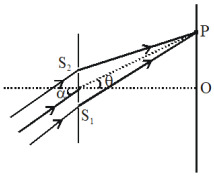
Shape of fringe pattern for pin hole is hyperbolic

Shape of fringe pattern for slit is straight line

Δx_{max} can't be greater than 'd' distance between the source in A, B & C & Δx_{min} can't be less than d-distance between the source in D.

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2. Ans (A->PRST,B->QRST,C->R,D->PRST)

(A) 

$S_1P - (S_2P + d\sin\alpha) = n\lambda$

$d\sin\theta - d\sin\theta = 0$

$\theta = \alpha$

After introduction of sheet

$-(\mu - 1)t + d\sin\theta - d\sin\alpha = 0$

when $q = 0, -(\mu - 1)t = d\sin\alpha$

not possible

$* \beta = \frac{D\lambda}{d}$

$* d\sin\theta - d\sin\alpha = (n + \frac{1}{2})\lambda$: Minima

* Put n = 0 in above equation

$d\sin\theta - d\sin\alpha = \frac{\lambda}{2}$: Least order minima

(B) $* S_1P + (\mu_1 - 1)t - [S_2P + (\mu_2 - 1)t] = n\lambda$

$d\sin\theta + (\mu_1 - \mu_2)t = n\lambda$

$n = 0$

$d\sin q = -(\mu_1 - \mu_2)t$

zero order max. lies below O

$* d\sin\theta + (\mu' - 1)t' - (\mu_2 - \mu_1)t = 0$

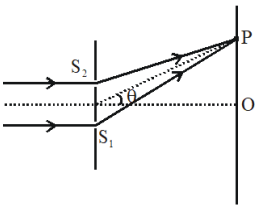
for $\theta = 0°$

$(\mu' - 1)t' = (\mu_2 - \mu_1)t$

$* \beta = \frac{D\lambda}{d}$

* Point O can be minima

* Point O can be least order minima

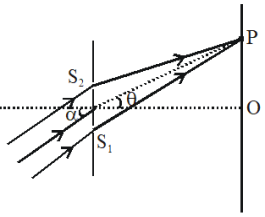
(C) 

$* S_1P - [S_2P + (\mu - 1)t] = n\lambda$

$d\sin\theta - (\mu - 1)t = 0$: CBF

$\sin\theta = \frac{(\mu - 1)t}{d}$: CBF

$* \beta = \frac{D\lambda}{d}$

(D) 

3. Ans (A->P,B->R,C->P,D->S)

Path difference for

$\Delta x_{AB}\frac{2d^2}{D}$ $\Delta x_{AC}\frac{9d^2}{2D}$

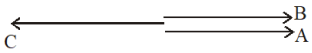
(A) If $\frac{d^2}{D} = \lambda$ then

$\Delta x_{AB} = 2\lambda$ $\Delta x_{AC} = \frac{9}{2}\lambda$

$\Delta\phi_{AB} = \frac{2\pi}{\lambda} \times 2\lambda$

$\Delta\phi_{AC} = \frac{2\pi}{\lambda} \times \frac{9}{2}\lambda = 9\pi$

$= 4\pi$



$I_R = I_0$

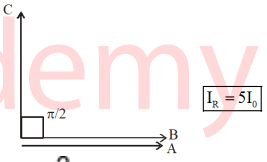
(B) If $\frac{d^2}{D} = \frac{\lambda}{2}$

$\Delta x_{AB} = 2 \times \frac{\lambda}{2} = \lambda$

$\Delta x_{AC} = \frac{9}{2} \times \frac{\lambda}{2} = \frac{9\lambda}{4}$

$\Delta\phi_{AB} = \frac{2\pi}{\lambda} \times \lambda = 2\pi$

$\Delta\phi_{AC} = \frac{2\pi}{\lambda}\Delta x = \frac{2\pi}{\lambda} \times \frac{9\lambda}{4} = \frac{9\pi}{2}$



$I_R = 5I_0$

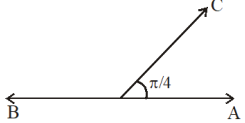
(C) If $\frac{d^2}{D} = \frac{\lambda}{4}$

$\Delta x_{AB} = 2 \times \frac{\lambda}{4} = \frac{\lambda}{2}$

$\Delta x_{AC} = \frac{9}{2} \times \frac{\lambda}{4} = \frac{9\lambda}{8}$

$\Delta\phi_{AB} = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi$

$\Delta\phi_{AC} = \frac{2\pi}{\lambda} \times \frac{9\lambda}{8} = \frac{9\pi}{4}$



$I_R = I_0$

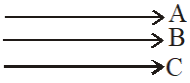
(D) If $\frac{d^2}{D} = 2\lambda$

$\Delta x_{AB} = 2 \times 2\lambda = 4\lambda$

$\Delta x_{AC} = \frac{9}{2} \times 2\lambda = 9\lambda$

$\Delta\phi = \frac{2\pi}{\lambda} \times 4\lambda = 8\lambda$

$\Delta\phi_{AC} = \frac{2\pi}{\lambda}9\lambda = 18\pi$



$I_R = 9I_0$

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SOLUTION

PHYSICS

SECTION-I (i)

1. Ans (D)

$E = a(1 + \cos 2\pi f_1 t)\cos 2\pi f_2 t = a\cos 2\pi f_2 t + a\cos 2\pi f_1 t \cos 2\pi f_2 t$

$\Rightarrow E = a\cos 2\pi f_2 t + \frac{1}{2} a\cos 2\pi(f_1 + f_2)t + \frac{1}{2} a\cos 2\pi(f_1 - f_2)t$

This is a complex vibration consisting of harmonic vibrations of frequencies f_2, $(f_1 + f_2)$ and $(f_1 - f_2)$

The highest is $(f_1 + f_2)$. So, $h\nu = \phi + T_{max}$

$T_{max} = h(f_1 + f_2) - \phi$

$= \frac{6.6 \times 10^{-34}}{1.6 \times 10^{-19}} (3.6 \times 10^{15} + 1.2 \times 10^{15}) - 2.35 =$ 17.45 eV

2. Ans (A)

$p = \frac{h}{\lambda}$

$K.E. = \frac{p^2}{2m} = \frac{h^2}{2m\lambda^2}$

If entire K.E. of electron is converted into photon then

$\frac{h^2}{2m\lambda^2} = \frac{hc}{\lambda_0}$

$\lambda_0 = \frac{2mc\lambda^2}{h}$

3. Ans (D)

Transition from n = 3 to 2 (first line OF Balmer Series corresponds to longest wavelength), photon emitted with momentum

$P_{photon} = \frac{h}{\lambda} = Rh\left(\frac{1}{3^2} - \frac{1}{2^2}\right) = \frac{5Rh}{36}$

Apply Momentum Conservation

$P_{photon} = P_{atom}$

$\frac{5Rh}{36} = MV$

$V = \frac{5Rh}{36M}$

4. Ans (D)

From Moseley's law, $\sqrt{\nu} = a(Z - b)$

$\frac{1}{\lambda} \propto k(Z - b)^2$ {k is a constant}

For K_α : b = 1

$\frac{\frac{1}{\lambda_{K_\alpha \text{copper}}}}{\frac{1}{\lambda_{K_\alpha \text{impurity}}}} = \left(\frac{Z_{Cu} - b}{Z_{imp} - b}\right)^2 \Rightarrow$

$\frac{\lambda_{K_\alpha \text{impurity}}}{\lambda_{K_\alpha \text{copper}}} = \left(\frac{Z_{Cu} - b}{Z_{imp} - b}\right)^2$ so $Z_{imp} = 26$

5. Ans (B)

$\lambda_{min} = \frac{hc}{eV}$

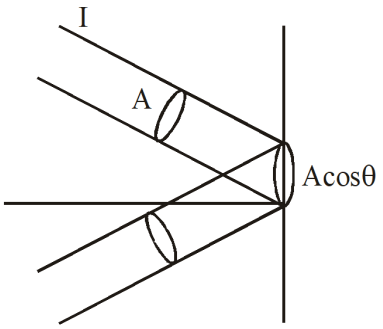
Cut off wavelength depends on the energy of the accelerated electrons and is independent of nature of target.

$\lambda_{K\alpha} \propto \frac{1}{(z-b)^2}$

characteristic wavelength depend on atomic no and cut off wavelength depend on energy of e^-.

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6. Ans (A)



$$\left(\frac{dp}{dt}\right)_{\text{incident}} = \frac{I}{C}A\cos^2\theta$$

$$\left(\frac{dp}{dt}\right)_{\text{reflected}} = \frac{-I}{C}A\,0.5\cos^2\theta$$

$$F = \frac{3}{2}\frac{I}{C}A\cos^2\theta$$

$$\frac{F}{A} = P = \frac{3I}{2C}\cos^2\theta = \frac{3I}{4C}$$

7. Ans (B)

Released energy = $2 \times 4 \times 7 - 2 \times 1 - 7 \times 5.4 =$ 16 MeV

8. Ans (A)

Equation of β^+-decay of

${}_6C^{11}$; ${}_6C^{11} \longrightarrow {}_5B^{11} + {}_{+1}\beta^0 + \nu + Q$

Q-value of reaction = Δmc^2

=

$$\left[m\left({}_6C^{11}\right) - 6m_e - m\left({}_5B^{11}\right) + 5m_e - m_e\right]c^2$$

$$= \left[m\left({}_6C^{11}\right) - m\left({}_5B^{11}\right) - 2m_e\right]c^2$$

=

$$[11.011434 - 11.009305 - 2 \times 0.000548]\,uc^2$$

$= [0.001033]\ uc^2 = 0.001033 \times 931.5$ MeV

$= 0.962$ MeV

9. Ans (C)

$\lambda_\alpha = \frac{1}{1620}$ per year and $\lambda_\beta = \frac{1}{405}$ per year and it is given that the fraction of the remained activity $\frac{A}{A_0} = \frac{1}{4}$

Total decay constant

$$\lambda = \lambda_\alpha + \lambda_\beta = \frac{1}{1620} + \frac{1}{405} = \frac{1}{324} \text{ per year}$$

We know that $A = A_0e^{-\lambda t} \Rightarrow t = \frac{1}{\lambda}\log_e\frac{A_0}{A}$

$$\Rightarrow t = \frac{1}{\lambda}\log_e 4 = \frac{2}{\lambda}\log_e 2$$

$= 324 \times 2 \times 0.693 = 449$ years

10. Ans (B)

$$6.9\,\% = \frac{6.9}{100} = \frac{1}{14.5} \approx \frac{1}{16} \approx \frac{1}{2^4}$$

Total time is little less than $4t_{1/2} = 4 \times 12.3$

$= 49.2$

So answer T slightly less than $4t_{1/2}$

$T = 47.5$

11. Ans (C)

Electrostatic energy = $BE_N - BE_O$

$= \left[[7M_H + 8M_n - M_N] - [8M_H + 7M_n - M_O]\right] \times C^2$

$= [-M_H + M_n + M_O - M_N]C^2$

$= [-1.007825 + 1.008665 + 15.003065 - 15.000109] \times 931.5$

$= +3.5359$ MeV

$$\Delta E = \frac{3}{5} \times \frac{1.44 \times 8 \times 7}{R} - \frac{3}{5} \times \frac{1.44 \times 7 \times 6}{R} = 3.5359$$

$$R = \frac{3 \times 1.44 \times 14}{5 \times 3.5359} = 3.42\text{fm}$$

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PHYSICS

SECTION-I (ii)

12. Ans (A,B,D)

$$E_{\text{excitation}} = \frac{1}{2}\frac{m}{2}u^2 = E_2 - E_1 = \frac{E_0}{2}$$

$$\lambda_0 = \frac{hc}{\sqrt{2m_e[(E_2 - E_1) - \phi]}}$$

$$= \frac{hc}{\sqrt{2m_e\left[\frac{E_0}{2} - \phi\right]}} = \frac{hc}{\sqrt{m_eE_0 - 2\sigma m_e}}$$

Also $\Rightarrow$ Energy transferred

$$\Rightarrow \quad \frac{1}{2}mu^2 - \frac{1}{2}m\left(\frac{u}{2}\right)^2 = mu^2\left(\frac{1}{2} - \frac{1}{8}\right)$$

$$= mu^2\left(\frac{3}{8}\right) = \frac{mu^2}{4}, \frac{3}{2} = \frac{E_0}{2} \cdot \frac{3}{2} = \frac{3E_0}{4}$$

Also For second case,

$$E_{\text{excitation}} = \frac{1}{2}\frac{4m}{5}u'^2 = E_2' - E_1' = 4(E_2 - E_1) = \frac{4}{5}E_0'$$

$$E_0' = 5(E_2 - E_1) = \frac{5E_0}{2}z$$

13. **Ans (B,C)**

$A = A_0e^{-\lambda t}$

$6.932 \times 10^{18} = A_0e^{-\frac{\ell n2}{40}(80)}$

$= A_0e^{-\ell n4} = \frac{A_0}{4}$

$A_0 = 4 \times 6.932 \times 10^{18}$

$\lambda N_0 = 4 \times 6.932 \times 10^{18}$

$\frac{0.6932}{40}N_0 = 4 \times 6.932 \times 10^{18}$

$N_0 = 160 \times 10^{19}$

$N_0 = 1.6 \times 10^{21}$

Also $\frac{3}{4}(1.6 \times 10^{21})E_0 = 6 \times 10^8$

$E_0 = 5 \times 10^{-13}$

14. **Ans (B,D)**

From definition of nuclear fussion and nuclear fission.

PHYSICS

SECTION-I (iii)

15. **Ans (B)**

$\frac{hc}{\lambda_0} = \phi_{wf} \Rightarrow \lambda_0 = 6200\text{Å}$

for emission $\lambda < \lambda_0 \Rightarrow 3000 + 40t = 6200 \Rightarrow t =$ 80 sec

16. **Ans (B)**

$\frac{P}{100} = \frac{dN}{dt}\frac{hc}{\lambda}$ for 3 minute $\Rightarrow \frac{dN}{dt} = \frac{\lambda}{hc} \times 1$

$\Rightarrow dN = \int_0^{80} \frac{\lambda}{hc}dt$; $N' = \frac{3.71 \times 10^{21}}{20}$

for one hour $N = N' \times 20 = 3.71 \times 10^{21}$

17. **Ans (B)**

$\left(\frac{dN}{dt}\right) = \frac{P\lambda}{hc} = \frac{100}{hc}(3000 + 40t)$

18. **Ans (A)**

$_{84}P_0^{210} \rightarrow {}_{82}Pb^{206} + {}_2He^4$

Mass converted to energy per reaction is

$\Delta m = m\left({}_{84}P_0^{210}\right) - \left[m\left({}_{82}Pb^{206}\right) + m\left({}_2He^4\right)\right]$

$= 5.25$ MeV $= 8.4 \times 10^{-13}$ Joule

Total amount of energy required $= 1.2 \times 10^7$ Joule

Input Energy $= 1.2 \times 10^7 \times 10 = 1.2 \times 10^8$ J

Number of reaction required per day

$n = \frac{dN}{dt} = \frac{1}{7} \times 10^{21}$ per day

If N is the number of polonium atoms required the

$-\frac{dN}{dt} = \lambda N = n = \lambda N$

$N = \frac{n}{\lambda} = \frac{nT}{0.693} = \frac{200}{7} \times 10^{21}$

Mass of each P_0 – atom = 210 amu

Mass of P_0 required after 693 days $= \left(\frac{200}{7}\right) \times 10^{21} \times 210\text{amu} = 1.0$ gm

Initial mass (m_0) of $P_0\left(\frac{m}{m_0}\right) = \left(\frac{1}{2}\right)^{t/T}$

$\frac{t}{T} = \frac{693}{138.6} = 5 \Rightarrow m_0 = (2)^5 \times 1g = 32gm$

Initial number of P_0 atoms

$= N_0 = (2)^{t/T.N} = \times 10^{23}$

Initial activity of P_0 :

$R = -\frac{dN}{dt} = \lambda N_0 = \frac{0.693}{T}N_0 = 4.57 \times 10^{21}$ per day

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19. Ans (A)

$_{84}P_0^{210} \rightarrow {}_{82}Pb^{206} + {}_2He^4$

Mass converted to energy per reaction is

$$\Delta m = m\left({}_{84}P_0^{210}\right) - \left[m\left({}_{82}Pb^{206}\right) + m\left({}_2He^4\right)\right]$$

$= 5.25 \text{ MeV} = 8.4 \times 10^{-13}$ Joule

Total amount of energy required $= 1.2 \times 10^7$ Joule

Input Energy $= 1.2 \times 10^7 \times 10 = 1.2 \times 10^8$ J

Number of reaction required per day

$$n = \frac{dN}{dt} = \frac{1}{7} \times 10^{21} \text{ per day}$$

If N is the number of polonium atoms required the

$$-\frac{dN}{dt} = \lambda N = n = \lambda N$$

$$N = \frac{n}{\lambda} = \frac{nT}{0.693} = \frac{200}{7} \times 10^{21}$$

Mass of each P_0 – atom = 210 amu

Mass of P_0 required after 693 days =

$$\left(\frac{200}{7}\right) \times 10^{21} \times 210\text{amu} = 1.0 \text{ gm}$$

Initial mass (m_0) of $P_0 \left(\frac{m}{m_0}\right) = \left(\frac{1}{2}\right)^{t/T}$

$$\frac{t}{T} = \frac{693}{138.6} = 5 \Rightarrow m_0 = (2)^5 \times 1g = 32gm$$

Initial number of P_0 atoms

$= N_0 = (2)^{t/T.N} = \times 10^{23}$

Initial activity of P_0 :

$$R = -\frac{dN}{dt} = \lambda N_0 = \frac{0.693}{T} N_0 = 4.57 \times 10^{21} \text{ per day}$$

20. Ans (A)

$_{84}P_0^{210} \rightarrow {}_{82}Pb^{206} + {}_2He^4$

Mass converted to energy per reaction is

$$\Delta m = m\left({}_{84}P_0^{210}\right) - \left[m\left({}_{82}Pb^{206}\right) + m\left({}_2He^4\right)\right]$$

$= 5.25 \text{ MeV} = 8.4 \times 10^{-13}$ Joule

Total amount of energy required $= 1.2 \times 10^7$ Joule

Input Energy $= 1.2 \times 10^7 \times 10 = 1.2 \times 10^8$ J

Number of reaction required per day

$$n = \frac{dN}{dt} = \frac{1}{7} \times 10^{21} \text{ per day}$$

If N is the number of polonium atoms required the

$$-\frac{dN}{dt} = \lambda N = n = \lambda N$$

$$N = \frac{n}{\lambda} = \frac{nT}{0.693} = \frac{200}{7} \times 10^{21}$$

Mass of each P_0 – atom = 210 amu

Mass of P_0 required after 693 days =

$$\left(\frac{200}{7}\right) \times 10^{21} \times 210\text{amu} = 1.0 \text{ gm}$$

Initial mass (m_0) of $P_0 \left(\frac{m}{m_0}\right) = \left(\frac{1}{2}\right)^{t/T}$

$$\frac{t}{T} = \frac{693}{138.6} = 5 \Rightarrow m_0 = (2)^5 \times 1g = 32gm$$

Initial number of P_0 atoms

$= N_0 = (2)^{t/T.N} = \times 10^{23}$

Initial activity of P_0 :

$$R = -\frac{dN}{dt} = \lambda N_0 = \frac{0.693}{T} N_0 = 4.57 \times 10^{21} \text{ per day}$$

21. Ans (D)

KE_{max} of β^{-1}.

$Q = 0.8 \times 10^6$ eV

$KE_P + KE_{\beta^-} + KE_{\bar{\nu}} = Q$

KE_P is almost zero

When $KE_\beta = 0$

Then, $KE_{\bar{\nu}} = Q - KE_P$

$\cong Q$

22. Ans (C)

$0 \leqslant KE_{\beta^-} \leqslant Q - KE_P - KE_{\bar{\nu}}$

$0 \leqslant KE_{\beta^-} < Q$

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PHYSICS

SECTION-I (v)

23. Ans (C)

Completing reaction in list II

(1) $^{15}_{8}O \rightarrow ^{15}_{7}N + {}_{1}^{0}e$ (β^+ decay Q)

(2) $^{238}_{92}U \rightarrow ^{234}_{90}Th + {}_{2}^{4}He$ (Alpha decay P)

(3) $^{185}_{83}Bi \rightarrow ^{184}_{82}Pb + {}_{1}^{1}H$ (Proton emission S)

(4) $^{239}_{94}Pu \rightarrow ^{140}_{57}La +$ .....(Fission R)

PHYSICS

SECTION-II (ii)

1. Ans (1.1)

Ans. 1.1×10^{12}

$\lambda_1 = 4144$ Å

$E_1 = \frac{12400}{4144}eV$, $E_1 = 3$ eV

$\lambda_2 = 4972$ Å

$E_2 = \frac{12400}{4972} = 2.5\,eV$

$\lambda_3 = 6216$

$E_3 = \frac{12400}{6216} = 2\,eV$

number of photo electron emitted in 2 second

$$= 2 \times \left(\frac{1.2 \times 10^{-3}}{3 \times 1.6 \times 10^{-19}} + \frac{1.2 \times 10^{-3}}{2.5 \times 1.6 \times 10^{-19}} \right) \times 10^{-4}$$

$$= 2 \times 10^{16} \left(\frac{1.2}{4.8} + \frac{1.2}{400} \right) \times 10^{-4}$$

$= 1.1 \times 10^{12}$

2. Ans (151.00)

Ans. KE ≅ 151 eV

Energy $= \frac{p^2}{2M}$

$$= \frac{h^2}{2\lambda^2 m_\alpha}$$

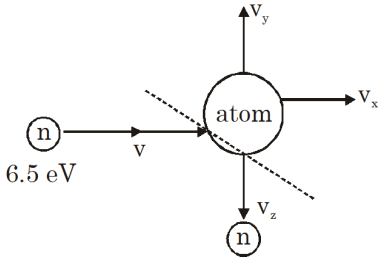
$$= \frac{\left(6.63 \times 10^{-34}\right)^2}{2 \times \left(0.5 \times 10^{-10}\right)^2 \times 9.11 \times 10^{-31}}$$

$$= \frac{9.65 \times 10^{-17}}{4} = 2.412 \times 10^{-17} = 151 eV$$

3. Ans (16.640 to 24.20)

Allowed values of energy of neutron = 6.36 eV and 0.312 eV ; Allowed values of energy of He atom = 17.84 eV and 16.328 eV ,

He+



$m_{He}v_y = m_n v_z \Rightarrow 4v_y = v_z$

$m_n v = m_{He} v_x \; v_x = \frac{v}{4}$

$\frac{1}{2}m_n v^2 = 65ev$

$\frac{1}{2}m_n v^2 = \frac{1}{2}m_{He}v_x^2 + \frac{1}{2}m_{He}v_y^2 + \frac{1}{2}m_n v_z^2$ + energy in excitation

$65ev = \frac{65ev}{4} + \frac{1}{2}4m_0 v_y^2 + \frac{1}{2}m_0 v_z^2$ + energy in excitation

$65 \times \frac{3}{4} = m_0\left(2v_y^2 + 8v_y^2\right)$ + energy in excitation

$65 \times \frac{3}{4} = 10m_0 v_y^2$ + energy in excitation

$K.E_{possible}$ for neutron $= (48.75 - 40.8) \times \frac{16}{20} = 6.36$

or $(48.75 - 12.09) \times \frac{16}{20} = 0.312$

$K.E_{atom} = \frac{65}{4} + \frac{6.36}{4}$ or $\frac{65}{4} + \frac{0.312}{4}$

= 17.84 or 16.328

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PHYSICS

SECTION-III

1. **Ans (2)**

K = U + K

$$\frac{p_1^2}{2m} = K_1 = 16eV$$

$$\frac{p_2^2}{2m} = K_2 = 4eV$$

$$\lambda_1^2 \times 16 = \lambda_2^2 \times 4$$

$$2 = \frac{\lambda_2}{\lambda_1}$$

2. **Ans (8)**

Given distance between electrode = 4×10^{-2}m

Average distance travelled by electrons between successive collision = 8×10^{-5} m.

Thus,

$$\frac{eV}{4 \times 10^{-2}} \times 8 \times 10^{-5} = 16eV \Rightarrow V = 8keV$$

3. **Ans (8)**

As we know,

$$F = qvB = m\frac{v^2}{R} \Rightarrow v = \frac{q}{m}BR$$

The kinetic energy of the photoelectron

$$\frac{1}{2}mv^2 = \frac{1}{2}\frac{e^2B^2R^2}{m}$$

$$= \frac{1}{2}\frac{\left(1.6 \times 10^{-19}\right)^2\left(2 \times 10^{-2}\right)^2\left(23 \times 10^{-3}\right)^2}{\left(9.1 \times 10^{-31}\right)}$$

$$= 2.97 \times 10^{-15} \text{ J}$$

$$= \frac{2.97 \times 10^{-15}}{1.6 \times 10^{-19}} = 18.4 \text{ KeV}$$

Energy of the incident photon = $\frac{hc}{\lambda}$

$$= \frac{12.4}{0.50} = 24.8keV$$

Therefore binding energy = 24.8 − 18.4 = 6.4 KeV

= 6400 eV

4. **Ans (3)**

Atom will exist if size of nucleus is less than the orbit of π-meson

$$\therefore 1.6 \times 10^{-15}Z^{1/3} < \frac{0.53}{265Z} \Rightarrow Z < 37.5$$

$$r = \frac{n^2h}{4\pi^2Kzem_\pi}$$

$$r = \frac{\varepsilon_0h^2}{\pi m_\pi e^2} \times \frac{n^2}{z} \Rightarrow \left(1^{st} \text{ orbit}\right)$$

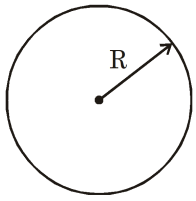
$$r = \frac{0.53}{265} \times \frac{1}{Z}$$

Atom will exist if size of nucleus is less than the orbit of π-meson

$$\therefore 1.6 \times 10^{-15} Z^{1/3} < \frac{0.53}{265Z} \Rightarrow Z < 37.5$$

$\therefore Z_{max} = 37$

5. **Ans (7)**



R = 1 cm

ϕ = 4.7 eV

$$\frac{hc}{\lambda} = \phi - eV$$

$$\frac{1240\,(ev)\,(nm)}{200\,(nm)} = 4.7\,(eV) + eV$$

$$\frac{1240}{200}e = 4.7eV + eV$$

6.2 − 4.7 = V ∴ V = 1.5 volt

$$\frac{1}{4\pi\epsilon_0}\frac{Q}{R} = 1.5$$

$$\left(9 \times 10^9\right)\frac{Ne}{\frac{1}{100}} = 1.5$$

9×10^{11} Ne = 1.5;

$$N = \frac{1.5}{9 \times 10^{11} \times 1.6 \times 10^{-19}} = \frac{15}{16} \times \frac{1}{9} \times 10^8$$

$$= \frac{5}{3 \times 16} \times 10^8 = \frac{50}{48} \times 10^7$$

Z = 7

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6. Ans (6)

$$5 \times \frac{\ell n 2}{15} = \ell n\left(\frac{3.7 \times 10^4}{296 \times v}\right)$$

$$2^{1/3} = \frac{3.7 \times 10^4}{296 \times v} \times 60$$

$v = 6000\ cm^3$

$v = 6$ litre

7. Ans (8)

$$\left|\frac{dN}{dt}\right| = \lambda N_0 e^{-\lambda t} \quad \therefore \ell n\left|\frac{dN}{dt}\right| = \ell n(N_0\lambda) - \lambda t$$

$$\therefore \ell n\left|\frac{dN}{dt}\right| = \ell n(N_0\lambda) - \lambda t \Rightarrow -\lambda = \text{slope} = -\frac{1}{2}\text{year}^{-1}$$

$$\Rightarrow \lambda = \frac{1}{2}yr^{-1}$$

$$\therefore t_{1/2} = \frac{0.693}{\lambda} = 1.386$$ given time is 3 times of $t_{1/2}$ $\therefore$ value of p is 8.

8. Ans (5)

$$A = \frac{A_0}{2^n} \Rightarrow 2^n = \frac{6800}{425}$$; n = No. of half life

$2^n = 16 = 2^4$

$\Rightarrow n = 4$

$\Rightarrow 4 \times t_{1/2} = 10$

$\Rightarrow t^{1/2} = 2.5h$

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SOLUTION

PHYSICS

SECTION-I (i)

1. Ans (C)

Least count = 1MSD − 1VSD = 0.5 mm $-\left(\frac{49}{50}\right)$ (0.5 mm) = 0.001 cm

Reading = 7.45 + 29 × 0.001 = 7.479 cm

2. Ans (C)

$m = \frac{F}{a} = 1kg$

$\frac{dm}{m} = \frac{dF}{F} + \frac{da}{a}$

$= \frac{0.1}{2} + \frac{0.1}{2} = 0.1$

3. Ans (B)

Least count = $\frac{\text{pitch}}{\text{No. of divisions}} = \frac{2}{200} = 0.01mm$

Reading = 3 × 2 + (46 − 5) (0.01) = 6.41 mm

4. Ans (B)

Least count = $\frac{\text{pitch}}{\text{No. of divisions}} = \frac{0.5}{50} = 0.01$ mm

Diameter of wire = 6 × 0.5 + 46 × 0.01 = 3.46 mm

5. Ans (A)

From screw gauge z.e. = + 0.20 mm

Dia of chalk = 3.5 − 0.2 = 3.3 mm = 0.33 cm

For vernier callipers, z.e. = − 0.05 cm

Thus it must show 0.33 − 0.05 = 0.28 cm

so (a) is correct

7. Ans (A)

$F_B = W - W'$

$= 1 \pm (0.1)$

$\frac{\rho_0 Vg}{\rho_w Vg} = \frac{5 \pm 0.05}{1 \pm 0.1}$

$\frac{\rho_0}{\rho_w} = \frac{5 \pm 0.05}{1 \pm 0.1}$

$= (5) \pm (\Delta\rho)$

$\frac{\Delta\rho}{5} = \frac{0.05}{5} + \frac{0.1}{1}$

$= 0.01 + 0.1 = 0.11$

8. Ans (C)

$\frac{\Delta u}{u} \times 100 = \left(\frac{2\Delta d}{d} + \frac{\Delta v}{v} + \frac{\Delta \ell}{\ell}\right) \times 100$

$\frac{\Delta \ell}{\ell} = \frac{30}{1800}$

$= \left[2 \times \frac{0.04}{1-6} + \left(\frac{30}{1800}\right)\right] \times 100$

= 8.166% ≈ 8.2%

9. Ans (C)

Least count = $\frac{\text{Pitch}}{\text{Number of division of circular scale}}$

$= \frac{1mm}{100} = 0.01mm = 0.001cm$

Zero error = −0.03 mm

Measurement = 3.76 − (−0.03) = 3.79 mm

10. Ans (C)

Chritina and Alice both measure error of 2 mm while bob measures error of 4 mm.

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11. **Ans (C)**

$$\frac{1}{2f\sqrt{\mu}} = 8 \times 10^{-3}$$

$$\frac{1}{2 \times 10^{-2} \times 8 \times 10^{-3}} = f$$

$$f = \frac{10^5}{16} = \frac{100 \times 100 \times 10}{4 \times 4}$$

$= 6250$ Hz

$$\frac{\Delta f}{f} = \frac{1}{2}\frac{\Delta \mu}{\mu} + \frac{\Delta \text{slope}}{\text{slope}}$$

$$= \frac{1}{10} + \frac{0.3}{80} = \frac{11}{80}$$

$$\Delta f = 6250 \times \frac{11}{80} = 859.8\text{Hz}$$

12. **Ans (D)**

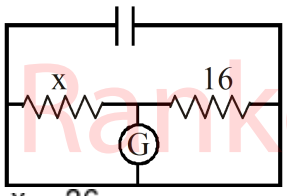
$R = \sqrt{AB}$

$R = r \pm \Delta r$

where, $r = \sqrt{ab}$

and $\frac{\Delta r}{r} = \frac{1}{2}\frac{\Delta a}{a} + \frac{1}{2}\frac{\Delta b}{b}$

13. **Ans (B)**



$$\frac{x}{16} = \frac{36}{64} = 9\Omega$$

$$\frac{dx}{x} = \frac{d\ell}{\ell(1-\ell)} = \frac{0.5 \times 10^{-1}}{36 \times 64}$$

$$dx = \frac{0.05 \times 9}{36 \times 64} = \frac{1}{5120}\Omega$$

$$x = 9 \pm \frac{1}{5120}\Omega$$

14. **Ans (B)**

1MSD = 1 mm, 1VSD = 0.4 mm L.C = 0.2 mm

Zero error = $-(5 \times 0.4 - 1)$mm = -1mm

Reading = $22 + (1 - 2 \times 0.4)$ − zero error

$= 22 + 0.2 + 1$

$= 23.2$ mm

$\Rightarrow$ report = $(23.2 \pm 0.2$ mm)

15. **Ans (B)**

$$Y = \frac{F/A}{\Delta L/L} = \frac{\frac{10}{\pi}(4 \times 10^{-8})}{\frac{0.8 \times 10^{-2}}{2}}$$

$$= \frac{20}{0.8 \times 4 \times \pi \times 10^{-11} \times 2} = 2 \times 10^{11}$$

$$\frac{\Delta y}{y} = 2\frac{\Delta d}{d} + \frac{\text{Error}(\Delta L)}{\Delta L}$$

$$= 2\frac{0.01}{0.4} + \frac{0.04}{0.8} = \frac{0.08}{0.8} = 0.1$$

16. **Ans (D)**

Lesser the least count, more precise the instrument is

$LC_1 = \frac{1}{200}$, $LC_2 = \frac{0.5}{400}$, $LC_3 = 0.1$ mm, $LC_4 = 0.2$ mm

17. **Ans (C)**

Voltmeter is connected in parallel & Ammeter in series as shown in option (C).

PHYSICS

SECTION-I (ii)

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18. **Ans (B)**

$$r = \left(\frac{1-a}{1+a}\right)$$

$$\frac{\Delta r}{r} = \frac{\Delta(1-a)}{(1-a)} + \frac{\Delta(1+a)}{(1+a)}$$

$$= \frac{\Delta a}{(1-a)} + \frac{\Delta a}{(1+a)}$$

$$= \frac{\Delta a(1+a+1-a)}{(1-a)(1+a)}$$

$$\therefore \Delta r = \frac{2\Delta a}{(1-a)(1+a)}\frac{(1-a)}{(1+a)} = \frac{2\Delta a}{(1+a)^2}$$

19. **Ans (C)**

$N = N_0 e^{-\lambda t}$

$\ell nN = \ell nN_0 - \lambda t$

$$\frac{dN}{N} = -d\lambda t$$

Converting to error,

$$\frac{\Delta N}{N} = \Delta\lambda t$$

$$\therefore \Delta\lambda = \frac{40}{2000 \times L} = 0.02$$ (N is number of nuclei left undecayed)

PHYSICS

SECTION-I (iii)

20. Ans (B ,D)

$\ell_1 + e = \frac{\lambda}{4}$ Case-1 :

$\frac{\Delta v}{v}\% = \frac{0.1 \times 100}{33} \approx 0.3\%$

$\ell_2 + e = \frac{3\lambda}{4}$ $e = \frac{50-51}{2} = -ve$

$\lambda = 2(\ell_2 - \ell_1)$ Case-2 :

$\frac{\Delta v}{v}\% = \frac{0.1 \times 100}{50.8} \approx 0.2\%$

$\Rightarrow v = f\lambda$ $\Rightarrow$

$e = \frac{76-75.6}{2} = 0.2 = +ve$

$\frac{\Delta v}{v} = \frac{(\Delta\ell_2 + \Delta\ell_1)}{\ell_2 - \ell_1}$ Case-3 :

$\frac{\Delta v}{v}\% = \frac{0.1 \times 100}{30.4} \approx 0.33\%$

$e = \frac{\ell_2 - 3\ell_1}{2}$ $e = \frac{46.0-46.8}{2} = -ve$

Case-4 : $\frac{\Delta v}{v}\% = \frac{0.1 \times 100}{31.5} \approx 0.32\%$

$e = \frac{48.2-48.0}{2} = 0.1cm$

21. Ans (A,C)

Using lens formula and data given in graph.

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ ...(1)

$\frac{1}{f} = \frac{1}{10} - \frac{1}{-10}$

$f = +5$

By differentiating eq. (1)

$\Rightarrow \frac{-\Delta f}{f^2} = \frac{-\Delta v}{V^2} + \frac{\Delta u}{u^2} = \frac{1}{+V^2}\Delta v + \frac{1}{u^2}\Delta u +$

$\Rightarrow +\frac{\Delta f}{5^2} = \frac{1 \times (0.1)}{+10^2} + \frac{1 \times (0.1)}{10^2}$

$\Delta f = \frac{0.2}{100} \times 25 = \frac{0.2}{4} = 0.05$

so, $f = 5 \pm 0.05$

Fractional error is 0.01

22. Ans (B,C)

Percentage errors in calculation of a, b and M are

$\frac{\Delta a}{a} \times 100\% = \frac{0.01}{4} \times 100\% = 0.25\%$

$\frac{\Delta b}{b} \times 100\% = \frac{0.01}{2} \times 100\% = 0.5\%$

$\frac{\Delta M}{M} \times 100\% = \frac{0.003}{0.3} \times 100\% = 1\%$

Moment of inertia about the given axis

$I = \frac{1}{2}M\left(a^2 + b^2\right)$

$\frac{\Delta I}{I} = \frac{\Delta M}{M} + \frac{2a\Delta a + 2b\Delta b}{a^2 + b^2}$

$= 0.01 + \frac{2(4)(0.01) + 2(2)(0.01)}{20}$

$\frac{\Delta I}{I} \times 100\% = 1 + 5\{0.08 + 0.04\} = 1.6\%$

24. Ans (A,C,D)

$\frac{R_x}{x} = \frac{R}{1-x}$

$\frac{R_x}{20} = \frac{20}{80}$

$R_x = 5$

$R_x = \frac{x}{1-x}[R]$

$\Delta R_x = R\left[\frac{\Delta x(1-x) - x(-\Delta x)}{(1-x)^2}\right]$

$\Delta R_x = 0.03\Omega$

error will be minimused when null-point is found at middle of wire.

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PHYSICS

SECTION-III

1. Ans (0)

$$dv = -\frac{1}{2}\frac{v_0 T_0^{1/2} e^{\left[\frac{2}{3}\frac{(T-T_0)}{T_0}\right]}}{T^{3/2}}dT + v_0\left(\frac{T_0}{T}\right)^{1/2} e^{\left[\frac{2}{3}\frac{(T-T_0)}{T_0}\right]} = \frac{2}{3T_0}dT$$

$$\frac{dv}{v} = -\frac{1}{2}\frac{dT}{T} + \frac{2dT}{3T_0} = \left(\frac{2}{3T_0} - \frac{1}{2T}\right)dT$$

At $T = \frac{3T_0}{4}, \frac{dv}{v} = 0$

$\therefore$ % error = 0

2. Ans (7)

$(7 \pm 1) \times 10^{-3}$ T

3. Ans (4)

$f = \frac{v}{2(\ell + 2e)}$ where e = end correction = 0.6 r

$\therefore f = \frac{v}{2(\ell + 2 \times 0.6r)} = \frac{v}{2(\ell + 1.2r)}$ $\therefore$ $\frac{\Delta f}{f} = \frac{\Delta v}{v} - \frac{\Delta(\ell + 1.2r)}{\ell + 1.2r} = \frac{\Delta v}{v} - \frac{\Delta\ell + 1.2\Delta r}{\ell + 1.2r}$

here $\frac{\Delta v}{v} = 0$ (given) $\frac{\Delta f}{f} \times 100 = -\frac{\Delta\ell + 1.2\Delta r}{\ell + 1.2r} \times 100$

for maximum % error : $\Delta\ell = 0.1$, $\Delta r = 0.05$

$\left(\frac{\Delta f}{f} \times 100\right)_{max} = \frac{0.1 + 1.2 \times 0.05}{94 + 1.2 \times 5} \times 100 =$ 0.16%

4. Ans (1)

$$\tan\theta = \frac{r\omega^2}{g} = \frac{R(\sin\theta)\omega^2}{g}$$

or $\cos\theta = \frac{g}{R\omega^2}$, i.e., $\frac{R-h}{h} = \frac{g}{R\omega^2}$

$h = R - (g/\omega^2)$

$g = \omega^2(R - h)$

$dg = -\omega^2 dh$, i.e., $(\Delta g)_{min} = -(\omega^2)_{min}(\Delta h)_{min}$

So substituting the value of ω_{min} and $(\Delta h)_{min}$

$(\Delta g)_{min} = -100 \times 10^{-4}$ m/s^2

5. Ans (3)

D = (1.25 mm ± 0.01 mm)

$Y = \frac{FL}{A\ell}$

ℓ = elongation.

% error in Y = $\frac{\Delta L}{L} \times 100 + \frac{2\Delta D}{D} \times 100 + \frac{\Delta\ell}{\ell} \times 100$

$= \left(\frac{0.01}{100} \times 100\right) + \left(\frac{0.02}{1.25} \times 100\right) + \left(\frac{0.001}{0.125} \times 100\right) = 2.41$

n = 3.

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